

Grounds for Industrial Growth^{*}

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Abstract

This paper examines the effects of land market liberalization on manufacturing firms through the staggered repeal of India's Urban Land Ceiling and Regulation Act (UL-CRA), which severely restricted urban land transactions. The Repeal enabled more productive firms to expand land use and scale up production, thereby reducing land misallocation. This shift in land use also fueled business dynamism: treated firms experienced significantly higher product turnover, and market selection improved through the exit of unprofitable firms and the entry of more dynamic startups. To quantify the aggregate implications of the Repeal, we develop a general equilibrium model of firm dynamics with land market frictions disciplined by our reduced-form evidence. Through the lens of this model, the Repeal increased long-run manufacturing productivity growth by 0.32 percentage points and led to a 7.30% improvement in consumption-equivalent welfare.

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“Often the most valuable asset of an unprofitable firm was land. Since the [Urban Land Ceiling and Regulation] Act prohibited the firm from selling it at the market price, the firm chose not to exit and held onto the precious piece of land.”

– Arvind Panagariya, *India: The Emerging Giant*, 2010

1 Introduction

Restrictions on land property rights are common in developing countries, often characterized by insecure tenure, limits on land ownership, and restrictions on the ability to freely exchange land (World Bank, 2003, 2007). While the development economics literature has long emphasized the importance of land property rights in *agricultural* production (de Janvry, 1981; Lin, 1992; Banerjee et al., 2002; Goldstein and Udry, 2008; Adamopoulos and Restuccia, 2020), there are few attempts to examine their consequences on *manufacturing* firms.¹

In this paper, we address this gap by studying the micro- and macroeconomic impacts of India’s urban land market liberalization on manufacturing firms, both empirically and theoretically. Prior to the reform, urban land markets were tightly controlled by the 1976 Urban Land Ceiling and Regulation Act (ULCRA), which imposed stringent restrictions on land transactions. These restrictions limited firms’ ability to expand, adjust their landholdings, and exit the market—an issue eloquently captured in the epigraph from Panagariya’s book on India’s economic reforms. Considered a policy failure, the ULCRA was repealed by the central government in 1999.

To identify the causal effects of the liberalization, we assemble hand-collected data on the staggered repeal of the ULCRA across states and on variation in its enforcement within states. Our empirical strategy is a Difference-in-Differences design that compares firms in 64 Urban Agglomerations (UAs)²—where these land-market restrictions applied—to firms outside UAs, before and after the state-level repeal. Using non-UA firms as a comparison group allows us to isolate the effects of the liberalization from other state-level shocks and contemporaneous policies. We show that UA and non-UA firms exhibit similar pre-trends prior to the Repeal, and that estimated effects on non-UA firms are small, which mitigates concerns about spillovers.

¹Three studies examine how land market frictions affect Indian manufacturing firms (Kimura, 2011; Duranton et al., 2017; Sood, 2022). We discuss these papers in detail below.

²An Urban Agglomeration (UA) comprises a core town and its surrounding suburban areas and may span multiple statutory towns and other urban entities such as railway colonies or university campuses. Conceptually, it is similar to a metropolitan statistical area in the United States.

Our results show that, following the Repeal, treated plants expanded their landholdings by about 22% on average, thickening the right tail of the plant-size distribution. This reflected both within-plant expansion and a decline in the number of operating plants. The reform also facilitated land transactions: treated plants increased both land purchases and land sales, while land revaluation did not rise, consistent with lower frictions in buying and selling land rather than with a pure price effect. In addition, the Repeal improved the allocation of land: initially more productive plants increased their landholdings by 16 to 19%, while low-productivity plants did not. Consequently, the liberalization of the land market unleashed industrial growth. Treated plants expanded production by acquiring 16% more fixed assets (excluding land) and hiring 7% more labor, leading to a 17% increase in sales and a 13% increase in physical productivity.

The land market liberalization also significantly reshaped business dynamism and market selection. Following the Repeal, the rates at which plants introduced new products and discontinued old ones increased by roughly 14 and 15 percent, respectively, reflecting intensified creative destruction.³ These margins are directly observed from plants' product portfolios, an unusual advantage of the ASI data. To our knowledge, this provides some of the first causal evidence that factor-market distortions affect product innovation (Aghion et al., 2023; Bertrand et al., 2021). Furthermore, we find that the Repeal improved the selection of plants: unprofitable or dormant plants became more likely to exit, while the composition of new entrants shifted toward high-productivity, high-growth plants.

To quantify the macroeconomic consequences of the Repeal on the Indian manufacturing sector, we develop a general equilibrium model of endogenous growth disciplined by our reduced-form causal estimates. Our model builds on the framework of Klette and Kortum (2004), which is particularly well suited to our setting as we directly observe plants' product portfolios and thus their product innovation decisions. The economy features two types of firms: *dynamic* firms that can expand into multi-product firms, and *stagnant* firms that operate a single product line throughout their lifecycle.⁴ We model the ULCRA as a size-dependent wedge on the effective rental cost of land. As discussed in Section 2, this formulation captures the way the Act raised the cost of adjusting landholdings, especially for larger firms facing greater regulatory scrutiny.

³The detailed product information also enables us to explore the broader consequences of the Repeal. The liberalization likely reduced marginal costs for treated plants, potentially allowing them to pass these gains on to consumers through lower prices. We find that treated plants reduced their product prices by roughly 10% without raising their markups, indicating that the gains were largely passed through to consumers.

⁴This distinction is motivated in part by the ASI evidence that more than half of plants never introduce a new product over their observed life cycle, while a smaller set of plants accounts for most product creation.

We calibrate the model to match cross-sectional moments of the Indian manufacturing sector together with our DiD estimates of the Repeal’s causal effects on the marginal revenue product of land (MRPLand), product turnover, and entry and exit. The model closely replicates the main treatment effects from our empirical analysis. These moments discipline two distinct aspects of the land market distortions. The decline in the MRPLand indicates that the Repeal lowered the effective cost of transacting land. In the model, this disciplines the average magnitude of the land wedge. The joint responses of product turnover, exit, and entry composition then determine whether the reform mattered disproportionately for more scalable plants. In the model, that pattern helps identify how sharply the land wedge rises with size. Beyond these targeted moments, the model also reproduces several untargeted features of the data, including the skewed product distribution and the flat life-cycle growth profile of Indian plants.

In the calibrated model, the counterfactual experiment eliminates the land market wedge. In the new steady state, the rate of creative destruction rises by 1.5 percentage points, as the higher return to expansion induces incumbent firms to introduce more new products and shifts entry toward dynamic firms. The model therefore predicts a compositional upgrading within the manufacturing sector: the dynamic share of entrants rises from 19.1% to 21.7%, and dynamic firms’ share of products rises from 40.1% to 44.5%. Together, these changes lead to a 0.32 percentage point acceleration in long-run aggregate productivity growth.

Accounting for transition dynamics, the Repeal is equivalent to a 7.30% permanent increase in consumption per capita. Growth acceleration is the largest component, contributing 5.96 percentage points to log welfare. Eliminating the government’s valueless use of final goods contributes a further 2.71 percentage points, while the change in resources allocated to product expansion and entry contributes -1.63 percentage points. The static gain from removing the cross-sectional dispersion of land distortions is negligible (+0.00 p.p.).

Literature review

Our study contributes to several strands of literature. First, it adds to the well-established literature on land market frictions and land reforms in developing countries (de Janvry, 1981; Lin, 1992; Besley and Burgess, 2000).⁵ In the agricultural sector, existing studies have

⁵India enacted the most extensive body of postcolonial land reforms compared to any other country, but empirical assessments are challenging due to state-specific variations in policies. Besley and Burgess (2000) categorize roughly 80 reforms into four broad categories and find mixed impacts of land reforms on economic growth. Using the same data, Ghatak and Roy (2007) find significant negative effects on agricultural productivity but emphasize the heterogeneity of effects across states and over time, cautioning

highlighted the important role of property rights, particularly tenancy rights, in enhancing agricultural investment (Banerjee et al., 2002; Goldstein and Udry, 2008), as well as the importance of well-functioning land markets and clear land titles in resource and human capital allocation (de Janvry et al., 2015; Adamopoulos and Restuccia, 2020; Chari et al., 2021; Acampora et al., 2022; Adamopoulos et al., 2022; Liu et al., 2023; Adamopoulos et al., 2024). We extend this body of research by identifying the causal effects of land market frictions on *manufacturing* firms, leveraging novel landholding and land adjustment data in the Indian manufacturing census and the gradual repeal of the ULCRA as a natural experiment.

Second, we contribute to the expanding body of literature on resource misallocation and productivity growth among manufacturing firms in developing countries (Hsieh and Klenow, 2009; Hsieh and Olken, 2014; Hopenhayn, 2014; Restuccia and Rogerson, 2017; Verhoogen, 2023). In the Indian context, manufacturing firms tend to be inefficiently small and remain so throughout their lifecycle due to muted creative destruction (Akcigit et al., 2021; Peters and Zilibotti, 2021). Several recent studies have examined various forces that may hinder manufacturing growth, such as the role of contract enforcement (Boehm and Oberfield, 2018), delegation (Akcigit et al., 2021), industrial licensing (Chari, 2011; Alfaro and Chari, 2015), reservation laws (Martin et al., 2017), FDI restrictions (Bau and Matray, 2023), priority-sector lending (Banerjee and Duflo, 2014), and labor regulations (Amirapu and Gechter, 2019; Bertrand et al., 2021). We provide new evidence on the sources of misallocation and weak business dynamism in developing countries by exploring the role of land market frictions.

Three papers on land market frictions and Indian manufacturing are especially relevant. Kimura (2011) uses the ULCRA repeal to study land collateralization and credit reallocation across urban firms, showing that land deregulation affected land transactions and borrowing capacity. Duranton et al. (2017) quantify the misallocation of land and other factors in Indian manufacturing and relate it to land and property-market distortions. Sood (2022) combines ASI land data with a structural model of land acquisition to study land aggregation costs, fragmentation, and firm growth. Relative to this literature, our contribution is to provide causal evidence on how land market liberalization affects manufacturing plants' land use, production, product turnover, and selection, and to discipline a model of endogenous economic growth through product innovation with those causal estimates.

A growing literature studies how distortionary regulations and market frictions shape innovation incentives and productivity dynamics (Mukoyama and Osotimehin, 2019; König et al., 2022; Aghion et al., 2023; Akcigit et al., 2023). Bento and Restuccia (2017) quantify

against simplistic interpretations.

how the elasticity of distortions with respect to physical productivity affects aggregate output and productivity growth in India and the United States. Their dynamic framework implies larger output losses from factor misallocation than standard static frameworks (Hsieh and Klenow, 2009), but the impact on productivity growth is limited because lower incumbent innovation is offset by greater firm entry. Similarly, Bertrand et al. (2021) develop a model of product innovation to study Indian labor market frictions and find that countervailing effects on incumbent expansion and entry leave the aggregate growth rate largely unchanged. Our results stand in contrast to this pattern: growth acceleration is the largest component of the welfare improvement from the repeal of the ULCRA, alongside a material direct gain from eliminating the government’s wasteful use of final goods.

The remainder of the paper is organized as follows: Section 2 provides background information on the policy context and land market institutions in India. Section 3 describes our datasets and outlines the identification strategy. Section 4 presents our main empirical findings, highlighting the static and dynamic consequences of the Repeal. Section 5 outlines our model and discusses some of its implications. Section 6 details the estimation of the model’s parameters and quantifies both the equilibrium effects and welfare consequences of the Repeal.

2 Institutional context

In this section, we provide a brief overview of how the ULCRA increased landholding and transaction costs by imposing ceilings on vacant land ownership and restricting land transactions. We also present essential background information on the policy context and the eventual repeal of the Act.

The Urban Land (Ceiling and Regulation) Act (ULCRA). After gaining independence, India struggled with a severe urban housing shortage, with a deficit of 3 million housing units in 1971 (Tiwari, 2016).⁶ This shortage became more pressing due to rapid urbanization: between 1951 and 1971, the urban population expanded from 62 million to 109 million. At the same time, policymakers expressed growing concern over speculative landholding, as large landowners retained extensive tracts of vacant urban land, constraining supply and deepening the perceived housing “crisis.” Despite the enactment of numerous rural land legislation, urban land markets remained largely unregulated, with no statutory

⁶The shortage of housing is measured as the number of households living in non-serviceable katcha (non-durable) houses, obsolescent structures, overcrowded dwellings, or without shelter.

limits on land ownership or possession. This confluence of accelerating urban growth, acute housing shortages, and weak urban land regulation culminated in the enactment of the Urban Land (Ceiling and Regulation) Act in 1976.

From its inception, the ULCRA was fundamentally aimed at socializing urban land, an objective that became politically feasible during the National Emergency period (1975–1977), when democratic institutions were effectively suspended. The Act sought to address what policymakers viewed as an “excessive” concentration of urban landholdings, to curb speculative behavior, and to promote a more equitable distribution of land. Its provisions applied to designated urban agglomerations (UAs), defined as core cities and their contiguous suburban areas with populations exceeding 200,000 according to the 1971 Census. A total of 73 UAs were notified under the ULCRA to guide state-level implementation. Ultimately, the Act was adopted by 17 states and applied to all union territories, covering 64 UAs in practice.⁷

Ceilings on vacant landholdings. The ULCRA imposed stringent ceilings on vacant urban landholdings to limit the concentration of land within urban agglomerations. These ceilings applied broadly to individuals, families, firms, companies, and associations, irrespective of their legal or incorporation status. Any land held in excess of the prescribed limits was subject to compulsory acquisition by the state, with compensation set at concessional rates.⁸ These ceiling limits were extremely restrictive, ranging from 500 to 2,000 square meters depending on the size of the urban agglomeration.⁹ Importantly, the Act defined landholdings broadly, encompassing not only owned land but also land held under lease or mortgage, thereby capturing virtually all forms of land possession within its scope.

The definition of “vacant” land, however, remained ambiguous and was frequently interpreted to include any land not physically covered by structures (Ridhi, 2023). This ambiguity expanded bureaucratic discretion and often pushed larger parcels into litigation, exemption

⁷Several states and union territories without urban agglomerations exceeding the 200,000 population threshold also adopted the Act, including Haryana, Himachal Pradesh, Manipur, Meghalaya, and Tripura. In contrast, Jammu & Kashmir, Kerala, and Tamil Nadu did not adopt the ULCRA despite having qualifying UAs.

⁸Compensation was determined as eight and one-third times the average annual net income over the preceding five years for income-generating land, and at statutory maximum rates of five to ten rupees per square meter for non-income-generating land, depending on the agglomeration category. In all cases, total compensation was capped at two lakh rupees, a nominal ceiling fixed in 1976 and worth approximately \$4,400 by the time of the Repeal at the then-prevailing exchange rate of about 45 rupees per U.S. dollar.

⁹Urban agglomerations were classified into four categories based on 1971 Census population figures: *Category A* (Delhi, Mumbai, Kolkata, and Chennai) imposed a ceiling of 500 sq. meters per person; *Category B* (population above 1 million) allowed 1,000 sq. meters; *Category C* (300,000–1,000,000) permitted 1,500 sq. meters; and *Category D* (200,000–300,000) allowed up to 2,000 sq. meters.

requests, or informal circumvention. In practice, the Act therefore increased the frictions associated with assembling, buying, selling, and developing urban land, particularly for larger firms and developers.

Transfer of property. The ULCRA also imposed stringent restrictions on the transfer of land and buildings, applying to all transfers regardless of whether the land exceeded the established ceiling limits. Under the Act, no transfer—whether by sale, mortgage, gift, lease, or other means—could occur without prior government approval. Moreover, upon receiving an application for transfer, the state retained the authority to compulsorily acquire the land or building in question, with compensation determined in accordance with the Act.¹⁰ These provisions exposed both buyers and sellers to approval delays and the possibility of expropriation at below-market compensation, sharply reducing land market activity. As a result, in some cities land transactions declined by as much as 75 percent, while land prices increased by 30 to 50 percent in the post-ULCRA period (Srinivas, 1991).

Repeal and aftermath. By the late 1990s, the shortcomings of the ULCRA had become increasingly apparent. Although the Act was intended to curb the concentration of urban land and promote equitable access, it largely failed to achieve these objectives. According to the Ministry of Urban Affairs, by 1999 only 20,654 hectares (approximately 80 square miles) of urban land had been acquired under the Act, and nearly half of this land remained undeveloped. A growing body of empirical evidence further documented the ULCRA’s failure to stimulate affordable housing construction, contributing to the central government’s decision to repeal the Act in 1999 (Srinivas, 1991; Joshi and Little, 1996; Sridhar, 2010; Siddiqi, 2013; Dutta and Gandhi, 2023).

The central government’s 1999 repeal was abrupt and largely unexpected, unlike the Maharashtra and Andhra Pradesh repeals a decade later, which were publicly anticipated for the reasons discussed below. In its 1998–1999 working report, the Ministry of Urban Affairs acknowledged the absence of any systematic assessment of the repeal’s potential impact on land prices, noting that “[T]he Government have not made any analysis with regard to impact of the repeal Act on the prices of land as also the trend of land prices after the Act was imple-

¹⁰These restrictions applied uniformly to all urban land and buildings, regardless of whether the structures were constructed before or after the Act’s enactment. The ULCRA also regulated building construction: where a structure contained a dwelling unit, the maximum permissible built-up area was capped at 300 sq. meters in *Category A* and *Category B* cities, and 500 sq. meters in *Category C* and *Category D* cities, ostensibly to control urban density. In addition, the state could order the demolition of buildings constructed after the Act if deemed contrary to the public interest.

mented in 1976.” Nevertheless, under India’s constitutional framework, legislative authority over urban land rests with the states. Thus, the central government’s repeal only meant states were no longer obligated to maintain urban land ceilings. To abolish these ceilings, each state had to adopt the central Repeal Act by a resolution of its own legislative assembly, the procedure required under Article 252 of the Constitution for a law that Parliament had enacted at the states’ request.

A majority of states and union territories adopted the Repeal Act in 1999, as summarized in Table B.12, while others followed in a staggered manner over the subsequent decade. Between 2000 and 2011, the Act was repealed in Chhattisgarh and Madhya Pradesh (2000), Odisha (2002), Assam (2003), Bihar (2006), Maharashtra (2007), Andhra Pradesh and Telangana (2008),¹¹ and Jharkhand (2011). West Bengal adopted the ULCRA but has still not adopted the Repeal Act, while Tamil Nadu never adopted the ULCRA in the first place and continues to operate its own land ceiling law; urban land ceilings therefore remain in effect in both states.¹²

The dates listed above and in Table B.12 are formal, calendar repeal years. Because the ASI reporting year runs from April to March, our regressions instead assign each state to the ASI year that contains its repeal date and code the state as post-repeal in the years *after* that year. The two conventions coincide for repeals enacted in mid-year, but not for those falling between January and March: the central Repeal Act, in force on 11 January 1999, is assigned to ASI year 1998, Madhya Pradesh’s February 2000 repeal to 1999, and the Andhra Pradesh repeal of 27 March 2008 to 2007—before the 2006 treatment date discussed above.

While the precise reasons for the delayed and uneven repeal across states are not fully understood, existing case studies suggest that political factors predominated over economic ones. The ULCRA generated substantial opportunities for rent-seeking and corruption, creating incentives for state-level elites to resist the Repeal (Siddiqi, 2013). In addition, partisan

¹¹The Maharashtra and Andhra Pradesh repeals took formal effect in late 2007 and early 2008, respectively (Maharashtra by notification of 1 December 2007; Andhra Pradesh with effect from 27 March 2008). Both states had, however, committed *statewide* to repeal the ULCRA in 2006 as a stated condition for central infrastructure funding under the Jawaharlal Nehru National Urban Renewal Mission (JNNURM), so the reform was publicly anticipated well before its formal enactment. We therefore date the onset of treatment in these two states to 2006 in our baseline specification, and separately assess JNNURM investment as a potential confounder in Appendix E. Because the JNNURM commitment is also what dates treatment in these two states, that appendix identifies the program’s effects from cross-sectional variation in implementation intensity across mission cities rather than from its timing. Telangana, which was part of Andhra Pradesh until 2014, is recorded as Andhra Pradesh throughout our sample.

¹²West Bengal, governed by the Left Front—a coalition of Marxist and socialist parties—from 1977 to 2011, that is, for nearly all of our sample period, is notable for its comparatively extensive land expropriation and redistribution (Raychaudhuri, 2004). These states are therefore treated as outliers and excluded from the analysis.

fragmentation and coalition politics often hindered the formation of legislative consensus necessary to dismantle the Act ([The Economic Times, 2007](#)).

3 Data and empirical strategy

This section describes the data and empirical strategy used in our analysis. We combine plant-level manufacturing data with a newly constructed dataset on the geographic coverage of the ULCRA and the staggered timing of its repeal across states. Our identification strategy exploits the interaction between repeal timing and local ULCRA exposure.

3.1 Data sources

ASI data. Our primary data source is the Annual Survey of Industries (ASI), which provides extensive information on manufacturing plants from 1993 to 2013 with district and plant identifiers.¹³ The ASI provides a representative sample of registered manufacturing plants in India, with large plants covered yearly and smaller ones surveyed on a sampling basis. Our empirical analysis therefore uses the plant-year as the unit of observation, except for a handful of exhibits that are estimated at the district level (plant entry, district-wide land totals, and the household-survey rent outcomes), and entry, exit, and count outcomes should be interpreted as pertaining to registered manufacturing plants observed in the ASI rather than the full universe of firms. Like other manufacturing surveys, the ASI provides detailed plant-level information, including fixed assets, wage bills, number of employees, consumption of raw materials, and sales revenue.¹⁴

¹³The reporting period for the ASI is the Indian fiscal year, which begins on April 1 and ends on March 31. Throughout the paper, when we refer to a survey year, we use the calendar year in which the fiscal year commences. While the official ASI data only provides plant identifiers after 1998, the dataset that was purchased separately contains plant identifiers starting from 1993, which is also used in, for example, [Bertrand et al. \(2021\)](#) and [Bollard et al. \(2013\)](#). The publicly available ASI dataset lacks district information, which is critical to our analysis. To address this, following [Martin et al. \(2017\)](#), we match the panel version of ASI with an older cross-sectional version, which contains district identifiers until 2009, based on time-invariant factors and open/close variables. Since the ASI does not provide district identifiers after 2009, we use the panel structure of our data to infer the district information for a subset of plants (approximately 2/3) that appear in the data before 2010. Our analysis is robust to only using the pre-2009 sample. Since district boundaries are rather volatile in India, we built a crosswalk that maps the district information to time-invariant 1991 districts.

¹⁴All financial amounts are expressed in 2004 Rupees following the procedures outlined in [Allcott et al. \(2016\)](#). Revenue is adjusted for inflation using 3-digit industry-level commodity price deflators. The deflators for materials reflect the average price changes in outputs from a given industry’s suppliers, as determined by the 1993-1994 input-output table. For capital, we use an implied national deflator based on capital formation provided by the Reserve Bank of India.

A distinctive feature of the ASI is its detailed disaggregation of plants’ capital stock into land, buildings, machinery, and other fixed assets, which allows us to separately identify land inputs. We use the average of the book value at the start and end of the year to measure land use, which reflects the acquisition cost of land rather than its current market value.¹⁵ This measure includes both owned and leased land that is recorded in the plant’s fixed-asset block and used in production, but excludes unused land.¹⁶ Land is also reported less consistently than the other asset categories: it is non-missing for about 58% of plant-years and is absent from the 1997 survey round altogether, which is why the samples for our land-based outcomes are smaller than those for capital, employment, and sales. A limitation of the ASI is that it does not report land prices or physical land quantities, an issue we address in greater detail in the empirical analysis.

Another unique feature of the ASI dataset is that it provides information on both total product sales and total quantity sold at the product level, as mandated by the 1956 Companies Act. Products are defined based on 5-digit product codes, encompassing over 12,000 distinct products. Leveraging the plant-level panel structure of the data, we directly observe product innovation through changes in a plant’s product portfolio, including the addition and deletion of products. These product-based measures of innovation are crucial for us to examine the effects of land restrictions on innovation, especially since firms in developing countries rarely file patents or report their R&D expenditure. Table B.13 reports summary statistics for the main ASI variables used in our analysis.

ULCRA data. We manually collected and compiled a novel ULCRA dataset that records the timing of repeal across states and the geographic coverage of the 1976 Act.¹⁷ The 1976 Act delineates numerous urban agglomerations (UAs) and their associated towns and villages. We systematically gathered coordinates for over six hundred listed locales through extensive online research. To ensure their accuracy, we used the Indian Village Directory to verify the existence and correct spellings of these places, especially when they were unavailable online. Historical name changes and alternate spellings presented additional challenges, which were

¹⁵Land value may be adjusted for market changes through land revaluation, though such adjustments are rare—fewer than 1% of firms report a positive revaluation. Additionally, land does not depreciate.

¹⁶Land rented under arrangements that are not capitalized in fixed assets does not enter this measure. It is instead reported in a separate category of other expenditures, where plants report the total amount of rent paid for land or royalties on mines, quarries, etc., but these amounts are not itemized. However, it is relatively rare—less than 10% of firms ever report a positive total rent expenditure.

¹⁷Collecting data on the timing of the repeal is challenging due to the lack of a centralized policy information platform in India, leading to discrepancies in previous studies regarding the timing of the repeal at the state level (Sridhar, 2010; Kimura, 2011; Duranton et al., 2017). We address this by consulting multiple legal databases and extracting repeal dates from public judicial records that explicitly mention these dates.

addressed through targeted searches. We then mapped all UAs to district boundaries defined in the 1991 district map.

We use districts as our unit of treatment: a plant is classified as a UA plant if it is located in a district containing any UA locale, and is treated only once its state has also repealed the Act.¹⁸ This choice is guided by several considerations. First, the ASI reports plant locations only at the district level, preventing us from precisely identifying whether plants are located strictly *within* UA boundaries. Second, defining treatment at the district level mitigates potential bias from plants relocating between cities and surrounding suburbs, a concern that is limited in practice given the rarity of cross-district moves.¹⁹ Finally, the effective reach of the ULCRA likely extended beyond formal UA boundaries. For example, following the Act’s introduction, developers reportedly hoarded land in adjacent areas, driving land prices up by 30–50% (Srinivas, 1991). In practice, using districts as the treatment unit effectively approximates the actual coverage of the ULCRA, as population census data show that over 60% of non-agricultural employment within districts is concentrated in UAs.²⁰

3.2 Empirical strategy

To estimate the causal effects of the ULCRA repeal, we implement a difference-in-differences (DiD) strategy that exploits both the staggered timing of repeal across states and the fact that the ULCRA only applied to Urban Agglomerations (UAs). The key comparison is thus between plants located in UA districts, where these land-market restrictions applied, and plants in non-UA districts, before and after repeal within the same state. Our baseline empirical specification is given by the following equation:

$$Y_{idst} = \alpha + \beta_1 \cdot PostRepeal_{st} + \beta_2 \cdot PostRepeal_{st} \times UA_d + \gamma \cdot X_{idst} + \delta_t + \delta_d + \epsilon_{idst} \quad (1)$$

where Y_{idst} represents the various outcome variables for plant i in district d of state s in year t . $PostRepeal_{st}$ is an indicator variable that equals 1 if state s has adopted the repeal act by year t , dated by the ASI year that contains the repeal; for Maharashtra and Andhra Pradesh we date the onset of treatment to their 2006 statewide JNNURM repeal commitment, as

¹⁸Almost all UAs can be uniquely matched to a single district, meaning that all locales within a UA are situated in the same district. However, there are a few exceptions, such as the Calcutta UA and Madras UA, which span multiple districts—both of which fall in the two states we exclude from the sample.

¹⁹Cross-district relocation is rare in our data as less than 3% of plants have ever changed districts. This low rate may even be inflated by entry errors in district information.

²⁰Figure B.8 plots village and town-level non-agricultural employment against their distance from the UAs’ centroids. Notably, employment decreases sharply to nearly zero outside a 5-kilometer radius from these centroids, suggesting that most non-agricultural activity occurs within or very close to UAs.

discussed in Section 2. UA_d is an indicator variable set to 1 if the plant resides in a district d that contains an UA; its main effect is absorbed by the district fixed effects. X_{idst} denotes the remaining controls, which we absorb as fixed effects: rural location, 2-digit industry, and the district’s pre-Repeal average plant land-size quartile interacted with year.²¹ δ_t and δ_d are year and district fixed effects, which account for any aggregate fluctuations and time-invariant cross-district heterogeneity. In specifications that focus on within-plant dynamics, we additionally include plant fixed effects δ_i . Plant-level regressions are weighted by the ASI sampling multiplier so that the estimates are representative of the population of registered plants. Standard errors are clustered at the state level—27 states and union territories—as the reform was adopted gradually by individual states and union territories.

Our primary coefficient of interest is β_2 (reported as ‘Repeal $\times$ UA’ in the tables, with β_1 reported as ‘Repeal’), which captures the differential response of plants in UA districts—those more directly exposed to ULCRA-related land frictions—relative to plants in non-UA districts, before and after repeal. This coefficient isolates the effect of relaxing those frictions in locations directly exposed to the policy. By contrast, β_1 measures the average post-repeal change for plants in non-UA districts. Because the ULCRA did not apply outside UAs, a nonzero estimate of β_1 may reflect contemporaneous state-level shocks, spillovers from treated UA districts to nearby non-UA areas, or biases associated with staggered adoption. For transparency, we report both coefficients in our baseline specification. We also show below that the estimates for β_2 are robust to including state-by-year fixed effects, which absorb β_1 and restrict identification to within-state-year differences between UA and non-UA districts: the production estimates are essentially unchanged, and the effect on the marginal revenue product of land, if anything, strengthens.

The identification of β_2 relies on a parallel-trends assumption: absent the Repeal, the difference in outcomes between plants in UA and non-UA districts would have evolved similarly across states that had already repealed and states that had not yet repealed the ULCRA. Equivalently, the UA–non-UA differential in treated states would have followed the same path as the corresponding differential in not-yet-treated states in the absence of the policy. This assumption rules out confounding shocks that differentially affect UA and non-UA districts and evolve systematically with repeal timing.

²¹Our industry classification is a time-consistent crosswalk between the 1987 and 1998 revisions of the National Industrial Classification, which yields 52 groups, roughly at the two-digit level. The land-size control is the quartile of the district’s average plant landholding in the pre-Repeal years, interacted with year fixed effects. Some specifications depart from this baseline set: the event studies additionally absorb ownership fixed effects; the productivity tables absorb ownership fixed effects and report specifications with and without the land-size control; and the exit specifications add plant-age fixed effects.

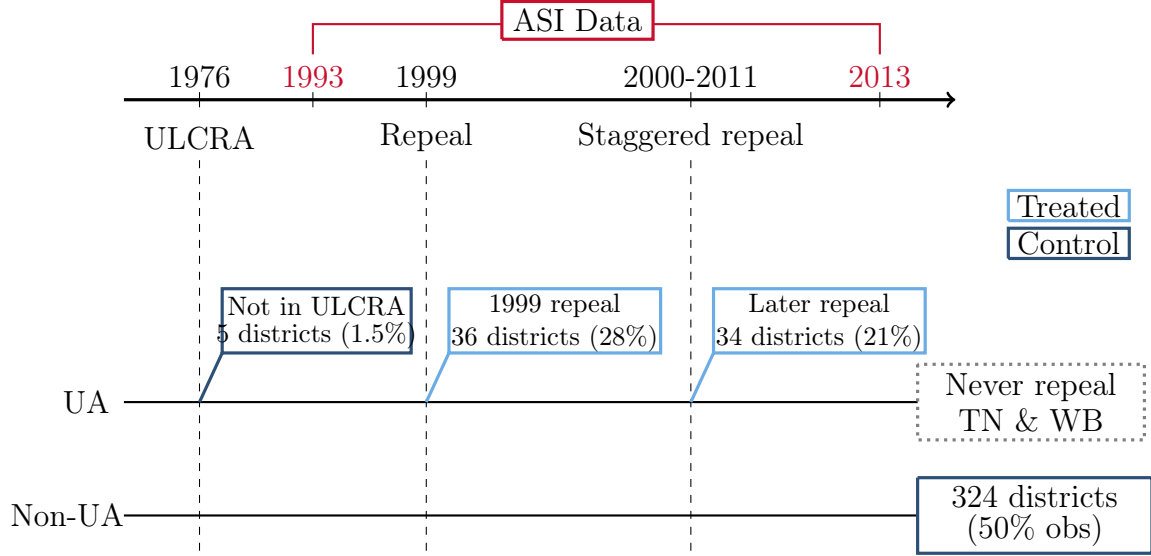


Figure 1: Timeline of the ULCRA and its Repeal

To illustrate our empirical design, we present the timeline of the ULCRA and its Repeal in Figure 1. The control group, non-UAs, consists of 324 districts, which constitutes approximately 50% of our ASI data. Jammu & Kashmir and Kerala, while having qualified UAs, never participated in the ULCRA, so the five UA districts in these two states have $PostRepeal_{st} = 0$ throughout and serve as controls as well. However, they constitute merely 1.5% of the data, and our results are robust to dropping them. A majority of the states repealed the ULCRA in 1999, as listed in Table B.12, encompassing 36 UA districts, or 28% of the observations. From 2000 to 2011, those lagging states implemented the Repeal gradually, including 34 UA districts, or 21% of the data. Tamil Nadu and West Bengal never adopted the Repeal Act and were excluded from the sample as discussed earlier. Therefore, our design essentially compares UAs vs. non-UAs, before and after these two episodes of repeal.

3.3 Threats to identification

In this section, we further elaborate on our identification strategy and address concerns about several potential sources of bias that could contaminate our results.

Non-random assignment of the Repeal. A potential concern is that firms in states that repealed the ULCRA earlier may differ systematically from those that repealed later, and that firms in UA districts may differ from firms in non-UA districts. However, district fixed effects (δ_d) account for any pre-existing, time-invariant differences (observed or unobserved)

between firms across UAs, non-UAs, early-repeal states, and late-repeal states. Importantly, our identification strategy does not require the timing of repeal to be random or unrelated to underlying state-level trends. Instead, the estimate of β_2 remains unbiased as long as the factors driving earlier repeal do not generate differential changes in outcomes between UA and non-UA districts. A violation would arise, for example, if states with limited urban land supply and rising land prices in UAs were more likely to repeal earlier. Although we do not observe land prices directly, pre-existing differences in average plant landholdings across districts, and flexible trends by those differences, are absorbed by the district pre-Repeal land-size quartile-by-year fixed effects.

Validation of identification assumptions. Although the identifying assumptions cannot be tested directly, we provide several pieces of evidence consistent with their validity. First, our event study plots for β_1 show no visible pre-existing trends between non-UA firms in states that had repealed and those in states that had not, and the estimates for β_1 remain close to zero after repeal. Second, event study estimates for β_2 show no evidence of differential pre-trends between firms in UA and non-UA districts prior to repeal across states. Finally, we perform a range of robustness checks to demonstrate that our estimates remain stable when we include additional controls for differential trends at the district, industry, and state levels.

We further assess the validity of our identification strategy by testing for differential pre-trends between UA and non-UA districts prior to the Repeal. Using firm-level data from the pre-1998 period, we examine a range of outcomes, including output, landholdings, fixed assets, employment, and the marginal revenue product of land (MRPLand).²² For each outcome, we regress the variable on a UA indicator, a linear time trend, and their interaction, controlling for state and industry fixed effects. As reported in Table B.14, the interaction terms are consistently small and statistically insignificant, providing no evidence of differential trends between UA and non-UA districts prior to the Repeal.

We also conduct additional checks to address concerns about the potential endogeneity of repeal timing, particularly whether early repeal could be driven by land shortages or unequal land access. To this end, we examine whether the timing of the repeal correlates with the pre-Repeal distribution of land and the degree of land productivity dispersion across firms. Specifically, we regress an indicator for repeal adoption on lagged (up to three years) measures of the standard deviation of landholdings, as well as the average and standard deviation of

²²MRPLand is measured using the average revenue product of land. Under a Cobb–Douglas production function, a higher average revenue product implies a higher marginal revenue product within industries.

MRPLand, constructed at the state–industry level using ASI data. The results, presented in Table B.18, show no systematic relationship between these pre-Repeal characteristics and the timing of repeal. We further show, in the discussion of staggered adoption below, that our results are robust to using only the first episode of the repeal.

Staggered DiD. Recent work has shown that two-way fixed-effects (TWFE) difference-in-differences estimators can be biased when treatment adoption is staggered across units and treatment effects are heterogeneous over time or across groups (Athey and Imbens, 2018; Borusyak et al., 2023; Callaway and Sant’Anna, 2020; Goodman-Bacon, 2021; Sun and Abraham, 2021; de Chaisemartin and D’Haultfœuille, 2020). In such settings, the TWFE estimand is a weighted average of underlying treatment effects, and those weights can be problematic when already-treated units serve as controls.²³ We therefore treat the baseline TWFE results as our main specification and use alternative staggered-adoption estimators as robustness checks. Specifically, we show that our estimates remain similar when using the stacked DiD estimator of Cengiz et al. (2019), the imputation estimator of Borusyak et al. (2023), and the group-time estimator of Callaway and Sant’Anna (2020). We also show that the results remain similar when restricting the sample to the first repeal episode in 1999. Because that episode involves a single adoption cohort triggered by the central government’s own repeal, it removes comparisons between early- and late-treated states and further limits the scope for cross-state differences in repeal timing to drive our estimates.

4 The microeconomic impacts of the Repeal

In this section, we present empirical evidence on the microeconomic effects of repealing the ULCRA. We show that the Repeal liberalized urban land markets, improved the allocative efficiency of land, and triggered firm expansion, productivity growth, and greater business dynamism.

²³In the context of staggered DiD TWFE estimates, biases can arise in at least two different ways, even when the parallel trends assumption is satisfied. First, the traditional TWFE estimator is a weighted average of each of the 2x2 ATTs (Goodman-Bacon, 2021). With time-varying treatment effects, this composition can lead to biases, particularly when early-treated groups serve as control groups, i.e., “forbidden comparisons”. As a result, treatment effects for some units and periods receive negative weights in the TWFE estimand, possibly leading to underestimated or even incorrectly signed TWFE estimates—e.g., the estimand could be negative even if all the treatment effects are positive. Second, the TWFE estimator disproportionately weighs units treated in the middle of the panel, which is often not relevant to the policy under study. If these units’ treatment effects significantly differ from those in other periods, the estimator may be biased and hard to interpret.

4.1 Did the Repeal relax urban land constraints?

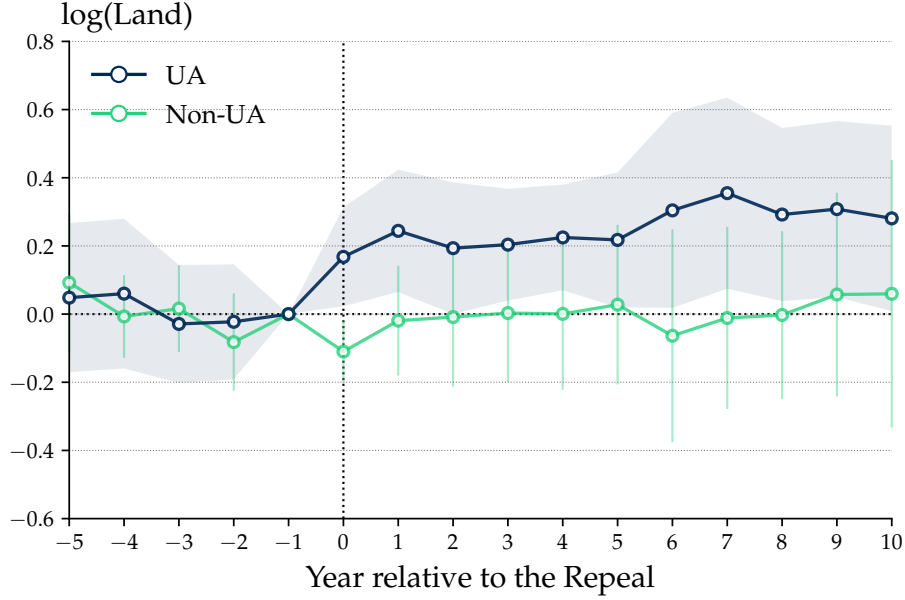
We begin by analyzing the impact of the liberalization on plants’ *average* land use. Using the event-study analogue of equation (1), which replaces the single $PostRepeal_{st}$ indicator with a full set of event-time indicators and their interactions with UA_d , we estimate dynamic effects on plants’ land use, measured by the stock of land employed in production. Figure 2 plots the estimated coefficients for plants located in non-UAs (β_1 , green) and the differential coefficients for plants located in UAs (β_2 , navy).

The event-study estimates show no visible differential effects prior to the Repeal for either group. The flat pre-Repeal path of β_2 indicates that, before liberalization, UA firms were not on obviously different land-use trajectories than non-UA firms. In addition, the pre-Repeal behavior of β_1 suggests no obvious differences in landholding trends between non-UA firms in earlier- and later-repealing states. Together, these patterns are consistent with the identifying assumption, as is the evidence in Appendix Table B.14.

Consistent with the ULCRA applying only to urban agglomerations, the reform has no discernible effect on plants’ land use in non-UAs. Pooling the post-Repeal years into the difference-in-differences specification of equation (1)—reported in Column (1) of Table 1—the non-UA effect is $\beta_1 = 0.005$ (standard error 0.057), statistically indistinguishable from zero and with a confidence interval that excludes an effect of the magnitude we estimate for UA plants. Together with the β_1 profile in the event study, which is statistically indistinguishable from zero throughout, this lessens concerns that our estimates are being driven by common state-level shocks that affect treated and untreated districts similarly, or by large spillovers from UAs to non-UAs.²⁴ In contrast, plants located in UAs experience a sizable post-Repeal expansion in land use: the differential effect is $\beta_2 = 0.202$ (standard error 0.056), so land used in production by UA plants rose by roughly 22 percent relative to non-UA plants after the Repeal, corresponding to an increase of about 1.25 million Indian rupees for the average UA plant.²⁵ The event-study profile in Figure 2 shows that this average effect is not driven by any single post-Repeal year: the response emerges immediately after the reform and remains stable thereafter.

²⁴Large spillovers are unlikely given the size of Indian districts—averaging roughly 5,000 square kilometers—and the substantial costs associated with cross-district firm relocation and migration (Kone et al., 2018).

²⁵Obtained by applying the estimated 22% effect to the mean pre-Repeal landholding of UA plants in repealing states (≈ 5.6 million deflated rupees, unweighted, over 61,043 plant-years). The landholding distribution is heavily right-skewed—the median is about 0.22 million—so this figure describes the mean plant rather than the typical one.



Note: This figure reports the event study graph for the treatment effect of the Repeal Act on plants' landholdings. The dependent variable is the logarithm of the book value of land. Coefficients are normalized 1 year before the reform. The green line with error bars indicates the effects on plants in non-UAs (β_1) and the navy line with the shaded band represents the differential effects on plants in UAs (β_2). The regression includes ownership dummies, an urban dummy, industry fixed effects, district-level land endowment-by-year fixed effects, district fixed effects, and year fixed effects. Standard errors are clustered at the state level. The confidence interval is at the 95% level.

Figure 2: The effect of the Repeal on plant-level land use

Why did average land use increase? At first glance, the post-Repeal increase in plants' average land use may seem puzzling, given that land supply is often thought to be inelastic in the short run. Two features of our empirical design help clarify this result. First, the event-study estimates in Figure 2 do not include plant fixed effects, so the observed increase reflects both within-plant expansion and changes in the composition of plants operating in UAs. Second, Appendix Table B.15 shows that we cannot reject that district-level total land used in production is unchanged after the Repeal, while the number of plants declines by roughly 10 percent, an effect significant at the 10 percent level. Taken together, these patterns suggest that the reform induced the exit of smaller plants and a thickening of the right tail of the land-use distribution among survivors. Appendix C.1 further shows that the Repeal increased the likelihood and scale of land purchases, and reports suggestive—though imprecisely estimated—evidence that treated plants became more responsive in adjusting

landholdings after profitability shocks.

A natural concern is that the measured increase in land use could be driven by rising land prices rather than by a relaxation of the constraint itself. The marginal revenue product of land speaks directly to this concern, and it is the estimate our model will target. A cost-minimizing plant equates the marginal revenue product of land to the effective cost of land—the market price multiplied by any wedge the plant faces. If land had simply become more expensive, plants would economize on it until MRPLand *rose* to meet the higher price. Column (2) of Table 1 shows the opposite: relative to non-UA plants, MRPLand among UA plants *fell* by about 10% after the Repeal. Because our land measure is deflated book value at acquisition cost rather than a current market valuation, a contemporaneous movement in land prices does not mechanically enter the denominator, and several further pieces of evidence point the same way.²⁶ The decline in MRPLand therefore points to a fall in the wedge, which is what makes this the natural moment to identify the land distortion in Section 5.²⁷

Robustness to staggered adoption. As a robustness check, Appendix Figure B.9 reports event-study estimates using several DiD estimators designed for staggered treatment timing. Panel (a) compares the stacked DiD approach of Cengiz et al. (2019), the imputation estimator of Borusyak et al. (2023) and the group-time estimator of Callaway and Sant’Anna (2020), alongside two-way fixed effects as a benchmark and a single-cohort design that eliminates early–late comparisons by construction. Pre-Repeal coefficients are flat for every estimator, and the post-Repeal paths agree in timing and are similar in magnitude. Panels (b) and (c) replicate Figure 2 by jointly estimating β_1 and β_2 using the stacked DiD and imputation

²⁶First, Appendix Table C.31 shows that the Repeal raised land turnover on both the extensive and the intensive margin, and for purchases and sales alike, while Appendix Table C.32 shows that land *purchases* increased significantly after the Repeal whereas land *revaluations* showed no detectable change; the additional land therefore reflects genuine acquisition rather than a pure appreciation of existing holdings. Second, as noted above, we cannot reject that total district-level nominal land use is unchanged post-Repeal, though that estimate is imprecise; if higher prices were the primary driver, we would have expected a clear increase in the aggregate *value* of land used in production. Third, while direct data on industrial land prices are unavailable, residential rent expenditures from the National Sample Surveys show no post-Repeal increase in rent expenditure per capita or in rent expenditure shares (Appendix Table D.35). Since residential and industrial land prices tend to co-move in urban areas, this further weakens the price-based explanation. Finally, in the next section we document real expansions in production—capital, labor, and sales—among UA plants, making it unlikely that the rise in land use reflects valuation effects alone.

²⁷The dependent variable is log MRPLand residualized on 2-digit-industry-by-year means, so the estimate is a within-industry comparison and is not driven by differences in land intensity across industries. Appendix Table B.16 re-estimates it under progressively finer fixed effects: the effect strengthens to -0.185 once state-by-year fixed effects absorb any contemporaneous state-level policy or shock, so the baseline estimate we take to the model is the most conservative of the three.

	(1)	(2)	(3)	(4)	(5)	(6)
	Land	MRPLand	Capital	Employment	Wage Bills	Sales
Repeal	0.005 (0.057)	0.017 (0.044)	0.081 (0.104)	−0.029 (0.038)	−0.005 (0.051)	−0.021 (0.049)
Repeal × UA	0.202*** (0.056)	−0.102** (0.048)	0.145** (0.058)	0.067** (0.026)	0.083*** (0.025)	0.160*** (0.046)
Observations	338,686	298,863	530,008	533,802	531,119	464,755
R^2	0.183	0.053	0.254	0.148	0.214	0.244
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
District FE	Yes	Yes	Yes	Yes	Yes	Yes
Size × Year FE	Yes	Yes	Yes	Yes	Yes	Yes

Note: All dependent variables are in logarithms and deflated, and all columns estimate the same specification (year, district, rural, 2-digit industry and landholding-quartile-by-year fixed effects). Column (1) is land used in production. Column (2) is the marginal revenue product of land, measured as log sales over land and residualized on 2-digit-industry-by-year means so that the estimate is a within-industry comparison; Appendix Table B.16 reports it under progressively finer fixed effects. Columns (3)–(6) are production outcomes; capital excludes land value, so the land and capital responses are measured on disjoint assets. ‘Land Size×Year’ are district average pre-treatment land-size bins interacted with year. Sample sizes differ across columns because the outcomes are reported by different sets of plants, so the coefficients are not algebraically comparable across columns. Robust standard errors clustered at the state level are in parentheses. Significance levels: * 10%, ** 5%, *** 1%.

Table 1: Treatment effect of the Repeal on land use, MRPLand, and plant production

approaches, respectively.²⁸ Taken together, these exercises show that the baseline findings are not sensitive to the particular estimator used to handle staggered adoption.

Land market allocative efficiency. We next examine whether the Repeal improved the allocation of land across firms. Following Chari et al. (2021), we study the differential impacts of the Repeal on plants based on their pre-Repeal total factor productivity (TFP), while including plant fixed effects to isolate within-plant adjustments in land use. The inclusion of plant fixed effects absorbs all time-invariant plant characteristics and restricts attention to firms that survive from the pre- to the post-Repeal period, allowing us to focus on reallocation among incumbents. We defer a discussion of changes in plant composition—including entry and exit responses—to Section 4.3.²⁹

²⁸These methods are particularly well suited to accommodate heterogeneous treatment effects between UA and non-UA firms.

²⁹An alternative approach is to examine heterogeneity by plants’ pre-Repeal marginal revenue product of land (MRPLand) before the Repeal, as suggested by Bau and Matray (2023). However, a limitation of this strategy is the slow adjustment of land: firms with high MRPLand may already be in the process of expanding landholdings and could exhibit faster land growth even absent the Repeal. We discuss this approach further

	(1)	(2)	(3)	(4)
	High TFPR	High TFPQ	Low TFPR	Low TFPQ
Repeal	0.028 (0.042)	0.039 (0.036)	0.011 (0.040)	0.006 (0.052)
Repeal $\times$ UA	0.151*** (0.051)	0.177*** (0.047)	0.047 (0.047)	0.020 (0.056)
Observations	60,084	57,794	55,659	56,199
R^2	0.888	0.892	0.894	0.891
District FE	Yes	Yes	Yes	Yes
Size $\times$ Year FE	Yes	Yes	Yes	Yes
Ind. $\times$ Year FE	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes

Note: The dependent variable is log-landholdings. TFPR and TFPQ are estimated for each 2-digit industry using the methodology of De Ridder, Grassi, and Morzenti (2022). The sample is split by whether a plant’s pre-Repeal average TFP lies above or below the district median. All regressions include plant fixed effects. Robust standard errors clustered at the state level are in parentheses. Significance levels: * 10%, ** 5%, *** 1%.

Table 2: Within-plant land use growth by pre-Repeal TFP

We measure productivity using both revenue-based TFP (TFPR) and quantity-based TFP (TFPQ) to account for the possibility that high TFPR reflects market power or product appeal rather than underlying technical efficiency. Both measures are estimated separately within each two-digit industry using the methodology of [De Ridder et al. \(2022\)](#). We then estimate the effect of the Repeal on within-plant changes in land use, splitting the sample based on whether a plant’s average pre-Repeal TFPR(Q) lies above or below the district median. [Table 2](#) reports the results: Columns (1)–(2) show effects for initially high-productivity plants, while Columns (3)–(4) correspond to less productive plants. We find that, following the Repeal, UA plants with higher initial TFPR(Q) expand their land use by 16–19 percent, whereas the effects for lower-productivity plants are small and statistically insignificant.

The precisely estimated null effect for non-UA plants further alleviates concerns that our findings reflect slow-moving land accumulation rather than the Repeal itself. If high-productivity plants were expanding landholdings even in the absence of liberalization, we would expect to observe similar increases among high-productivity firms in non-UAs. Instead, we find no such effect. [Appendix Table B.20](#) shows that these results are robust across alternative measures of productivity. [Figure 3](#) presents event-study estimates for high- and low-productivity plants in UAs, revealing no systematic differential pre-trends—

in [Appendix C.1](#).

the one exception is a marginally significant negative coefficient two years before the reform in the high-TFPQ panel—and reinforcing the interpretation that the post-Repeal expansion reflects a causal response, although the year-by-year estimates are individually imprecise in the early post-Repeal years. Taken together, these findings suggest that the Repeal improved the allocative efficiency of land markets by enabling more productive firms to expand.³⁰

At first glance, it may seem surprising that less productive plants did not experience a decrease in their land use after the Repeal. This raises the question: where did these new plots of land come from? It is important to clarify that our analysis here, by including plant fixed effects, focuses only on the group of plants that have survived from the pre-Repeal to the post-Repeal periods—typically larger plants included in ASI’s census scheme.³¹ Therefore, these *less* productive plants were not the *least* productive ones. In Section 4.3, we explore how the Repeal released resources from dormant plants (defined there as plants that still held onto their land and buildings but produced nothing) by driving them out of the market.

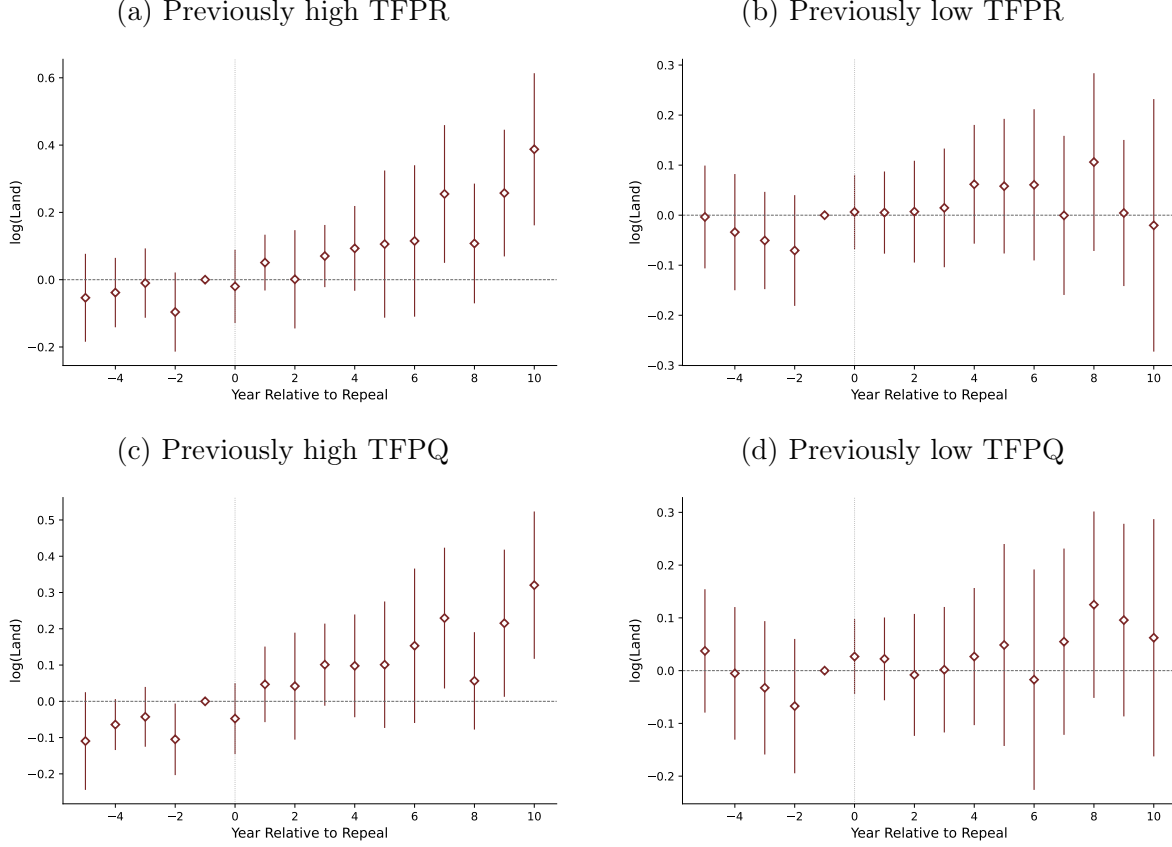
4.2 Did the Repeal unleash industrial growth?

Manufacturing production. We now examine whether this land-market liberalization translated into increased manufacturing output. Because firms may have acquired land for speculative purposes rather than production, we test whether the Repeal relaxed economically meaningful land frictions for urban manufacturers. Using equation (1), we estimate the Repeal’s effect on fixed assets excluding land, employment, wage bills, and sales, alongside the land-use and MRPLand estimates of the preceding subsection. This allows us to distinguish between productive expansion and mere asset accumulation.

Table 1 shows that the Repeal had no detectable effect on plants in non-UA districts, consistent with our previous findings. Columns (1)–(2) repeat the land and MRPLand responses discussed above; Columns (3)–(6) report the production outcomes, estimated on the same specification. In contrast to non-UAs, the average plant in a UA experienced a 16% increase in its stock of buildings and machinery following the reform—an expansion in capital that is distinct from the land response, since our capital measure excludes land value. This capital expansion did not occur at the expense of labor; rather, employee counts and wage bills grew by roughly 7 and 9 percent, respectively, and, as a result, UA plants saw a 17% rise in total revenues. Collectively, these findings provide strong evidence that ULCRA-related

³⁰Appendix C.1 provides additional discussion of the Repeal’s implications for the dispersion of the marginal revenue product of land (MRPLand) and land misallocation.

³¹The ‘census scheme’ in the ASI refers to plants with more than 100 employees that are surveyed every year, as opposed to a ‘sample scheme’ which may include smaller plants surveyed less frequently.



Note: The dependent variable is log-landholdings. We estimate TFPQ and TFPR for each 2-digit industry using the methodology proposed by [De Ridder et al. \(2022\)](#). The regression sample is segmented by whether firms' pre-Repeal average TFPR(Q) was above or below the district median. All regressions include plant fixed effects, year fixed effects, and controls incorporate both district pre-Repeal land endowment size bins and firm-specific land size bins, interacted with year fixed effects. Standard errors are clustered at the state level. The confidence interval is at the 95% level.

Figure 3: The effect of the Repeal on land use by pre-Repeal TFP

land frictions materially constrained urban manufacturing prior to the reform.

These effects are not uniform across the plant size distribution: Appendix C.2 shows that they peak among upper-middle-sized plants, which also face the highest returns to land and therefore had the most to gain from cheaper land adjustment.

Appendix Figure B.10 presents the event study graphs for the treatment effects on firm production. Consistent with our DiD estimates, the Repeal did not affect plants in non-UAs, while plants in UAs ramped up production after the Repeal. Importantly, the UA differential β_2 —the parameter that identifies the effect—shows no differential pre-Repeal trend for any outcome, supporting the causal interpretation of our findings; the non-UA series β_1 for employment drifts modestly downward before the reform, which does not enter β_2 . Addi-

tionally, Appendix Table B.21 and Appendix Table B.22 show that our results are robust to adding state-by-year or sector-by-year fixed effects to flexibly control for any state-level or 2-digit-sector-level policy changes or shocks. Appendix Table B.25 and Appendix Table B.26 further report the imputation and Stacked DiD estimators, which address potential biases in staggered DiD: both deliver point estimates close to the baseline, though the imputation estimates are less precise, with only the land and capital effects remaining individually significant.

Manufacturing productivity. Having shown that the Repeal Act eased land-market frictions and enabled plants to expand production, we now turn to its implications for plant-level physical productivity (TFPQ). We measure TFPQ using several approaches, with our baseline estimate following the methodology of De Ridder et al. (2022). Column (2) of Table 3, which uses the same district, year, and size-by-year fixed effects as our baseline, documents a 13% increase in TFPQ for UA plants following the Repeal, suggesting a substantial improvement in physical productivity; the effect is 11% under the most saturated specification, which adds industry-by-year fixed effects (Column 3). This finding is robust across three alternative TFPQ measures (Appendix Table B.23) and a range of model specifications. In addition, the event-study estimates displayed in Appendix Figure B.13 reveal no evidence of differential pre-trends, reinforcing a causal interpretation of the productivity gains.

One may question why the estimated effects on physical productivity are so large, given that sales and input usage rose by similar magnitudes in Table 1. This finding is particularly striking in light of Bollard et al. (2013), who found little evidence of reform-led productivity growth in Indian manufacturing. A plausible explanation is that efficiency gains translate primarily into lower prices rather than higher revenues, dampening observed changes in revenue-based productivity (TFPR). As emphasized by Garcia-Marin and Voigtländer (2019), more efficient producers may optimally charge lower prices, which compresses TFPR even as physical productivity improves. If the Repeal reduced marginal costs by lowering effective land use costs—and these reductions were fully passed through to prices—TFPR would respond only weakly despite substantial gains in TFPQ.

Consistent with this mechanism, Appendix Table B.24 shows that TFPR increases only modestly following the Repeal, with economically small point estimates that are significant in only two of the three specifications (Columns 1–3), while average output prices—measured relative to comparable products—decline by roughly 10%, though this last estimate is significant only at the 10 percent level. The reduction in prices suggests that firms responded primarily by lowering their marginal cost of production rather than by upgrading the quality

	(1)	(2)	(3)
	TFPQ	TFPQ	TFPQ
Repeal	-0.004 (0.077)	0.002 (0.073)	-0.020 (0.052)
Repeal $\times$ UA	0.118* (0.063)	0.121** (0.055)	0.104** (0.044)
Observations	365,284	365,225	365,211
R^2	0.488	0.489	0.532
District FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Size $\times$ Year FE	No	Yes	Yes
Ind. $\times$ Year FE	No	No	Yes

Note: The dependent variable is TFPQ, estimated within 2-digit industries following De Ridder et al. (2022). ‘Land Size $\times$ Year’ are district average pre-treatment land-size bins interacted with year; ‘Industry $\times$ Year’ are 2-digit-industry-by-year fixed effects. Additional controls: urban dummy, ownership fixed effects, and 2-digit industry fixed effects. Robust standard errors clustered at the state level are in parentheses. Significance levels: * 10%, ** 5%, *** 1%.

Table 3: Treatment effect of the Repeal on productivity

of their products, which would command higher prices. Moreover, we find no statistically significant change in markups, implying that the productivity gains delivered by the Repeal were largely passed through to consumers rather than captured by firms as higher rents.

4.3 Selection and innovation after the Repeal

Weeding out “dormant” plants. As documented by Akcigit et al. (2021), India’s manufacturing sector exhibits weak selection, allowing stagnant plants with limited growth potential to survive for many years. In line with this evidence, Chatterjee et al. (2023) show that exit rates among Indian manufacturing plants are markedly lower than even those observed in other developing economies. By restricting land transfers, the ULCRA likely amplified the costs of divestment and exit, further entrenching productive resources in unprofitable—or even dormant—plants (Panagariya, 2010). In this section, we put Panagariya’s conjecture to a formal test by examining how the Repeal affected plant exit behavior.

A valuable feature of the ASI data is that, since 1998, it distinguishes between plants

that are no longer producing but still exist—classified as *dormant*³²—and plants that have ceased to exist and are therefore removed from the survey, which we define as *exit*.³³ This measurement window carries an important limitation for what follows. Because the status codes that identify exit begin only in 1998, and because a plant cannot be observed exiting in the first year of the panel, no informative pre-Repeal exit variation exists for the seventeen states and union territories that repealed the Act in 1999: exit is zero by construction in the single pre-Repeal year available for that cohort. A parallel-trends check on exit is therefore unavailable for those states. Our estimates below rely on the design we use throughout, which compares UA with non-UA plants as repeal timing varies across states, rather than on the raw comparison of exit rates between early- and late-repealing states. Within that design, the identifying variation for exit comes disproportionately from the later-repealing cohorts, which are the ones with observable pre-Repeal years.

Table 4 reports these estimates, using our baseline empirical equation. Column (1) indicates that, after the Repeal, exit rates among UA plants rose by about 1.1 percentage points relative to non-UA plants (β_2)—roughly 31% of the unweighted sample mean exit rate of 3.6%. The response is much stronger among plants that had ever been in dormancy (Column (3)), which exhibit a 3.5 percentage point increase in exit rates, or roughly 64% of their unweighted mean exit rate of 5.5%.³⁴ By comparison, exit responses among non-dormant plants (Column (2), about 1.1 percentage points) are similar to those in the overall sample. That the response is three times larger among plants with a history of dormancy is the sharpest form of Panagariya’s conjecture: the reform did not merely raise turnover, it fell disproportionately on plants that had already stopped producing while continuing to hold land and buildings.³⁵ Column (4) makes the same point from the opposite direction by

³²Before 1998, the *status of units* variable in the ASI was limited to two codes: (1) open and (2) closed. Since 1998, the ASI has expanded this categorization to provide specific reasons behind closures. We categorize plants as “dormant” if, during the survey, it is found that the unit was present at the specified location and had employed some staff during the reference period but either did not commence production or produced nothing (coded as 2 in the variable *status of units*). Alternatively, a plant is also considered dormant if it was present at the location but did not employ any staff or produce anything during the reference period (coded as 3 in the variable *status of units*).

³³Plants are labeled as exited if they meet any of the following criteria: status code 3 for three years or more, non-existence (absence of any fixed assets), deregistration, falling out of survey coverage (i.e., no longer operating as manufacturing plants), or any other reason deemed appropriate by the National Statistical Office for removal from the survey (ASI *status of units* codes 4, 6, 11, or 12). Our definition of exit excludes cases where plants still exist but no response is received from the owners. However, it does include cases where plants remain under code 3 for more than three years.

³⁴The regressions are weighted by the ASI sampling multipliers whereas these base rates are unweighted plant-year averages; normalizing instead by the corresponding weighted means (4.7% and 6.6%) gives 23% and 53%.

³⁵Disproportionately, but not exclusively: plants with a dormancy history are only 13% of the sample, so in absolute terms the majority of the additional exits still occur among plants that were never recorded as

using dormancy itself as the dependent variable. The estimate is economically small and statistically indistinguishable from zero (-0.002 , standard error 0.005 , against a base dormancy rate of 4.4%), so we find no evidence that the Repeal changed the probability that a plant *becomes* dormant; what it changed is the rate at which plants already in that state leave the market.³⁶ The reform therefore operated on the exit margin rather than on the incidence of dormancy, which is what the resource-release interpretation requires.³⁷

Appendix Figure B.14 provides descriptive corroboration from a different contrast built on the same repeal timing—comparing exit rates in states that repealed in 1999 with those in Andhra Pradesh and Maharashtra, the last large states to repeal. Exit rates run persistently higher in the early-repealing states through the 2000s and the differential closes after 2009, once the late-repealing states have themselves liberalized. Appendix Figure B.15 shows the corresponding series for dormancy rates. We treat both as suggestive context rather than as tests: they pool UA and non-UA plants, they use only two control states—the same two that supply much of the regression’s identifying pre-Repeal variation—and, for the reason given above, the exit figure cannot display a pre-Repeal year for the early cohort. Dormancy, by contrast, is observed contemporaneously from 1998, so the dormancy figure does normalize on a pre-Repeal year.

These exit responses also vary systematically with pre-Repeal profitability. Appendix Figure B.16 shows that plants with lower profitability prior to the liberalization experienced significantly larger increases in exit rates after the Repeal. Taken together, these results provide strong evidence that the Repeal intensified the process of creative destruction by easing the exit of unprofitable and dormant firms, in line with Panagariya’s observation.

Creative destruction through product turnover. Beyond plant exit, the Repeal also fueled business dynamism among surviving plants through product innovation. We measure the latter using the ASI’s detailed product-level data.³⁸ In particular, we construct two plant-year measures of product innovation: (i) the number of new products added to a dormant.

³⁶Column (4) excludes the 2010 and 2011 rounds, in which the ASI did not record the dormancy codes and the indicator is identically zero for every plant; retaining them would impose a mechanical zero on part of the post-Repeal window.

³⁷The estimation sample is restricted to plants with the landholding and employment information required by our controls, which excludes many non-reporting plants; its exit rate (3.6%) is accordingly lower than the rate in the unrestricted plant panel (about 9%). The estimates should therefore be read as effects among plants observed in production, a conservative sample for detecting exit.

³⁸The ASI dataset provides product-level sales and quantities, as mandated by the 1956 Companies Act. Products are categorized by 5-digit product codes, with over 12,000 distinct products in our sample, differentiating items like men’s shirts (28232) from women’s shirts (28234).

	Exit			Dormant
	(1) Overall	(2) Non-dormant	(3) Dormant	(4) All plants
Repeal	−0.005 (0.008)	−0.007 (0.008)	−0.033 (0.019)	0.009 (0.005)
Repeal × UA	0.011** (0.004)	0.011** (0.004)	0.035** (0.016)	−0.002 (0.005)
Observations	287,316	249,183	38,128	253,368
R^2	0.060	0.055	0.154	0.037
Year FE	Yes	Yes	Yes	Yes
District FE	Yes	Yes	Yes	Yes
Firm controls	Yes	Yes	Yes	Yes
Size × Year FE	No	No	No	No
Ind. × Year FE	No	No	No	No

Note: In columns (1)–(3) the dependent variable is a plant exit dummy (ASI ‘status of units’ codes 4, 6, 11, or 12): column (1) uses all plants, column (2) restricts to plants never flagged as dormant, and column (3) to plants ever flagged as dormant (status codes 2 or 3). Column (4) instead uses dormancy itself as the dependent variable, and asks whether the Repeal changed the probability of *being* dormant rather than of exiting; it excludes the 2010 and 2011 rounds, in which the ASI did not record the dormancy codes and the indicator is identically zero for every plant. Controls include a rural dummy, 2-digit industry fixed effects, firm age fixed effects, and landholding quartile-by-year fixed effects. Robust standard errors clustered at the state level are in parentheses. Significance levels: * 10%, ** 5%, *** 1%.

Table 4: Treatment effect of the Repeal on plant exit and dormancy

plant’s portfolio and (ii) the number of discontinued products.³⁹ In our data, more than 50% of plants never introduce a new product over their entire life cycle. This is consistent with estimates in [Akcigit et al. \(2021\)](#) and [Peters and Zilibotti \(2021\)](#), which suggest that the majority of Indian plants are subsistence plants with low growth potential.

Table 5 reports the impact of the Repeal on product innovation within UAs and non-UAs, estimated by PPML on the 1995–2009 product panel using our baseline controls. The Repeal led to a 14% increase in the addition of new products (Column 1) and a 15% increase in the deletion of old products (Column 2) among plants in UAs, indicating intensified creative

³⁹Our measure of product innovation has several advantages over more conventional measures of innovation, such as patent counts and R&D spending, which are rarely pursued by firms in developing countries. For example, the ASI only began including questions about R&D activities in 2015, and merely 3% of plants reported having spent anything on R&D.

	(1)	(2)
	#New Products	#Del. Products
Repeal	0.004 (0.049)	-0.035 (0.085)
Repeal $\times$ UA	0.132*** (0.051)	0.140** (0.058)
Observations	264,437	272,666
R^2		
Year FE	Yes	Yes
District FE	Yes	Yes
Size $\times$ Year FE	Yes	Yes

Note: #New Products and #Del. Products are counts of 5-digit products added to / removed from a plant's portfolio. Both columns are PPML estimates, based on data from 1995–2009. Robust standard errors clustered at the state level are in parentheses. Significance levels: * 10%, ** 5%, *** 1%.

Table 5: Treatment effect of the Repeal on creative destruction

destruction. Appendix Table B.28 shows that these effects remain positive under a more conservative definition of products based on 4-digit codes, significant at the 5 and 10 percent levels for product additions and deletions respectively,⁴⁰ suggesting that the effects reflect meaningful product turnover rather than reporting noise.

In contrast, Appendix Table B.29 shows no effect of the Repeal on the total number of products produced by a firm, suggesting that the liberalization reshaped product portfolios through higher turnover rather than through an expansion in product scope. Event-study estimates in Appendix Figure B.17 confirm the absence of differential pre-trends, despite somewhat noisier estimates. Overall, the evidence suggests that the land market liberalization increased business dynamism by inducing firms to refresh their product portfolios more frequently, adding new products while shedding old ones.

Entry and the selection of dynamic plants. The effect of the Repeal on plant entry is theoretically ambiguous. On the one hand, by easing land market restrictions, the Repeal reduced barriers to starting and scaling production, potentially attracting entrants

⁴⁰Using 4-digit codes mitigates concerns that measurement error or minor misclassification—such as a change from men's shirts (28232) to women's shirts (28234)—may be spuriously recorded as innovation. Products defined at the 4-digit level typically correspond to distinct subclasses produced with different techniques (e.g., knitted versus non-knitted apparel), and thus capture more substantive product changes.

with higher growth potential. At the same time, the intensification of creative destruction increased competitive pressures and the associated risk of failure, which may have deterred entry by less productive plants.

To study the impact of the Repeal on entry, we aggregate the ASI to the district–year level using sampling weights and estimate our DiD specification using Poisson Pseudo–Maximum Likelihood (PPML), which is well suited to count data with a high prevalence of zeros; district–year observations with fewer than 30 plants are excluded. Consistent with the presence of offsetting forces, the Repeal has no statistically significant effect on the *total* number of new entrant plants (Table 6, Column 1). Beneath this aggregate null, however, the Repeal reshaped the *composition* of entrants toward more dynamic plants.

We classify entrants by their *realized* dynamism—as *dynamic* if they introduced any new product over their life cycle, and as *stagnant* otherwise. Table 6 shows that the number of dynamic entrants rises significantly in UAs ($\text{Repeal} \times \text{UA} = 0.243$, Column 3), while total and stagnant entry are unchanged (Columns 1–2); the corresponding event study appears in Appendix Figure B.18. The Repeal thus shifted the composition of new entrants toward greater dynamism, drawing in more dynamic plants without changing the total inflow.⁴¹

Because realized dynamism is measured over a plant’s life cycle and could in principle respond to the Repeal itself, we corroborate this result with an *ex-ante* measure that cannot be an endogenous response to the reform. We train a LASSO on pre-Repeal data to predict dynamism from plant characteristics observed at entry—ownership and organizational form, importing activity, computer use, and the number of initial products, together with their interactions and quadratic terms—and this predicted measure closely tracks realized dynamism both before and after the Repeal (Appendix Figure B.19). Sorting entrants into quartiles of predicted dynamism yields the same qualitative pattern: entry declines among the least dynamic quartile, changes little in the middle quartiles, and rises among the most dynamic entrants (Appendix Figure B.20), and a binary LASSO classification confirms it (Appendix Table B.30). Both the realized and predicted classifications thus deliver the same conclusion—that the Repeal draws in more dynamic entrants—mitigating concerns that the result is an artifact of an endogenous ex-post classification.

⁴¹This pattern also explains why we observe a null effect on total entry in the ASI sample. The ASI data contains only formally registered manufacturing plants; informal plants and household enterprises are, therefore, outside the scope of the analysis. In other words, the ASI sample is likely to be left-truncated, with many less dynamic plants that experienced a decline in entry rate missing from our data. The absence of the left tail should not bias our estimate of the effect on dynamic plant entry, as those informal plants are unlikely to have high growth potential.

	(1)	(2)	(3)
	All	Stagnant	Dynamic
Repeal	0.054 (0.118)	−0.047 (0.133)	0.056 (0.114)
Repeal $\times$ UA	−0.020 (0.125)	−0.028 (0.163)	0.243** (0.115)
Observations	5,410	5,379	5,198
R^2			
Year FE	Yes	Yes	Yes
District FE	Yes	Yes	Yes
Size $\times$ Year FE	Yes	Yes	Yes

Note: District-year Poisson Pseudo-Maximum Likelihood (PPML) estimates using ASI sampling weights. District-years with fewer than 30 plants are excluded. Robust standard errors clustered at the state level are in parentheses. Significance levels: * 10%, ** 5%, *** 1%.

Table 6: Treatment effect of the Repeal on plant entry

4.4 Further remarks on the impacts of the Repeal

While our empirical analysis in this paper focuses on the impacts of the Repeal Act on manufacturing firms, the ramifications of this legislative change extended well beyond the manufacturing sector. The land ceiling regulation, a key element of the ULCRA, might influence aggregate productivity through both agglomeration and congestion forces, which have been extensively discussed in the urban literature. Furthermore, the expansion of urban manufacturing firms could potentially act as a catalyst for rural-to-urban migration and structural changes, thereby reshaping the spatial distribution of labor. Apart from these economic implications, the redevelopment of urban land holds significant relevance for the living conditions of urban inhabitants. For instance, [Gechter and Tsivanidis \(2023\)](#) provides a case study of the redevelopment of 60 textile mills in Mumbai. Their study highlights that while such urban redevelopment projects can benefit the city’s residents by increasing housing and floor space supply, they can concurrently impose substantial costs on slum dwellers who are forcibly relocated. These dimensions present interesting avenues for future research, yet fall outside the scope of this study.

Socioeconomic impacts of the Repeal. For interested readers, Appendix Section [D](#) documents the broader socioeconomic effects of the Repeal using village-level employment

and amenity data from three rounds of the Population Census, household wealth data from the 2011 Socio-Economic and Caste Census, and household expenditure data from the National Sample Surveys. The evidence shows that the Repeal accelerated the reallocation of agricultural labor into non-agricultural sectors. Notably, villages even 25 kilometers away from UAs experienced significant structural changes after the Repeal.

At the same time, we find no evidence of short- to medium-term improvements in local amenities—such as roads, electricity, or schools—around Urban Agglomerations. The null effects on amenities also mitigate concerns that our firm results are driven by infrastructure investments made after the Repeal. Additionally, we find no post-Repeal increase in residential rents. In the single round of the Socio-Economic and Caste Census, which permits only a cross-sectional comparison of municipalities closer to and farther from UAs rather than our difference-in-differences design, we likewise find no deterioration in housing conditions, and we find higher household incomes and lower poverty rates closer to UAs.

Jawaharlal Nehru National Urban Renewal Mission (JNNURM). One potential confounding policy is the Jawaharlal Nehru National Urban Renewal Mission (JNNURM), a nationwide urban development initiative launched by the federal government from December 2005 to March 2011. The JNNURM had two missions: Urban Infrastructure and Governance (UIG), which aimed to improve water supply and sanitation, solid waste management, road networks, urban transport, and the redevelopment of old city areas, and Basic Services to the Urban Poor (BSUP), which focused on slum development. One of the conditions for states to receive JNNURM funding was the repeal of the ULCRA if it had not already been repealed. The JNNURM targeted 65 cities in total, which included cities with populations exceeding one million in 2001, as well as cities of religious, historic, and touristic significance. The list of JNNURM cities had a partial overlap with the Urban Agglomerations (UAs) affected by the ULCRA.

This overlap raises the concern that our estimates may reflect the effects of JNNURM-funded infrastructure investments rather than land market liberalization. However, JNNURM implementation progressed slowly over our study period, which ends in 2013. According to the government’s audit report, by 2011, only 17.8% of approved UIG projects and 1.5% of BSUP projects had been completed ([Comptroller and Auditor General of India, 2012](#)). Using detailed JNNURM implementation data, we demonstrate that it is unlikely to bias our production results, as discussed in [Appendix E](#).

5 Theoretical framework

In this section, we present a general equilibrium model of firm growth through product innovation with land market frictions. The model builds on the framework of [Klette and Kortum \(2004\)](#) and introduces frictions to land transactions that distort firms' expansion strategies.

5.1 The economic environment

Preferences. Consider a continuous-time economy in which time is indexed by $t \in \mathbb{R}_+$. The economy is populated by an infinitely-lived representative household of unit measure with logarithmic preferences over aggregate consumption C_t :

$$U_0 = \int_0^\infty e^{-\rho t} \ln(C_t) dt \quad (2)$$

where $\rho > 0$ is the household's rate of time preference. This household inelastically supplies $H > 0$ unit of labor and $L > 0$ units of land per period.

Production technology. The economy is composed of a final sector producing a manufacturing final good according to a Cobb–Douglas technology that combines differentiated products indexed by $i \in [0, 1]$:

$$\ln(Y_t) = \int_0^1 \ln(y_{it}) di \quad (3)$$

where y_{it} is the quantity of product i used in production. Each product is manufactured by a single firm using labor and land according to the following Cobb–Douglas technology:

$$y_{it} = z_{it} h_{it}^\alpha \ell_{it}^{1-\alpha} \quad (4)$$

where z_{it} is the process efficiency at which product i is manufactured, h_{it} and ℓ_{it} are the quantity of labor and land used in production, respectively, and $\alpha \in (0, 1)$.

Expansion technology. A firm $f \in \mathcal{F}_t$ consists of a portfolio of products collected in the set $\mathcal{P}_{ft}$ with cardinality $n_{ft} = |\mathcal{P}_{ft}| \in \mathbb{N}$ and corresponding process efficiencies $\{z_{it}\}_{i \in \mathcal{P}_{ft}}$. Firms can grow over time by adding more products to their portfolio. Specifically, a firm holding n_{ft} products must allocate y_{ft}^X units of final goods to expand into new product lines

at rate X_{ft} . The expansion technology is given by:

$$y_{ft}^X = \frac{X_{ft}^2 Y_t}{2\theta_X n_{ft}} \quad (5)$$

where $\theta_X > 0$ measures the efficiency of the expansion technology of incumbent firms. In this specification, the quadratic cost of expansion is scaled by the firm's current number of active product lines. This scaling reflects the idea that larger firms may leverage their organizational infrastructure to expand more efficiently into new markets. Upon successful expansion, the newly operational product line displaces a randomly selected incumbent product through creative destruction. That new product improves on the displaced incumbent's process efficiency by a factor of $\lambda > 1$.

Entry, exit, and firm type. At every instant, a unit mass of potential entrants direct Y_t^E units of final goods toward entry to achieve an entry rate of e_t :

$$Y_t^E = \frac{e_t^2 Y_t}{2\theta_E} \quad (6)$$

where $\theta_E > 0$ measures the efficiency of the entry technology. Upon successful entry, each entrant endogenously selects a type that determines its growth prospects. There are two types of firms: *dynamic* firms have access to the expansion technology in equation (5) and can grow into multi-product firms, while *stagnant* firms operate a single product line for the remainder of their lifecycle. To achieve a probability d_t of drawing the dynamic type, an entrant must pay a type-selection cost of Y_t^T units of final goods, given by:

$$Y_t^T = -\frac{[\ln(1 - d_t) + d_t]Y_t}{\theta_T} \quad (7)$$

where $\theta_T > 0$ determines the efficiency of the type-selection technology. The cost of type selection is zero when $d_t = 0$ and diverges as $d_t \rightarrow 1$. When an entrant successfully introduces a new product, it displaces a randomly selected incumbent product through creative destruction and improves on its process efficiency by the same factor of $\lambda > 1$. Finally, firms exit when they lose all their products to creative destruction.

Resource constraints. Land is in fixed (inelastic) supply:

$$\int_0^1 \ell_{it} di \leq L. \quad (8)$$

The resource constraint for labor is given by:

$$\int_0^1 h_{it} di \leq H. \quad (9)$$

The final good's resource constraint is given by:

$$C_t + Y_t^X + Y_t^E + e_t \cdot Y_t^T + Y_t^G \leq Y_t \quad \text{where} \quad Y_t^X \equiv \int_{f \in \mathcal{F}_t} y_{ft}^X df. \quad (10)$$

Here, Y_t^G is the government's wasteful use of final goods, financed by the revenue collected from the land wedge and yielding no utility or productive benefit.

5.2 The market equilibrium allocation

We now characterize the agents' decision problems, which jointly determine equilibrium prices and quantities on the product, labor, and land markets. While the labor and land markets are perfectly competitive, product markets feature monopolistic competition.

Representing the ULCRA. A literal representation of the ULCRA as a binding ceiling on landholdings would be misleading. The statutory ceiling applied only to *vacant* land, since land already developed for industrial use did not count toward the cap. Moreover, enforcement was notoriously weak: by 1999, only 20,654 hectares had been acquired nationally, and firms routinely circumvented the ceiling through bribes and political connections. The ceiling rarely bound as a hard constraint on landholdings.

The more economically significant friction came from the Act's *transfer restrictions*. All land transactions required prior government approval, and the state retained authority to compulsorily acquire land during the application process at below-market compensation. These provisions exposed every buyer and seller to expropriation risk. The ceiling compounded these frictions by giving officials additional bureaucratic leverage: the low statutory thresholds (500 to 2,000 square meters) and the ambiguous definition of "vacant" land provided a legal pretext to scrutinize firms, initiate acquisition proceedings, and extract rents.

We capture these frictions as a size-dependent wedge $\tau(n) \geq 0$ on the effective rental cost of land for a firm operating n product lines. A firm with n products pays a factor $1 + \tau(n)$ above the market rental rate per unit of land. The government collects the excess payment $\tau(n)q_t\ell_{it}$ and uses the proceeds to purchase final goods that yield neither household utility nor productive services. This valueless use captures approval delays, mandatory disclosures,

expropriation risk, and the corruption costs that pervaded land transactions under the Act. Both dynamic and stagnant firms face this wedge, but since stagnant firms always hold a single product, they face $\tau(1)$.

The government's budget is balanced each instant:

$$Y_t^G = \sum_{n=1}^{\infty} \tau(n) q_t \ell_t(n) n m_t^D(n) + \tau(1) q_t \ell_t(1) M_t^S. \quad (11)$$

The revenue is not rebated to the household, so the household budget below is unchanged.

The wedge is increasing in n , reflecting the fact that the ULCRA's regulatory burden fell disproportionately on larger firms. The statutory ceilings on vacant landholdings and the requirement of prior government approval for all land transactions created frictions that scaled with a firm's land footprint—and therefore with its product scope, since each product line requires land as a factor of production. We parameterize this wedge as a saturating function $\tau(n) = \bar{\tau}(1 - e^{-\kappa n})$, where $\bar{\tau} > 0$ is the asymptotic wedge level and $\kappa > 0$ governs the rate of saturation. This functional form nests a constant wedge ($\kappa \rightarrow \infty$) and ensures that the wedge is bounded.

The household's problem. Taking prices as given, the household's problem is to choose a time path of consumption to maximize lifetime utility:

$$\max_{\{C_t\}_{t \geq 0}} \int_0^{\infty} e^{-\rho t} \ln(C_t) dt \quad \text{s.t.} \quad \dot{A}_t + P_t C_t \leq r_t A_t + w_t H + q_t L$$

where P_t denotes the price of the final good, w_t the wage, q_t the land rental rate, A_t is the value of corporate assets, and r_t is the rate of return on those assets:

$$A_t = \int_{f \in \mathcal{F}_t} V_{ft} df \quad \text{where} \quad \lim_{t \rightarrow \infty} e^{-\int_0^t r_{t'} dt'} A_t = 0.$$

Here, V_{ft} denotes the value of firm f . This problem delivers the usual Euler equation:

$$\dot{C}_t = (r_t - \rho) C_t.$$

The final sector's problem. The final good is produced competitively according to the technology in equation (3). Taking prices as given, the final sector chooses its demand for

each product to maximize profits:

$$\max_{\{y_{it}\}_{i \in [0,1]}} P_t Y_t - \int_0^1 p_{it} y_{it} di \quad \text{s.t.} \quad \ln(Y_t) = \int_0^1 \ln(y_{it}) di.$$

This problem delivers the following demand schedule for each product:

$$y_{it} = \frac{P_t Y_t}{p_{it}}, \quad \forall i \in [0, 1]$$

where the price of the final good is given by:

$$\ln(P_t) \equiv \int_0^1 \ln(p_{it}) di.$$

The final good is chosen as the numéraire such that $P_t = 1$ for all t .

The firm's problem. Each product line features a competitive fringe that uses the same production technology (4) as the leader but at its previous process efficiency level (i.e., the fringe's process efficiency for product i is z_{ift}/λ). The leader faces the ULCRA wedge on land costs, while the fringe does not: because the fringe represents the competitive outside option of small, single-product producers who operate on existing plots and need not acquire additional land through the regulated transfer process, its costs are undistorted. The fringe prices competitively, so the leader captures the entire market by pricing just below the fringe's marginal cost:

$$p_{ift} = \frac{\lambda \varsigma_t}{z_{ift}} \quad \text{where} \quad \varsigma_t \equiv \left(\frac{w_t}{\alpha}\right)^\alpha \left(\frac{q_t}{1-\alpha}\right)^{1-\alpha}.$$

The final sector's Cobb–Douglas aggregator implies that every product commands the same revenue. Since the leader's own unit cost is $\varsigma_t[1 + \tau(n_{ft})]^{1-\alpha}/z_{ift}$, where n_{ft} is the number of products held by the leading firm, per-product flow profit is:

$$\pi_t(n_{ft}) = Y_t \left[1 - \frac{T(n_{ft})}{\lambda}\right] \quad \text{where} \quad T(n_{ft}) \equiv [1 + \tau(n_{ft})]^{1-\alpha}.$$

Since $T(n_{ft}) \geq 1$ for all n_{ft} while the fringe is undistorted, the wedge strictly reduces per-product profits for every firm, including stagnant firms, which face $\tau(1) > 0$. Recall that the wedge $\tau(n)$ is parameterized as a saturating function:

$$\tau(n) = \bar{\tau}(1 - e^{-\kappa n}).$$

That is, the wedge rises steeply for small firms and saturates near $\bar{\tau}$ for large ones. Consequently, the profit penalty $T(n)/\lambda$ is increasing in n —each additional product line lowers the per-product margin—so the ULCRA acts as a progressive tax on product scope.

Let δ_t denote the aggregate rate of creative destruction. The Hamilton–Jacobi–Bellman (HJB) equation for a dynamic firm with $n \geq 1$ products is:

$$\begin{aligned} r_t V_t^D(n) &= n\pi_t(n) + \dot{V}_t^D(n) \\ &\quad + \max_{X_t \geq 0} \{X_t[V_t^D(n+1) - V_t^D(n)] - X_t^2 Y_t / (2\theta_X n)\} \\ &\quad - n\delta_t[V_t^D(n) - V_t^D(n-1)] \end{aligned}$$

with boundary condition $V_t^D(0) = 0$. The three terms on the right-hand side are (1) flow profits plus capital gains, (2) the option value of expansion net of its cost in final goods, and (3) the expected loss in value from creative destruction. The first-order condition for the per-product expansion rate $x_t(n) \equiv X_t/n$ yields:

$$x_t(n) = \frac{\theta_X [V_t^D(n+1) - V_t^D(n)]}{Y_t}. \quad (12)$$

By contrast, a stagnant firm holds a single product ($n = 1$), has no access to the expansion technology, and has value V_t^S satisfying:

$$(r_t + \delta_t)V_t^S = \pi_t(1) + \dot{V}_t^S.$$

The entrant’s problem. Potential entrants direct final goods toward entry to maximize expected profits net of entry costs:

$$\max_{e_t \geq 0, d_t \in [0,1]} \{e_t[d_t V_t^D(1) + (1 - d_t)V_t^S - Y_t^T] - Y_t^E\}.$$

The first-order condition for the dynamic share d_t balances the value premium of being dynamic against the type-selection cost:

$$d_t = \frac{\theta_T[V_t^D(1) - V_t^S]}{\theta_T[V_t^D(1) - V_t^S] + Y_t}.$$

Since the ULCRA depresses the value of dynamic firms more than that of stagnant firms (dynamic firms expect to grow into higher- n states where the wedge binds more severely), the premium $V_t^D(1) - V_t^S$ is compressed under the Act, reducing d_t and thereby the expected

value of entry. The first-order condition for the entry rate e_t equates the marginal entrant's expected value to its marginal cost:

$$e_t = \theta_E \cdot \left\{ \frac{d_t V_t^D(1) + (1 - d_t) V_t^S}{Y_t} + \frac{\ln(1 - d_t) + d_t}{\theta_T} \right\}.$$

Equilibrium definition. Now that all decision problems have been described, we can define the concept of a market equilibrium allocation.

Definition 5.1. Given initial conditions for the process efficiency of each product $\{z_{i0}\}_{i \in [0,1]}$, the distribution of product across dynamic firms $\{m_0^D(n)\}_{n \geq 1}$, and the measure of stagnant firms M_0^S , a market equilibrium allocation consists of time paths for allocations and prices such that the following conditions hold:

1. $\{C_t\}_{t \geq 0}$ solve the household's problem.
2. $\{\{y_{it}\}_{i \in [0,1]}\}_{t \geq 0}$ solve the final sector's problem.
3. $\{\{\{h_{ift}, \ell_{ift}, p_{ift}\}_{i \in \mathcal{P}_{ft}}, X_{ft}\}_{f \in \mathcal{F}_t}\}_{t \geq 0}$ solve the firm's problem.
4. $\{e_t, d_t\}_{t \geq 0}$ solve the entrant's problem.
5. $\{\{p_{it}\}_{i \in [0,1]}\}_{t \geq 0}$ clear the product markets.
6. $\{w_t\}_{t \geq 0}$ clear the labor market.
7. $\{q_t\}_{t \geq 0}$ clear the land market.
8. $\{r_t\}_{t \geq 0}$ clear the asset market.

Aggregate output and misallocation. Despite the rich heterogeneity in firm size and land distortion exposure, the market equilibrium allocation of our model admits a clean aggregation. As derived in appendix A.2, aggregate output at every instant satisfies:

$$Y_t = \frac{Z_t H^\alpha L^{1-\alpha}}{\mathcal{T}_t} \quad \text{where} \quad \mathcal{T}_t \equiv \mathcal{H}_t^\alpha \cdot \mathcal{L}_t^{1-\alpha} \quad (13)$$

where Z_t is the geometric mean of process efficiencies across the product continuum, which we refer to as the economy's total factor productivity (TFP):

$$\ln(Z_t) \equiv \int_0^1 \ln(z_{it}) di.$$

The second term, $\mathcal{T}_t$, is an aggregate misallocation wedge that satisfies $\mathcal{T}_t \geq 1$ with equality if and only if τ is identical across all firms (see appendix A.2). This wedge is composed of two product-weighted averages:

$$\begin{aligned}\mathcal{H}_t &\equiv \sum_{n \geq 1} [1 + \tau(n)]^{1-\alpha} n m_t^D(n) + [1 + \tau(1)]^{1-\alpha} M_t^S \\ \mathcal{L}_t &\equiv \sum_{n \geq 1} [1 + \tau(n)]^{-\alpha} n m_t^D(n) + [1 + \tau(1)]^{-\alpha} M_t^S.\end{aligned}$$

In the absence of the ULCRA (when $\tau(n) = 0$ for all n), the aggregate wedge equals one, which implies that aggregate output is equal to the productivity frontier Z_t . The structure of the wedge reveals the source of the factor-misallocation output loss: if τ were identical across all firms, then $\mathcal{T}_t = 1$ regardless of its level, because the exponents on $1 + \tau$ cancel exactly. This cross-sectional distortion arises entirely from the *size-dependence* of $\tau(n)$: larger firms face a higher effective land cost and use too little land per product, while smaller firms use too much. Since $\mathcal{T}_t$ depends on the current firm-size distribution, this aggregate distortion evolves endogenously as the distribution changes. A uniform positive wedge still has a direct resource cost through the government's valueless use Y_t^G , even though it does not generate cross-sectional factor misallocation.

TFP growth and creative destruction. Having identified Z_t as TFP, we now characterize its growth rate. At every instant, the rate of creative destruction (the rate at which existing products are displaced by innovations) is the sum of incumbent expansion and entry:

$$\delta_t = \sum_{n=1}^{\infty} x_t(n) n m_t^D(n) + e_t = \bar{x}_t^D N_t^D + e_t \quad (14)$$

where $\bar{x}_t^D$ is the product-weighted mean expansion rate of dynamic firms, N_t^D is the share of products held by dynamic firms, and e_t is the entry rate. As shown in appendix A.2, each displacement raises process efficiency by a factor of λ , so growth rate of TFP satisfies:

$$g_t = \delta_t \cdot \ln(\lambda) \quad (15)$$

at all times. This decomposition highlights the two margins through which the ULCRA may suppress growth. First, the land distortion reduces per-product profits:

$$\pi_t(n) = Y_t \left[1 - \frac{T(n)}{\lambda} \right]$$

lowering the value of expanding into new product lines and thereby depressing the expansion rate $\bar{x}_t^D$. Second, lower firm values reduce the entry rate e_t and distort the composition of entrants toward the stagnant type.

Disciplining the land wedge with MRPLand and dynamic responses. The aggregate wedge $\mathcal{T}_t$ in equation (13) summarizes cross-sectional factor misallocation, while ω_t^G measures the direct resource cost of the wedge revenue. Their common structural parameters $\bar{\tau}$ and κ are not directly observable. The model links these parameters to two distinct sets of empirical moments. The first is the marginal revenue product of land (MRPLand) difference-in-differences (DiD) coefficient estimated in Section 4. In the model, a product manufactured by a firm with n products uses:

$$\ell_t(n) = \frac{(1 - \alpha)Y_t}{\lambda q_t [1 + \tau(n)]^\alpha}$$

units of land (see appendix A.1), such that its log marginal revenue product of land is:

$$\ln(\text{MRPLand}) = \underbrace{\ln\left(\frac{\lambda q_t}{1 - \alpha}\right)}_{\text{absorbed by FEs}} + \alpha \ln[1 + \tau(n)].$$

In the DiD estimation exercise, the constant is absorbed by fixed effects and the treatment consists of eliminating the wedge. Therefore, the model counterpart of the MRPLand DiD coefficient is:

$$\beta_{DiD} = -\alpha \cdot \mathbb{E}_{\text{firm}}[\ln(1 + \tau(n))] \quad (16)$$

where the expectation is firm-weighted among dynamic firms on the pre-repeal balanced growth path. Since the land share $1 - \alpha$ is externally calibrated, the MRPLand DiD estimate directly disciplines the average log-wedge $\mathbb{E}_{\text{firm}}[\ln(1 + \tau(n))]$, which in turn pins down its overall magnitude $\bar{\tau}$. MRPLand alone, however, does not determine how the wedge varies with firm size. That information comes from the joint response of product creation, product destruction, exit, and dynamic entry documented in Section 4. These moments help distinguish a roughly uniform decline in effective land costs from one that disproportionately relaxes constraints on more scalable firms. Within the model, they therefore help pin down how sharply the wedge rises with product scope, as governed by κ .

Welfare decomposition. The preceding results identify three broad channels through which the ULCRA reduces welfare: cross-sectional factor misallocation through $\mathcal{T}_t$, direct

government use of final goods through ω_t^G , and dynamic effects on R&D, entry, and TFP growth. Let γ denote the consumption-equivalent welfare gain, defined as the factor by which pre-repeal consumption must be permanently scaled to make the household indifferent between the pre-repeal stationary allocation and the post-repeal path. As a useful steady-state benchmark, comparing the pre- and post-repeal balanced growth paths yields (see appendix A.4):

$$\ln(\gamma) = \underbrace{\ln(\mathcal{T})}_{\text{Misallocation}} + \underbrace{\frac{g^{\text{post}} - g^{\text{pre}}}{\rho}}_{\text{Growth}} + \underbrace{\ln\left(\frac{1 - \Gamma^{\text{post}}}{1 - \Gamma^{\text{pre}}}\right)}_{\text{R\&D/entry allocation}} + \underbrace{\ln\left(\frac{1 - \Gamma^{\text{pre}}}{1 - \Gamma^{\text{pre}} - \omega^{G,\text{pre}}}\right)}_{\text{Government waste}} \quad (17)$$

where $\Gamma \equiv (Y^X + Y^E + eY^T)/Y$ is the R&D and entry expenditure share and $\omega^G \equiv Y^G/Y$ is the government's wasteful-use share, so that $C/Y = 1 - \Gamma - \omega^G$. The first term captures the static gain from eliminating cross-sectional factor misallocation. The second term is the dynamic gain from permanently faster TFP growth. The third term reflects the change in final goods allocated to expansion and entry. The fourth term is the direct consumption gain from eliminating the government's valueless use of final goods. Post-repeal, $\omega^{G,\text{post}} = 0$.

5.3 Calibration

The model features six structural parameters $(\lambda, \theta_X, \theta_E, \theta_T, \bar{\tau}, \kappa)$ that we jointly calibrate by matching ten empirical moments. Two additional parameters are set externally: the discount rate $\rho = 0.05$ and the labor share $\alpha = 0.81$, following Sood (2022) for Indian manufacturing. Our calibration adopts a dual-sample approach that mirrors the model's two firm types: dynamic firms correspond to the organized sector covered by the Annual Survey of Industries (ASI), while stagnant firms correspond to the unorganized sector covered by the National Sample Survey (NSS). We estimate the six parameters by minimizing the weighted sum of squared percentage deviations between model and data moments. Appendix A.5 derives the model counterpart of each targeted moment.

Targeted moments. The ten moments fall into two distinct groups. The first five are cross-sectional features of the pre-repeal economy. The economy-wide TFP growth rate of 3.5% is taken from Bollard et al. (2013) for 1980–1992. The average number of products per firm is computed twice: once across all firms (ASI and NSS combined, capturing both organized and unorganized manufacturing) and once among dynamic (ASI) firms only. The entry rate is the share of organized-sector ASI plants observed in their first year of operation,

so it captures entry within registered manufacturing rather than economy-wide firm entry. The old-to-young ratio is the number of dynamic firms aged 40–45 relative to those aged 1–5, capturing the shape of the age distribution.

The remaining five moments are difference-in-differences coefficients from Section 4, measuring the causal effect of the Repeal on treated firms. The product creation and destruction DiDs capture the change in new products introduced and existing products discontinued, estimated by Poisson pseudo-maximum likelihood. The entry DiD measures the change in dynamic entry. The exit DiD captures the change in firm exit rates from a linear probability model. The MRPLand DiD measures the change in the log marginal revenue product of land.

Identification. Although all parameters are jointly calibrated, some moments are especially informative about particular parameters. The process efficiency step size $\lambda > 1$ scales TFP growth through $g = \delta \cdot \ln(\lambda)$. The expansion efficiency θ_X governs how many products dynamic firms accumulate, primarily disciplined by $\mathbb{E}[n \mid \text{dynamic}]$. The entry efficiency θ_E is disciplined by the entry rate, while the type-selection efficiency parameter θ_T shapes the age distribution through its effect on the dynamic share of entrants, as reflected in the old-to-young ratio. The ULCRA parameters $\bar{\tau}$ and κ are disciplined by the causal moments from the Repeal. The MRPLand DiD is informative about the average magnitude of the effective land-cost distortion through equation (16). The product creation, destruction, exit, and dynamic-entry DiDs then help pin down how strongly that distortion rose with firm size, because they reveal whether the Repeal primarily relaxed constraints on incumbent expansion, market selection, or both.

Calibrated parameters and model fit. Table 7 reports the externally set and calibrated parameter values. The calibrated step size $\lambda = 1.231$ implies that each innovation raises process efficiency by roughly 23.1%. The expansion and entry efficiencies $\theta_X = 0.073$ and $\theta_E = 0.100$ generate modest rates of product creation, consistent with the slow firm growth documented in Indian manufacturing. The type-selection parameter $\theta_T = 0.887$ governs the cost of drawing a dynamic type upon entry. The ULCRA parameters $\bar{\tau} = 0.231$ and $\kappa = 1.374$ imply a saturating wedge schedule: a single-product firm faces a wedge of 0.173, rising to 0.231 for a five-product firm and 0.231 for a ten-product firm.

Table 8 compares the targeted moments in the data and the model. The model reproduces the innovation DiD within about 5% and the destruction DiD within about 10% of their empirical counterparts. The entry DiD is about 20% below the data, while the exit DiD

Parameter	Description	Value	Target / Source
<i>Panel A. Externally calibrated</i>			
ρ	Discount rate	0.05	Standard
$1 - \alpha$	Land share	0.19	Sood (2022)
<i>Panel B. Internally calibrated</i>			
λ	Innovation step size	1.231	TFP growth rate
θ_X	Expansion efficiency	0.073	Avg. products (dynamic)
θ_E	Entry efficiency	0.100	Entry rate
θ_T	Type-selection efficiency	0.887	Old-to-young ratio
$\bar{\tau}$	Asymptotic land wedge	0.231	MRPLand DiD
κ	Wedge saturation rate	1.374	Product turnover and entry-exit DiDs

Note: Panel A parameters are set externally. Panel B parameters are jointly calibrated by minimizing the weighted sum of squared percentage deviations between model and data moments. The “Target / Source” column indicates the primary identifying moment for each parameter.

Table 7: Summary of parameters

is close in absolute terms. The model overstates the magnitude of the MRPLand decline by about 39%. Among the cross-sectional moments, the average product counts are well matched (within 1–6%), and the old-to-young ratio is within about 29%. The TFP growth rate and entry rate deviate by about 24% and 43%, respectively. These discrepancies reflect a tension between matching the pre-repeal level of firm dynamics and the DiD treatment effects with only six free parameters.

Untargeted moments. The model also matches several patterns not directly targeted in the calibration. The model implies a pre-repeal average markup of 1.231 over marginal cost. Estimating markups directly from the ASI data using the production function approach of [De Ridder et al. \(2022\)](#) yields a pre-repeal average of 1.34, which is reasonably close. The product distribution among dynamic firms is heavily right-skewed, with 62.0% of dynamic firms holding a single product (compared to 62.5% in the data). The model generates a flat lifecycle employment profile consistent with the well-documented pattern of slow firm growth in India ([Hsieh and Klenow, 2014](#)). Firm survival rates decline steeply with age, driven by the high rate of creative destruction faced by single-product firms. The age distribution of firms is also well-matched.

Moment	Data	Model	Sample
<i>Panel A. Cross-sectional moments</i>			
TFP growth rate	0.035	0.027	Economy-wide
$\mathbb{E}[n \mid \text{all}]$	1.143	1.210	Economy-wide
$\mathbb{E}[n \mid \text{dynamic}]$	1.748	1.760	Dynamic firms
Entry rate	0.052	0.075	Dynamic firms
Old-to-young ratio	0.077	0.099	Dynamic firms
<i>Panel B. Difference-in-differences estimates</i>			
Product creation DiD	0.132	0.138	Dynamic firms
Product destruction DiD	0.140	0.126	Dynamic firms
Entry DiD	0.243	0.194	Dynamic firms
Exit DiD	0.011	0.009	Dynamic firms
MRPLand DiD	-0.102	-0.142	Dynamic firms

Note: “Economy-wide” moments use ASI and NSS combined data; “Dynamic firms” moments use ASI (organized sector) only. Panel A moments are pre-repeal cross-sectional features. Panel B moments are DiD coefficients from Section 4. The TFP growth rate is from [Bollard et al. \(2013\)](#).

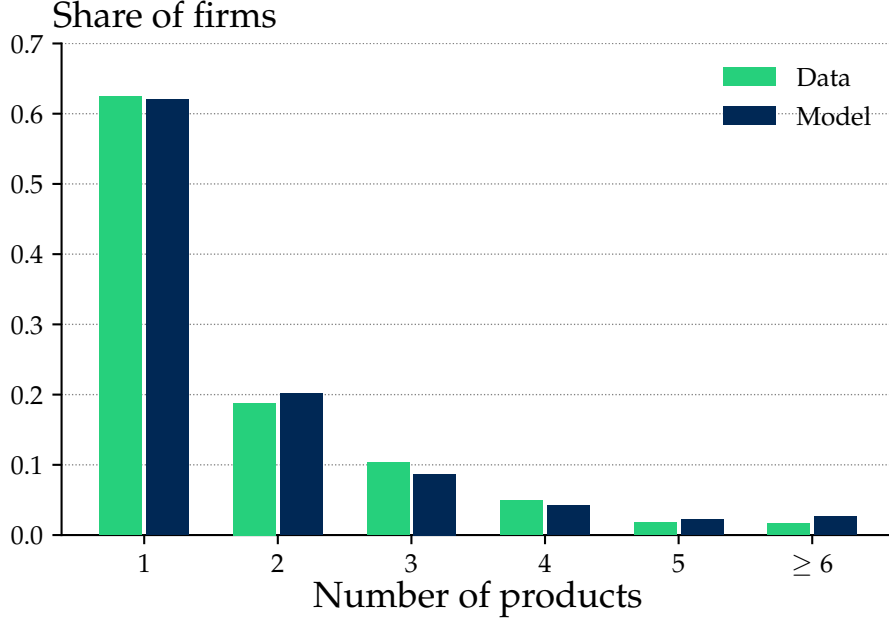
Table 8: Targeted moments: data vs. model

6 The macroeconomic impacts of the Repeal

We now use the calibrated model to quantify the aggregate effects of the Repeal. The counterfactual experiment sets the land market wedge to zero while holding all structural parameters at their calibrated values. This exercise should be interpreted as translating the causal microeconomic pattern documented in the data—lower effective land costs, greater product turnover, and stronger selection among scalable plants—into aggregate dynamics. We first examine how the Repeal reshapes firm dynamics in the new steady state, then compute the welfare gain along the transition path.

6.1 The long-run effects of the Repeal

Table 9 presents different moments of the pre- and post-repeal stationary market equilibrium allocations. Panel A focuses on product innovation and growth. The Repeal raises the growth rate of TFP by 0.32 percentage points, from 2.66% to 2.97%. Since $g = \delta \cdot \ln(\lambda)$ and λ is a structural parameter, the entire growth effect operates through an increase in the rate of creative destruction δ .



Note: Data: ASI pre-1999 urban-area sample (organized sector). Model: dynamic firm distribution $f^D(n)$ on the pre-repeal balanced growth path.

Figure 4: Product distribution

Why does creative destruction accelerate? Recall from equation (14) that $\delta = \bar{x}^D N^D + e$. Removing the wedge $\tau(n)$ raises per-product profits for all firms, increasing the marginal value of adding a product line. Dynamic firms respond by expanding faster, with the product-weighted expansion rate $\bar{x}^D$ rising by 1.1 percentage points. The entry rate e also rises modestly (+0.7 p.p.), as higher expected firm values make entry more attractive.

At first glance, the simultaneous increase in both expansion and creative destruction may seem paradoxical: in the standard framework of Klette and Kortum (2004), higher δ shortens the expected lifetime of each product line, reducing the return to expansion. This business-stealing force is present and quantitatively important, but it does not reverse the sign. Because dynamic firms hold only about two-fifths of all products before the Repeal, faster expansion raises the creative destruction rate by less than one for one. Meanwhile, the higher per-product profit flow anchors the franchise value of each product line independently of δ , preventing the business-stealing feedback from spiraling.

Why does the growth effect survive the business-stealing offset, when prior work on Indian factor market distortions finds near-zero or ambiguous effects on productivity growth? A key mechanism is the presence of stagnant firms, which hold roughly three-fifths of all product lines before the Repeal but perform no product expansion. When dynamic firms further

	Pre	Post	Change
<i>Panel A. Innovation and growth</i>			
Creative destruction rate (δ)	12.8%	14.3%	+1.5 (p.p.)
Product-weighted expansion rate ($\bar{x}^D$)	8.3%	9.4%	+1.1 (p.p.)
Entry rate (e)	9.5%	10.1%	+0.7 (p.p.)
TFP growth rate (g)	2.7%	3.0%	+0.32 (p.p.)
<i>Panel B. Selection</i>			
Dynamic product share (N^D)	40.1%	44.5%	+4.4 (p.p.)
Dynamic share of entrants (d)	19.1%	21.7%	+2.6 (p.p.)
$\mathbb{E}[n \mid \text{dynamic}]$	1.760	1.785	+1.4 (%)

Note: “Pre” and “Post” denote the balanced growth paths with and without the ULCRA wedge $\tau(n)$, respectively. All structural parameters are held fixed at their calibrated values.

Table 9: Equilibrium effects of the Repeal on firm dynamics

expand after the Repeal, they displace products held by both dynamic and stagnant firms. But stagnant firms do not respond to displacement by creating new products—they simply exit when they lose their only product. This asymmetry weakens the usual feedback loop: in a model where all firms expand, each displaced product triggers a competitive response that partially undoes the initial expansion impulse. Here, a substantial share of the products captured by dynamic firms comes from a non-reactive segment of the economy, allowing the growth acceleration to persist.

These product expansion effects are further reinforced by a compositional shift (Panel B). The dynamic product share N^D rises by 4.4 percentage points (from 40.1% to 44.5%), as dynamic firms expand into a larger fraction of the product space. This shift is driven in part by the dynamic share of entrants d rising from 19.1% to 21.7%: the Repeal increases the value premium of being dynamic, since the option to expand is more valuable when land costs do not penalize growth. The average product count among dynamic firms, $\mathbb{E}[n \mid \text{dynamic}]$, rises only modestly (from 1.760 to 1.785), as new single-product entrants dilute the firm-size distribution. This compositional upgrading is consistent with the empirical evidence in Section 4: the Repeal increased the entry of dynamic firms while leaving subsistence entry roughly unchanged.

6.2 Welfare gains from the Repeal

To quantify the welfare improvement from the Repeal, we solve for the full transition path of the reform. On the date of the Repeal, the wedge $\tau(n)$ is immediately removed, but the pre-determined firm-size distribution adjusts only gradually. As the distribution converges to the new steady state, the creative destruction rate δ_t , the product expansion rates $x_t(n)$, and the consumption share $\mathcal{C}_t$ of output all adjust along the transition. We define the welfare gain γ as the permanent proportional increase in pre-repeal consumption that would make the household indifferent between the pre-repeal stationary allocation and the post-repeal transition path. As derived in appendix A.4, $\ln(\gamma)$ can be decomposed into four components:

$$\begin{aligned} \ln(\gamma) = & \underbrace{\ln(\mathcal{T})}_{\text{misallocation}} + \underbrace{\ln\left(\frac{1 - \Gamma^{\text{pre}}}{1 - \Gamma^{\text{pre}} - \omega^{G,\text{pre}}}\right)}_{\text{government waste}} + \underbrace{\ln(\lambda) \int_0^\infty e^{-\rho t} (\delta_t^{\text{post}} - \delta^{\text{pre}}) dt}_{\text{growth acceleration}} \\ & + \underbrace{\int_0^\infty \rho e^{-\rho t} \ln\left(\frac{1 - \Gamma_t^{\text{post}}}{1 - \Gamma^{\text{pre}}}\right) dt}_{\text{R\&D and entry allocation}}, \end{aligned} \quad (18)$$

where Γ is the share of output devoted to expansion, entry, and type selection, and $\omega^G = Y^G/Y$ is the government's wasteful-use share. Table 10 reports this decomposition. The total improvement in welfare is equivalent to a 7.30% permanent increase in consumption per capita. Growth acceleration contributes 5.96%, while eliminating the government's valueless use contributes 2.71%. The R&D and entry allocation effect is -1.63%: faster growth changes the share of final goods devoted to product expansion and entry along the transition. As a robustness exercise, the second column of Table 10 assumes instead that the government rebates all wedge revenue lump-sum to the household. The welfare gain remains 4.43%, because the rebate removes only the direct government-waste channel and leaves firms' marginal land wedge, innovation, and entry decisions unchanged. Figure 6 plots the transition paths of the TFP growth rate and the consumption share of output.

The gain from removing cross-sectional factor misallocation is negligible (+0.00%). This component equals $\ln(\mathcal{T}^{\text{pre}})$. Since labor and land are inelastically supplied, a uniform wedge leaves $\mathcal{T} = 1$; only cross-sectional dispersion in $\tau(n)$ affects this component. This does not imply that a uniform wedge is costless: even a uniform wedge finances the government's valueless use of final goods, which is captured separately by the government-waste term in (18). A second-order approximation to the misallocation component yields $\ln(\mathcal{T}) \approx \alpha(1 - \alpha)\mathbb{V}(\tau)/[2(1 + \mathbb{E}[\tau])^2]$, where both moments are computed under the product distribution.

Component	Wasteful use	Lump-sum rebate
Static misallocation removal	+0.00	+0.00
Government waste removal	+2.71	0.00
R&D and entry allocation	−1.63	−1.63
Growth acceleration	+5.96	+5.96
Total log-welfare gain	+7.04	+4.33
Consumption-equivalent gain	+7.30	+4.43

Note: Component rows and the total log-welfare row report contributions to $\ln(\gamma)$ multiplied by 100. The final row converts the total into a permanent consumption-equivalent percentage gain. In the baseline, wedge revenue finances valueless government purchases. In the robustness column, it is rebated lump-sum to the household. See appendix A.4 for the derivation.

Table 10: Welfare under alternative uses of land-wedge revenue

The importance of the growth acceleration is the central quantitative finding of this exercise. Several recent studies have quantified the welfare effects of factor market distortions in India: Bertrand et al. (2021) estimate a 5.9% static gain from contract labor reform with zero dynamic gain; Akcigit et al. (2021) find that improving the managerial environment to U.S. levels would close approximately 11% of the India–U.S. income gap, with roughly equal static and dynamic components; Bau and Matray (2023) estimate 3–17% static gains from FDI liberalization. In contrast, the ULCRA repeal produces a welfare gain of comparable magnitude but primarily through a *dynamic* mechanism. This difference arises because the ULCRA distorts the returns to expansion, directly suppressing the product expansion engine that drives long-run growth.

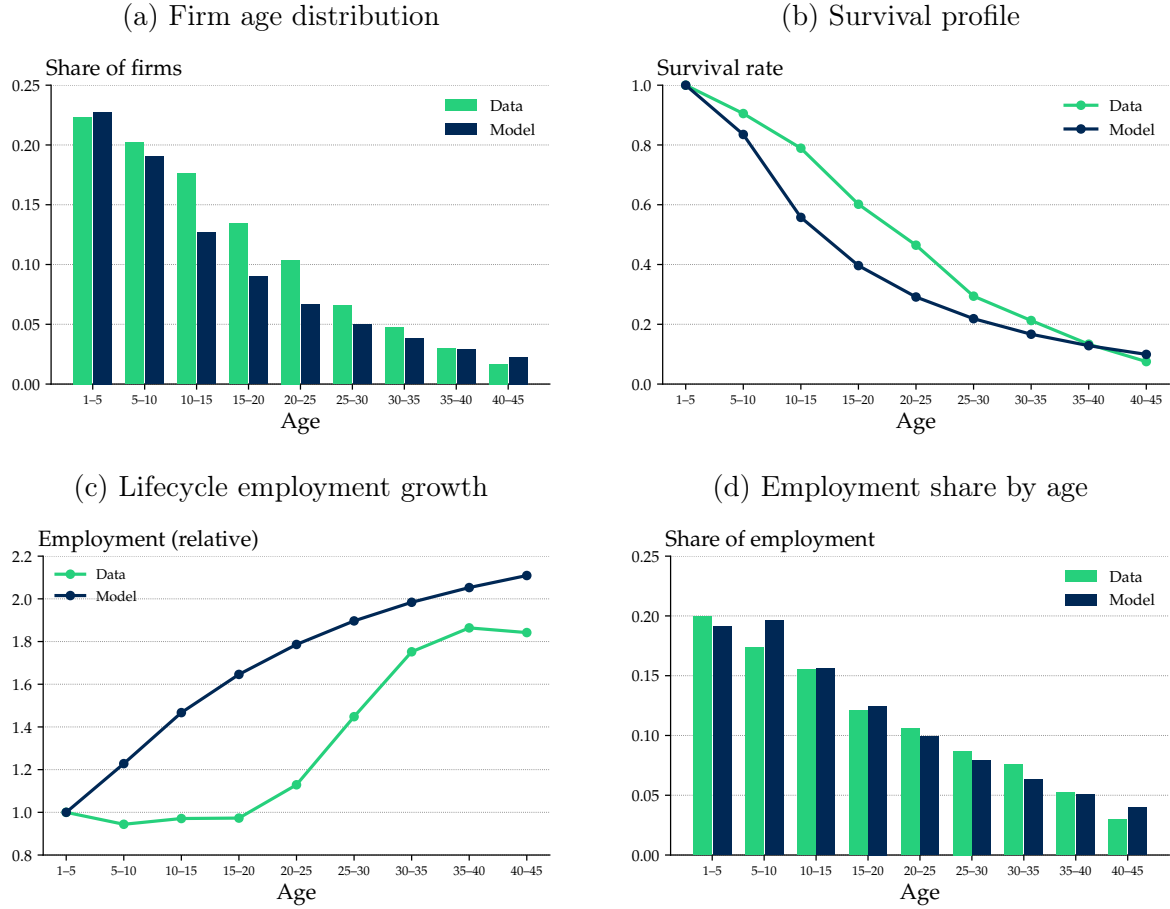
7 Conclusion

This paper studies the micro- and macroeconomic consequences of land market liberalization on manufacturing firms, leveraging the staggered repeal of India’s Urban Land Ceiling and Regulation Act as a natural experiment. Using a Difference-in-Differences framework, we show that the Repeal liberalized urban land markets, improved allocative efficiency, and unleashed industrial growth: treated plants expanded their land use, capital, employment, sales, and productivity. Beyond these static effects, we provide, to our knowledge, some of the first causal evidence that land market frictions affect firms’ product innovation decisions. The ULCRA’s repeal intensified creative destruction—both the creation and destruction of product lines accelerated—and shifted the composition of entrants toward more dynamic,

high-growth plants.

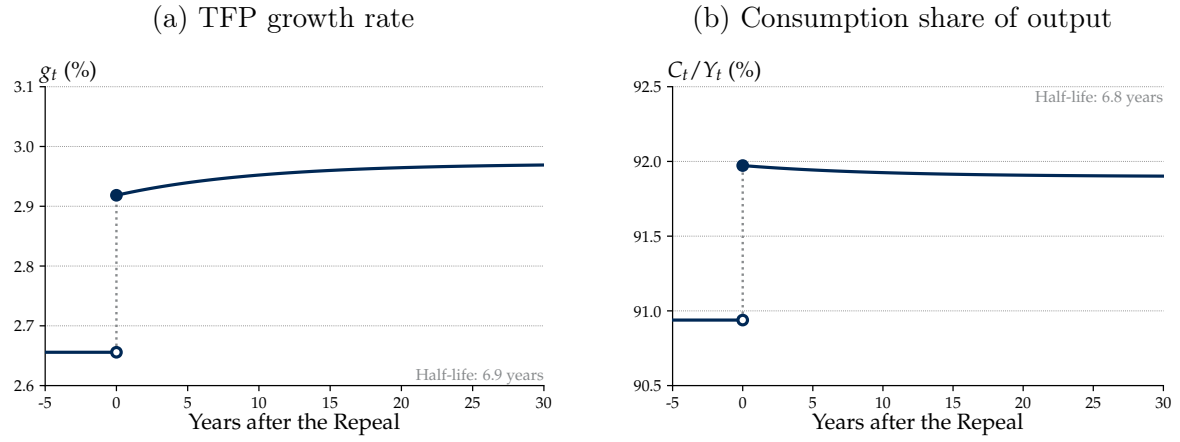
To quantify the aggregate implications of the Repeal, we developed and calibrated a general equilibrium model of endogenous growth in which the ULCRA acts as a size-dependent wedge on the effective cost of land. This formulation captures the land-market frictions created by the Act and their disproportionate burden on larger firms. The calibrated model indicates that the Repeal raised the growth rate of Indian manufacturing by 0.32 percentage points and generated a 7.30% improvement in consumption-equivalent welfare. Growth acceleration is the largest component of this gain. Eliminating the government’s valueless use of the land-wedge revenue also makes a material contribution, while the static productivity gain from removing cross-sectional factor misallocation is negligible.

Several directions remain for future research. Our analysis focuses on the manufacturing sector, but the ramifications of land market policies extend to urbanization, structural transformation, and housing markets; we provide preliminary evidence on some of these broader consequences in the appendix. More fundamentally, the interaction between land market reforms and other economic frictions—such as labor regulations, credit constraints, and infrastructure provision—could amplify or attenuate the growth effects we document. Understanding these complementarities is essential for designing effective reforms in developing economies.



Note: Data: ASI pre-1999 urban-area sample (organized sector only). Model: dynamic firms only. In all four panels, the ASI-to-dynamic mapping is maintained: no NSS (unorganized sector) data enters. Panel (a): share of firms in each age bin. Panel (b): cross-sectional age distribution normalized by the youngest bin, which equals the survival function in steady state. Panel (c): average employment per firm by age, normalized by the youngest cohort; the model counterpart is average products n per firm. Panel (d): share of total employment by age; the model counterpart is the share of total products.

Figure 5: Untargeted moments: model vs. data



Note: Transition paths after the unexpected permanent repeal of the ULCRA wedge $\tau(n)$ at $t = 0$. Panel (a): TFP growth rate $g_t = \delta_t \cdot \ln(\lambda)$. Panel (b): consumption share of output C_t/Y_t .

Figure 6: Transition dynamics

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A Theoretical appendix

This appendix provides complete derivations of all optimality conditions, aggregation results, the balanced growth path system, and the welfare decomposition stated in the main text. Each derivation proceeds from the optimization problem defined in Section 5 to the closed-form condition used in the calibration.

A.1 Derivation of optimality conditions

The household's problem. The representative household chooses a consumption path $\{C_t\}_{t \geq 0}$ to maximize lifetime utility subject to a flow budget constraint:

$$\int_0^\infty e^{-\rho t} \ln(C_t) dt \quad \text{s.t.} \quad \dot{A}_t = r_t A_t + w_t H + q_t L - C_t.$$

Form the current-value Hamiltonian with costate variable μ_t :

$$\mathcal{H}_t = \ln(C_t) + \mu_t(r_t A_t + w_t H + q_t L - C_t). \quad (\text{A.1})$$

The necessary conditions for an interior optimum are:

$$\text{FOC w.r.t. } C_t: \quad \frac{\partial \mathcal{H}_t}{\partial C_t} = \frac{1}{C_t} - \mu_t = 0, \quad (\text{A.2})$$

$$\text{Costate equation:} \quad \dot{\mu}_t = \rho \mu_t - \frac{\partial \mathcal{H}_t}{\partial A_t} = (\rho - r_t) \mu_t. \quad (\text{A.3})$$

Combining both conditions gives the Euler equation:

$$\frac{\dot{C}_t}{C_t} = r_t - \rho. \quad (\text{A.4})$$

The final sector's problem. The competitive final sector takes prices as given and chooses its demand for products to maximize profits subject to the Cobb–Douglas technology:

$$\max_{\{y_{it}\}_{i \in [0,1]}} P_t Y_t - \int_0^1 p_{it} y_{it} di \quad \text{s.t.} \quad \ln(Y_t) = \int_0^1 \ln(y_{it}) di$$

where P_t is the price of the final good. The first-order condition for each product i implies:

$$p_{it} y_{it} = P_t Y_t, \quad \forall i \in [0, 1].$$

To derive the price index, substitute $y_{it} = P_t Y_t / p_{it}$ into the technology $\ln(Y_t) = \int_0^1 \ln(y_{it}) di$:

$$\ln(Y_t) = \int_0^1 [\ln(P_t Y_t) - \ln(p_{it})] di = \ln(P_t Y_t) - \int_0^1 \ln(p_{it}) di.$$

Rearranging gives:

$$\ln(P_t) = \int_0^1 \ln(p_{it}) di. \quad (\text{A.5})$$

That is, the price of the final good is the geometric mean of product prices. Setting $P_t = 1$ (the numéraire normalization) imposes $\int_0^1 \ln(p_{it}) di = 0$. With this normalization:

$$p_{it} y_{it} = Y_t, \quad \forall i \in [0, 1]. \quad (\text{A.6})$$

The firm's problem. Consider a leading n -product firm with process efficiency z_{it} manufacturing product i . The leader minimizes total cost subject to meeting demand y_{it} :

$$\min_{h_{it}, \ell_{it} \geq 0} \{w_t h_{it} + [1 + \tau(n)] q_t \ell_{it}\} \quad \text{s.t.} \quad z_{it} h_{it}^\alpha \ell_{it}^{1-\alpha} \geq y_{it}.$$

Let μ_{it} denote the Lagrange multiplier on the production constraint. The first-order conditions are:

$$\text{FOC w.r.t. } h_{it}: \quad w_t = \mu_{it} \alpha z_{it} h_{it}^{\alpha-1} \ell_{it}^{1-\alpha} \quad (\text{A.7})$$

$$\text{FOC w.r.t. } \ell_{it}: \quad [1 + \tau(n)] q_t = \mu_{it} (1 - \alpha) z_{it} h_{it}^\alpha \ell_{it}^{-\alpha}. \quad (\text{A.8})$$

Taking the ratio of equations (A.7) and (A.8) eliminates μ_{it} :

$$\frac{h_{it}}{\ell_{it}} = \frac{\alpha}{1 - \alpha} \cdot \frac{[1 + \tau(n)] q_t}{w_t}. \quad (\text{A.9})$$

Substituting the optimal factor ratio back into the production function and solving for the cost-minimizing inputs gives:

$$w_t h_{it} = \alpha \cdot \frac{\varsigma_t(n)}{z_{it}} \cdot y_{it}, \quad (\text{A.10})$$

$$[1 + \tau(n)] q_t \ell_{it} = (1 - \alpha) \cdot \frac{\varsigma_t(n)}{z_{it}} \cdot y_{it} \quad (\text{A.11})$$

where the distorted unit cost index is:

$$\varsigma_t(n) \equiv \varsigma_t \cdot T(n). \quad (\text{A.12})$$

Here, we have defined:

$$\varsigma_t \equiv \left(\frac{w_t}{\alpha}\right)^\alpha \left(\frac{q_t}{1-\alpha}\right)^{1-\alpha} \quad \text{and} \quad T(n) \equiv [1 + \tau(n)]^{1-\alpha}.$$

The competitive fringe uses the same Cobb–Douglas technology as the leader but operates at process efficiency z_{it}/λ and faces no ULCRA wedge. Its marginal cost is therefore:

$$\text{MC}_{\text{fringe}} = \frac{\lambda \varsigma_t}{z_{it}}. \quad (\text{A.13})$$

The leader sets price equal to the fringe's marginal cost (limit pricing):

$$p_{it} = \frac{\lambda \varsigma_t}{z_{it}}. \quad (\text{A.14})$$

From equations (A.6), (A.11) and (A.14), solving for ℓ_{it} :

$$\ell_{it} = \frac{(1-\alpha)\varsigma_t(n)y_{it}}{[1+\tau(n)]q_t z_{it}} = \frac{(1-\alpha)Y_t}{\lambda q_t [1+\tau(n)]^\alpha}. \quad (\text{A.15})$$

Similarly, from equations (A.6), (A.10) and (A.14), solving for h_{it} :

$$h_{it} = \frac{\alpha \varsigma_t(n)y_{it}}{w_t z_{it}} = \frac{\alpha [1+\tau(n)]^{1-\alpha} Y_t}{\lambda w_t}. \quad (\text{A.16})$$

Revenue per product equals Y_t by equation (A.6). Production cost per product is the leader's marginal cost times quantity:

$$\frac{\varsigma_t \cdot T(n)}{z_{it}} \cdot y_{it} = \frac{\varsigma_t \cdot T(n)}{z_{it}} \cdot \frac{Y_t z_{it}}{\lambda \varsigma_t} = \frac{T(n)Y_t}{\lambda}.$$

Per-product profit is revenue minus cost:

$$\pi_t(n) = Y_t \left[1 - \frac{T(n)}{\lambda} \right]. \quad (\text{A.17})$$

Because per-product profit depends on n but not on the individual z_{it} 's, the dynamic firm's value $V_t^D(n)$ depends only on the product count n . The corresponding HJB equation is:

$$\begin{aligned} r_t V_t^D(n) = & n\pi_t(n) + \dot{V}_t^D(n) \\ & + \max_{X_t \geq 0} \{X_t[V_t^D(n+1) - V_t^D(n)] - X_t^2 Y_t / (2\theta_X n)\} \\ & - n\delta_t[V_t^D(n) - V_t^D(n-1)] \end{aligned} \quad (\text{A.18})$$

with boundary condition $V_t^D(0) = 0$. The three groups of terms on the right-hand side are: (1) flow profits plus capital gains, (2) the option value of expansion net of its cost in final goods, and (3) the expected loss from creative destruction. The maximand in the second line is concave in X_t , so the first-order condition is necessary and sufficient:

$$\frac{\partial}{\partial X_t} \left\{ X_t[V_t^D(n+1) - V_t^D(n)] - \frac{X_t^2 Y_t}{2\theta_X n} \right\} = V_t^D(n+1) - V_t^D(n) - \frac{X_t Y_t}{\theta_X n} = 0.$$

Solving for X_t and defining the per-product expansion rate $x_t(n) \equiv X_t/n$:

$$x_t(n) = \frac{\theta_X[V_t^D(n+1) - V_t^D(n)]}{Y_t}. \quad (\text{A.19})$$

A stagnant firm holds instead a single product, has no access to the expansion technology, and its value function satisfies:

$$(r_t + \delta_t)V_t^S = \pi_t(1) + \dot{V}_t^S.$$

The entrant's problem. The unit mass of potential entrants maximizes expected profit net of entry costs:

$$\max_{e_t \geq 0, d_t \in [0,1]} \{e_t[d_t V_t^D(1) + (1-d_t)V_t^S - Y_t^T] - Y_t^E\} \quad (\text{A.20})$$

where $Y_t^E = e_t^2 Y_t / (2\theta_E)$ and $Y_t^T = -[\ln(1-d_t) + d_t]Y_t / \theta_T$ from equation (6). Taking the first-order condition with respect to d_t :

$$e_t[V_t^D(1) - V_t^S] - \frac{e_t d_t Y_t}{\theta_T(1-d_t)} = 0.$$

The factor e_t cancels (assuming $e_t > 0$). Defining $\Delta V_t \equiv V_t^D(1) - V_t^S$ and solving for d_t :

$$d_t = \frac{\theta_T \cdot \Delta V_t}{\theta_T \cdot \Delta V_t + Y_t}. \quad (\text{A.21})$$

An interior solution $d_t > 0$ requires $\Delta V_t > 0$: the dynamic type must be strictly more valuable than the stagnant type. Taking the first-order condition with respect to e_t :

$$d_t V_t^D(1) + (1 - d_t) V_t^S - Y_t \left[\frac{e_t}{\theta_E} - \frac{\ln(1 - d_t) + d_t}{\theta_T} \right] = 0.$$

Solving for e_t , we obtain:

$$e_t = \theta_E \cdot \left\{ \frac{V_t^S + d_t \cdot \Delta V_t}{Y_t} + \frac{\ln(1 - d_t) + d_t}{\theta_T} \right\}. \quad (\text{A.22})$$

A.2 Aggregation

Labor market clearing. The labor market clearing condition yields:

$$\int_0^1 h_{it} di = H. \quad (\text{A.23})$$

Substituting equation (A.16) for h_{it} , we obtain:

$$\int_0^1 h_{it} di = \frac{\alpha Y_t \mathcal{H}_t}{\lambda w_t} = H \quad (\text{A.24})$$

where the aggregate labor distortion $\mathcal{H}_t$ is defined as:

$$\mathcal{H}_t \equiv \left\{ \sum_{n=1}^{\infty} [1 + \tau(n)]^{1-\alpha} n m_t^D(n) + [1 + \tau(1)]^{1-\alpha} M_t^S \right\}. \quad (\text{A.25})$$

Here, $m_t^D(n)$ is the measure of dynamic firms with n products and M_t^S is the measure of stagnant firms. The aggregate labor distortion $\mathcal{H}_t$ is a product-weighted average of the ULCRA wedges across firms. We can solve for the wage rate in equation (A.24) as:

$$w_t = \frac{\alpha Y_t \mathcal{H}_t}{\lambda H}.$$

Land market clearing. The land market clearing condition yields:

$$\int_0^1 \ell_{it} di = L. \quad (\text{A.26})$$

Substituting equation (A.15) for ℓ_{it} , we obtain:

$$\int_0^1 \ell_{it} di = \frac{(1-\alpha)Y_t \mathcal{L}_t}{\lambda q_t} = L \quad (\text{A.27})$$

where the aggregate land distortion $\mathcal{L}_t$ is defined as:

$$\mathcal{L}_t \equiv \left\{ \sum_{n=1}^{\infty} [1 + \tau(n)]^{-\alpha} n m_t^D(n) + [1 + \tau(1)]^{-\alpha} M_t^S \right\}. \quad (\text{A.28})$$

We can solve for the land rent in equation (A.27) as:

$$q_t = \frac{(1-\alpha)Y_t \mathcal{L}_t}{\lambda L}.$$

Government's wasteful use of final goods. The government collects the wedge payment $\tau(n)q_t \ell_t(n)$ from each product operated by an n -product firm and spends the revenue on final goods that provide no utility or productive services. Its balanced budget is:

$$Y_t^G = \sum_{n=1}^{\infty} \tau(n) q_t \ell_t(n) n m_t^D(n) + \tau(1) q_t \ell_t(1) M_t^S. \quad (\text{A.29})$$

Substituting land demand from (A.15) gives:

$$\omega_t^G \equiv \frac{Y_t^G}{Y_t} = \frac{1-\alpha}{\lambda} \left\{ \sum_{n=1}^{\infty} \tau(n) [1 + \tau(n)]^{-\alpha} n m_t^D(n) + \tau(1) [1 + \tau(1)]^{-\alpha} M_t^S \right\} = \frac{1-\alpha}{\lambda} (\mathcal{H}_t - \mathcal{L}_t). \quad (\text{A.30})$$

The last equality uses $(1+\tau)^{1-\alpha} - (1+\tau)^{-\alpha} = \tau(1+\tau)^{-\alpha}$. Since the revenue is not rebated, it does not enter household income; the household budget remains unchanged.

Final goods market clearing. Aggregate output is split between consumption, expansion, entry, type-selection spending, and the government's wasteful use. Total incumbent expansion spending is:

$$Y_t^X = \int_{f \in \mathcal{F}_t} \frac{X_{ft}^2 Y_t}{2\theta_X n_{ft}} df = \frac{Y_t}{2\theta_X} \cdot \sum_{n=1}^{\infty} x_t(n)^2 n m_t^D(n).$$

Total entry spending is:

$$Y_t^E = \frac{e_t^2 Y_t}{2\theta_E}.$$

Type-selection spending per entrant is:

$$Y_t^T = -\frac{[\ln(1 - d_t) + d_t]Y_t}{\theta_T}.$$

Therefore, the final goods market clearing condition is:

$$Y_t = C_t + \frac{Y_t}{2\theta_X} \cdot \sum_{n=1}^{\infty} x_t(n)^2 n m_t^D(n) + \frac{e_t^2 Y_t}{2\theta_E} - e_t \cdot \frac{[\ln(1 - d_t) + d_t]Y_t}{\theta_T} + Y_t^G.$$

Creative destruction. The aggregate rate of creative destruction is the sum of two flows: incumbent expansion and entry. Each dynamic firm with n products expands into new product lines at total rate $x_t(n)n$, displacing a randomly selected incumbent product. Entry creates new products at rate e_t , and each entrant product displaces a randomly selected incumbent product. Therefore, the aggregate creative destruction rate is:

$$\delta_t = \sum_{n=1}^{\infty} x_t(n) n m_t^D(n) + e_t.$$

Kolmogorov-Forward (KF) equation. Let $m_t^D(n)$ denote the measure of dynamic firms with n products. The KF equation for $m_t^D(n)$ at $n \geq 2$ is:

$$\begin{aligned} \dot{m}_t^D(n) &= x_t(n-1)(n-1)m_t^D(n-1) \\ &\quad - [x_t(n)n + \delta_t n]m_t^D(n) \\ &\quad + \delta_t(n+1)m_t^D(n+1). \end{aligned} \tag{A.31}$$

The first term is the inflow from expansion of $(n-1)$ -product firms, the second is the outflow from firms at n that either expand or lose a product, and the third is the inflow from $(n+1)$ -product firms that lose a product. At $n = 1$, there are two sources of inflow: entry of dynamic firms at rate $e_t d_t$ and two-product firms losing a product at rate $2\delta_t$. The outflow consists of expansion to $n = 2$ at rate $x_t(1)$ and exit upon losing the sole product at rate δ_t . Therefore, the KF equation at $n = 1$ is:

$$\dot{m}_t^D(1) = e_t d_t - [x_t(1) + \delta_t]m_t^D(1) + 2\delta_t m_t^D(2). \tag{A.32}$$

The measure of stagnant firms simply satisfies:

$$\dot{M}_t^S = e_t(1 - d_t) - \delta_t M_t^S. \tag{A.33}$$

Product shares. The total share of products held by dynamic and stagnant firms must exhaust the unit product continuum. Therefore, we have:

$$N_t^D + N_t^S = 1 \quad \text{where} \quad N_t^D \equiv \sum_{n=1}^{\infty} n m_t^D(n) \quad \text{and} \quad N_t^S \equiv M_t^S. \quad (\text{A.34})$$

Aggregate output. The numéraire normalization implies that the price of the final good is $P_t = 1$, which yields:

$$\int_0^1 \ln(p_{it}) di = 0.$$

Substituting the limit price from equation (A.14) gives:

$$\ln(\lambda_{\varsigma_t}) = \int_0^1 \ln(z_{it}) di.$$

Define the geometric mean of process efficiencies as:

$$\ln(Z_t) \equiv \int_0^1 \ln(z_{it}) di \quad (\text{A.35})$$

and substitute the definition of ς_t from equation (A.12):

$$\lambda \left(\frac{w_t}{\alpha} \right)^{\alpha} \left(\frac{q_t}{1-\alpha} \right)^{1-\alpha} = Z_t.$$

Using the wage and land rent from equations (A.24) and (A.27) gives:

$$Y_t = \frac{Z_t H^{\alpha} L^{1-\alpha}}{\mathcal{T}_t} \quad \text{where} \quad \mathcal{T}_t \equiv \mathcal{H}_t^{\alpha} \cdot \mathcal{L}_t^{1-\alpha}. \quad (\text{A.36})$$

In the absence of ULCRA wedges, $\mathcal{H}_t = \mathcal{L}_t = 1$ and aggregate output is equal to:

$$Y_t = Z_t H^{\alpha} L^{1-\alpha}.$$

With ULCRA wedges, aggregate output is distorted by the factor $\mathcal{T}_t$, which we now show is weakly greater than one.

Bound on the aggregate wedge.

Proposition A.1. *For any $\alpha \in (0, 1)$, any product weights $\{w_i\}_{i \in I}$ with $w_i \geq 0$ and*

$\sum_{i \in I} w_i = 1$, and any wedges $\{\tau_i\}_{i \in I}$ with $\tau_i \geq 0$:

$$\mathcal{T} \equiv \left[\sum_i w_i (1 + \tau_i)^{1-\alpha} \right]^\alpha \left[\sum_i w_i (1 + \tau_i)^{-\alpha} \right]^{1-\alpha} \geq 1$$

with equality if and only if τ_i is constant for all $i \in I$.

Proof. Let $x_i \equiv 1 + \tau_i$, drawn with probability w_i , and define the log-moment function $b(t) \equiv \ln(\mathbb{E}[x^t])$. Since $x_i > 0$, b is well-defined for all $t \in \mathbb{R}$. The function b is convex: for any $t_1, t_2 \in \mathbb{R}$ and $\nu \in [0, 1]$, Hölder's inequality with exponents $1/\nu$ and $1/(1-\nu)$ gives:

$$\mathbb{E}[(x^{t_1})^\nu \cdot (x^{t_2})^{1-\nu}] \leq \mathbb{E}[x^{t_1}]^\nu \cdot \mathbb{E}[x^{t_2}]^{1-\nu}$$

which is $b(\nu t_1 + (1-\nu)t_2) \leq \nu b(t_1) + (1-\nu)b(t_2)$. Now observe that:

$$\ln(\mathcal{T}) = \alpha b(1-\alpha) + (1-\alpha)b(-\alpha) \geq b(\alpha(1-\alpha) + (1-\alpha)(-\alpha)) = b(0) = 0$$

where the inequality comes from the convexity of b and the last step uses $b(0) = \ln(\mathbb{E}[x^0]) = 0$. Hölder's inequality holds with equality if and only if x_i is constant for all i , i.e., all τ_i 's are equal. $\square$

Second-order approximation of $\ln(\mathcal{T})$. Proposition A.1 shows that a uniform wedge $\tau_i = \mu$ for all i yields $\mathcal{T} = 1$ exactly, regardless of the common level μ . This result is a consequence of inelastic factor supply: since aggregate labor are fixed, a uniform wedge merely scales factor prices without distorting aggregate output. When the wedge varies across product lines, only its *cross-sectional dispersion* generates a loss. The following second-order approximation makes this precise.

Write $\tau_i = \mu + \varepsilon_i$ where $\mu \equiv \mathbb{E}[\tau_i]$ is the product-weighted mean and $\mathbb{E}[\varepsilon_i] = 0$, $\mathbb{E}[\varepsilon_i^2] = \sigma^2 \equiv \mathbb{V}(\tau)$. For a smooth function h , the second-order expansion of $\ln(\mathbb{E}[h(\tau)])$ around the degenerate distribution $\tau_i = \mu$ is:

$$\ln(\mathbb{E}[h(\tau)]) \approx \ln(h(\mu)) + \frac{\sigma^2}{2} \cdot \frac{h''(\mu)}{h(\mu)}.$$

Applying this to the two components of $\ln(\mathcal{T}) = \alpha \ln(\mathcal{H}) + (1 - \alpha) \ln(\mathcal{L})$:

$$\begin{aligned} h(\tau) = (1 + \tau)^{1-\alpha} &\implies \frac{h''(\mu)}{h(\mu)} = \frac{-\alpha(1 - \alpha)}{(1 + \mu)^2} \\ g(\tau) = (1 + \tau)^{-\alpha} &\implies \frac{g''(\mu)}{g(\mu)} = \frac{\alpha(1 + \alpha)}{(1 + \mu)^2}. \end{aligned}$$

The zeroth-order terms cancel: $\alpha \ln(h(\mu)) + (1 - \alpha) \ln(g(\mu)) = [\alpha(1 - \alpha) - \alpha(1 - \alpha)] \ln(1 + \mu) = 0$, confirming that $\mathcal{T} = 1$ when $\tau_i = \mu$. Collecting the second-order terms:

$$\ln(\mathcal{T}) \approx \frac{\sigma^2}{2(1 + \mu)^2} \cdot [\alpha \cdot (-\alpha)(1 - \alpha) + (1 - \alpha) \cdot \alpha(1 + \alpha)] = \frac{\alpha(1 - \alpha)}{2(1 + \mu)^2} \cdot \mathbb{V}(\tau) \quad (\text{A.37})$$

where $\mu = \mathbb{E}[\tau]$ and $\mathbb{V}(\tau)$ are both computed under the product distribution.

Growth of aggregate productivity. Let $\phi_t(s)$ denote the density of the logarithm of process efficiency $s \equiv \ln(z)$ across the product continuum. When creative destruction hits a product at rate δ_t , its log-productivity jumps by $\ln(\lambda)$. The KF equation for ϕ_t is:

$$\dot{\phi}_t(s) = \delta_t \cdot \phi_t(s - \ln(\lambda)) - \delta_t \cdot \phi_t(s). \quad (\text{A.38})$$

The first term is the inflow from products whose log-productivity was $s - \ln(\lambda)$ before being hit; the second is the outflow from products at s that are displaced. Taking the derivative of $\ln(Z_t)$ with respect to time:

$$\begin{aligned} \frac{\dot{Z}_t}{Z_t} &= \int s \cdot \dot{\phi}_t(s) ds \\ &= \int s \cdot [\delta_t \cdot \phi_t(s - \ln(\lambda)) - \delta_t \cdot \phi_t(s)] ds \\ &= \delta_t \int s \phi_t(s - \ln(\lambda)) ds - \delta_t \ln(Z_t). \end{aligned}$$

Substituting $u = s - \ln(\lambda)$ in the first integral gives:

$$\frac{\dot{Z}_t}{Z_t} = \delta_t \int [u + \ln(\lambda)] \phi_t(u) du - \delta_t \ln(Z_t) = \delta_t \cdot \ln(\lambda). \quad (\text{A.39})$$

This holds at all times: aggregate productivity grows at rate $\delta_t \ln(\lambda)$.

The differential-algebraic system. Equations (A.4), (A.17)–(A.19), (A.24)–(A.33), and (A.39), together with the final goods market clearing condition and the stagnant firm’s HJB, form a differential-algebraic system in the time-varying unknowns:

$$\{m_t^D(n), M_t^S, \delta_t, x_t(n), e_t, d_t, V_t^D(n), V_t^S, w_t, q_t, r_t, Y_t, C_t\}_{t \geq 0}.$$

From (A.39) and the price normalization, the output growth rate equals $\delta_t \ln(\lambda)$ whenever the firm-size distribution—and hence the product-weighted cost aggregates in equations (A.24) and (A.27)—is stationary.

A.3 Balanced growth path

Normalization. On the balanced growth path (BGP), Y_t , C_t , w_t , q_t , $V_t^D(n)$, and V_t^S all grow at the common constant rate $g \equiv \delta \ln(\lambda)$, so their ratios to C_t are time-invariant. Define the normalized quantities:

$$\mathcal{V}_t^D(n) \equiv \frac{V_t^D(n)}{C_t}, \quad \mathcal{V}_t^S \equiv \frac{V_t^S}{C_t}, \quad \mathcal{Y}_t \equiv \frac{Y_t}{C_t}, \quad \omega_t^H \equiv \frac{w_t}{C_t}, \quad \omega_t^L \equiv \frac{q_t}{C_t}.$$

On the BGP, these are all constant such that we suppress the time subscript when no ambiguity arises. The interest rate satisfies $r = \rho + g$ by the Euler equation. To derive the normalized HJB, note that $\dot{V}_t^D(n) = gV_t^D(n)$ on the BGP, so dividing equation (A.18) by C_t and using $r - g = \rho$ gives:

$$\rho \mathcal{V}^D(n) = n\mathcal{Y}[1 - T(n)/\lambda] + x(n)^2 n\mathcal{Y}/(2\theta_X) - n\delta[\mathcal{V}^D(n) - \mathcal{V}^D(n-1)]$$

where we have substituted the optimal per-product expansion rate from equation (A.19):

$$x(n) = \frac{\theta_X[\mathcal{V}^D(n+1) - \mathcal{V}^D(n)]}{\mathcal{Y}}.$$

A stagnant firm’s normalized value function instead satisfies:

$$\mathcal{V}^S = \frac{\mathcal{Y}[1 - T(1)/\lambda]}{\rho + \delta}.$$

The BGP system. The stationary system of equations is:

1. *Dynamic firm's HJB equation:*

$$\rho \mathcal{V}^D(n) = n\mathcal{Y}[1 - T(n)/\lambda] + x(n)^2 n\mathcal{Y}/(2\theta_X) - n\delta[\mathcal{V}^D(n) - \mathcal{V}^D(n-1)]. \quad (\text{A.40})$$

2. *Optimal expansion rate:*

$$x(n) = \theta_X[\mathcal{V}^D(n+1) - \mathcal{V}^D(n)]/\mathcal{Y}. \quad (\text{A.41})$$

3. *Stagnant firm's value function:*

$$\mathcal{V}^S = \mathcal{Y}[1 - T(1)/\lambda]/(\rho + \delta). \quad (\text{A.42})$$

4. *Optimal type selection:*

$$d = \frac{\theta_T \Delta \mathcal{V}}{\theta_T \Delta \mathcal{V} + \mathcal{Y}} \quad \text{where} \quad \Delta \mathcal{V} \equiv \mathcal{V}^D(1) - \mathcal{V}^S. \quad (\text{A.43})$$

5. *Optimal entry rate:*

$$e = \frac{\theta_E}{\mathcal{Y}} \cdot \left\{ \mathcal{V}^S + d\Delta \mathcal{V} + \frac{\mathcal{Y}[\ln(1-d) + d]}{\theta_T} \right\}. \quad (\text{A.44})$$

6. *Stationary firm-product distribution:*

$$0 = \begin{cases} (ed)/M^D - [\delta + x(1)]f^D(1) + 2\delta f^D(2) & \text{if } n = 1, \\ x(n-1)(n-1)f^D(n-1) - [x(n)n + \delta n]f^D(n) & \\ \quad + \delta(n+1)f^D(n+1) & \text{if } n \geq 2 \end{cases} \quad (\text{A.45})$$

where $f^D(n)$ is the fraction of dynamic firms with n products; $\sum_{n=1}^{\infty} f^D(n) = 1$.

7. *Product shares:*

$$N^S = e(1-d)/\delta \quad \text{and} \quad N^D = M^D \sum_{n=1}^{\infty} n f^D(n) = 1 - N^S. \quad (\text{A.46})$$

8. *Creative destruction:*

$$\delta = \bar{x}^D N^D + e \quad \text{where} \quad \bar{x}^D \equiv \frac{\sum_{n=1}^{\infty} x(n) n f^D(n)}{\sum_{n=1}^{\infty} n f^D(n)}. \quad (\text{A.47})$$

9. *Growth rate:*

$$g = \delta \cdot \ln(\lambda). \quad (\text{A.48})$$

10. *Goods market clearing:*

$$\frac{1}{\mathcal{Y}} + M^D \sum_{n=1}^{\infty} \frac{x(n)^2 n f^D(n)}{2\theta_X} + \frac{e^2}{2\theta_E} - e \cdot \frac{[\ln(1-d) + d]}{\theta_T} + \omega^G = 1. \quad (\text{A.49})$$

11. *Labor and land market clearing:*

$$\omega^H = \frac{\alpha \mathcal{YH}}{\lambda H} \quad \text{and} \quad \omega^L = \frac{(1-\alpha) \mathcal{YL}}{\lambda L}. \quad (\text{A.50})$$

Aggregate scale of the economy. The BGP system pins down the ratios of all variables to consumption C_t , but not the aggregate scale of the economy. To determine the scale, we can rewrite equation (A.36) as:

$$C_t = \frac{Z_0 e^{gt} H^\alpha L^{1-\alpha}}{\mathcal{YT}}.$$

Therefore, the aggregate scale is determined by the initial level of aggregate productivity.

A.4 Welfare

Define the household's lifetime utility when their consumption is perpetually multiplied by a factor of $\gamma > 0$ as:

$$U_0(\gamma) = \int_0^\infty e^{-\rho t} \ln(\gamma \cdot C_t) dt.$$

The consumption-equivalent welfare gain from the Repeal of the ULCRA is the value of γ that makes the household indifferent between the pre-repeal stationary allocation and the post-repeal transition path:

$$U_0^{\text{pre}}(\gamma) = U_0^{\text{post}}(1).$$

In other words, γ is the factor by which we must scale up consumption in the pre-repeal stationary allocation to make the household as well off as they would be under the post-

repeal transition path. The pre-repeal lifetime utility is a simple exponential integral:

$$\begin{aligned}
U_0^{\text{pre}}(\gamma) &= \int_0^\infty e^{-\rho t} \ln(\gamma C_t^{\text{pre}}) dt \\
&= \int_0^\infty e^{-\rho t} \ln \left(\frac{\gamma Z_0 e^{gt} H^\alpha L^{1-\alpha}}{\mathcal{T} \mathcal{Y}^{\text{pre}}} \right) dt \\
&= \frac{1}{\rho} \cdot \left[\ln \left(\frac{\gamma Z_0 H^\alpha L^{1-\alpha}}{\mathcal{T} \mathcal{Y}^{\text{pre}}} \right) + \frac{\delta^{\text{pre}} \ln(\lambda)}{\rho} \right].
\end{aligned}$$

Since $\tau(n) = 0$ for all n after the repeal, $\mathcal{T}_t^{\text{post}} = 1$ for all $t \geq 0$. The post-repeal lifetime utility is therefore:

$$\begin{aligned}
U_0^{\text{post}}(1) &= \int_0^\infty e^{-\rho t} \ln(C_t^{\text{post}}) dt \\
&= \int_0^\infty e^{-\rho t} \ln \left(\frac{Z_t^{\text{post}} H^\alpha L^{1-\alpha}}{\mathcal{Y}_t^{\text{post}}} \right) dt \\
&= \int_0^\infty e^{-\rho t} \ln \left(\frac{Z_0 e^{\ln(\lambda) \int_0^t \delta_s^{\text{post}} ds} H^\alpha L^{1-\alpha}}{\mathcal{Y}_t^{\text{post}}} \right) dt \\
&= \frac{1}{\rho} \cdot \left[\ln(Z_0 H^\alpha L^{1-\alpha}) - \int_0^\infty \rho e^{-\rho t} \ln(\mathcal{Y}_t^{\text{post}}) dt + \ln(\lambda) \int_0^\infty \rho e^{-\rho t} \int_0^t \delta_s^{\text{post}} ds dt \right].
\end{aligned}$$

Combining the two expressions and solving for γ gives:

$$\ln(\gamma) = \ln(\mathcal{T}) + \int_0^\infty \rho e^{-\rho t} \ln(\mathcal{C}_t^{\text{post}} / \mathcal{C}^{\text{pre}}) dt + \ln(\lambda) \int_0^\infty e^{-\rho t} (\delta_t^{\text{post}} - \delta^{\text{pre}}) dt$$

where we have defined the consumption share of aggregate output as:

$$\mathcal{C}^{\text{pre}} \equiv \frac{1}{\mathcal{Y}^{\text{pre}}} \quad \text{and} \quad \mathcal{C}_t^{\text{post}} \equiv \frac{1}{\mathcal{Y}_t^{\text{post}}}.$$

Let Γ^{pre} and Γ_t^{post} denote the shares of output devoted to expansion, entry, and type selection. Then $\mathcal{C}^{\text{pre}} = 1 - \Gamma^{\text{pre}} - \omega^{G,\text{pre}}$ and, because repeal is immediate, $\mathcal{C}_t^{\text{post}} = 1 - \Gamma_t^{\text{post}}$. The consumption-path term admits the exact decomposition:

$$\begin{aligned}
\int_0^\infty \rho e^{-\rho t} \ln \left(\frac{\mathcal{C}_t^{\text{post}}}{\mathcal{C}^{\text{pre}}} \right) dt &= \ln \left(\frac{1 - \Gamma^{\text{pre}}}{1 - \Gamma^{\text{pre}} - \omega^{G,\text{pre}}} \right) \\
&\quad + \int_0^\infty \rho e^{-\rho t} \ln \left(\frac{1 - \Gamma_t^{\text{post}}}{1 - \Gamma^{\text{pre}}} \right) dt.
\end{aligned}$$

Therefore, the welfare gain has four components: (1) removal of cross-sectional factor misallocation $\mathcal{T}$, (2) elimination of the government's wasteful use, (3) reallocation of final goods to expansion and entry along the transition, and (4) faster productivity growth.

Lump-sum-rebate robustness. Consider an alternative fiscal closure in which the government rebates the wedge revenue lump-sum to the representative household. Let B_t^G denote the transfer. The government budget is $B_t^G = Y_t^G$, where the right-hand side is the wedge revenue in (A.29), and B_t^G enters household income. Consolidating the household and government budgets eliminates this transfer from aggregate resources, so final goods market clearing becomes:

$$Y_t = C_t + \frac{Y_t}{2\theta_X} \cdot \sum_{n=1}^{\infty} x_t(n)^2 n m_t^D(n) + \frac{e_t^2 Y_t}{2\theta_E} - e_t \cdot \frac{[\ln(1 - d_t) + d_t] Y_t}{\theta_T}.$$

The rebate does not alter the effective land price $(1 + \tau(n))q_t$ faced by a firm and therefore does not undo the land distortion. It also leaves the real innovation equilibrium unchanged. To see this, divide each consumption-normalized firm value by $\mathcal{Y} = Y/C$. The HJB equations, expansion and entry conditions, and Kolmogorov forward equations are then independent of $\mathcal{Y}$; goods market clearing only determines the consumption-output ratio. Because none of the calibration targets is a consumption-share moment, the calibrated parameters and model fit are unchanged under the alternative closure.

Pre-repeal consumption is now $C^{\text{pre}}/Y^{\text{pre}} = 1 - \Gamma^{\text{pre}}$, rather than $1 - \Gamma^{\text{pre}} - \omega^{G,\text{pre}}$. The post-repeal allocation is identical under the two fiscal closures because wedge revenue is zero after repeal. Hence the rebate welfare gain satisfies:

$$\ln(\gamma^{\text{rebate}}) = \ln(\gamma) - \ln\left(\frac{1 - \Gamma^{\text{pre}}}{1 - \Gamma^{\text{pre}} - \omega^{G,\text{pre}}}\right). \quad (\text{A.51})$$

Thus the rebate robustness removes exactly the government-waste component from log welfare while preserving the static-misallocation, R&D-and-entry, and growth components.

A.5 Calibration

This section derives the model counterpart of each empirical moment targeted in the calibration. We adopt a *dual-sample* approach that reflects the structure of Indian manufacturing data. Dynamic firms correspond to the organized sector (Annual Survey of Industries, ASI), while stagnant firms correspond to the unorganized sector (National Sample Survey,

NSS). Some moments are computed over all firms (ASI+NSS), while cross-sectional and DiD moments condition on dynamic firms only (ASI). Superscripts “pre” and “post” denote the pre-repeal and post-repeal BGPs, respectively.

TFP growth rate.

From equation (A.48), the economy-wide TFP growth rate is:

$$g = \delta \cdot \ln(\lambda). \quad (\text{A.52})$$

Average number of products per firm (all firms).

The total mass of products is equal to one and the total measure of firms is $M^D + M^S$, so the economy-wide average (ASI+NSS) is:

$$\mathbb{E}[n \mid \text{all firms}] = \frac{1}{M^D + M^S}. \quad (\text{A.53})$$

Average number of products per dynamic firm.

Among dynamic (organized-sector) firms, the average number of products is:

$$\mathbb{E}[n \mid \text{dynamic}] = \frac{N^D}{M^D} = \frac{\sum_{n=1}^{\infty} n m^D(n)}{\sum_{n=1}^{\infty} m^D(n)}. \quad (\text{A.54})$$

This moment, together with $\mathbb{E}[n \mid \text{all}]$, pins down the stagnant share $\omega \equiv M^S/(M^D + M^S) = (\mathbb{E}[n \mid \text{dynamic}] - \mathbb{E}[n \mid \text{all}])/(\mathbb{E}[n \mid \text{dynamic}] - 1)$.

Entry rate (dynamic firms).

The empirical entry rate is the share of ASI firms observed in their first year of operation. Since the ASI covers organized-sector (dynamic) firms, its model counterpart is the age share among dynamic firms:

$$s^D[0, 1] = \frac{\int_0^1 m^D(a) da}{M^D} \quad (\text{A.55})$$

where $m^D(a)$ is the age measure density of dynamic firms derived in the old-to-young section below.

Old-to-young firm ratio.

When the expansion rate $x(n)$ depends on n , a firm's current product count is correlated with its age. To compute the old-to-young ratio, we must therefore track the *joint* distribution of firm age and product count. The derivation proceeds in four steps: (1) define the age-product density, (2) write its KF equation, (3) sum the KF equation to obtain an equation for total survival, and (4) incorporate stagnant firms.

Let $m^D(n, a)$ denote the measure density of dynamic firms with exactly $n \in \{1, 2, 3, \dots\}$ products at age $a \in \mathbb{R}_+$, defined so that $m^D(n, a)da$ is the measure of such firms with age in $[a, a+da)$. (To distinguish from the product-count measure $m^D(n) \equiv \int_0^\infty m^D(n, a)da$ defined earlier, the age-product density always carries two arguments in this section.) On the BGP, all flow rates are time-invariant, and the entry flow of dynamic firms is $e \cdot d$ per unit time. Since every dynamic firm enters with exactly one product at age zero, the initial condition is:

$$m^D(n, 0) = \begin{cases} e \cdot d & \text{if } n = 1, \\ 0 & \text{if } n \geq 2. \end{cases} \quad (\text{A.56})$$

As a cohort of dynamic firms ages from a to $a + da$, its product count changes through two channels: *expansion* (gaining one product at total rate $n \cdot x(n)$ for a firm currently at state n) and *creative destruction* (losing one product at total rate $n \cdot \delta$). A firm at $n = 1$ that loses its sole product exits permanently. The KF equation governing the evolution of $m^D(n, a)$ in age a is:

$$\partial_a m^D(1, a) = 2\delta m^D(2, a) - [x(1) + \delta]m^D(1, a) \quad (\text{A.57})$$

for $n = 1$, and for $n \geq 2$:

$$\begin{aligned} \partial_a m^D(n, a) = & (n-1)x(n-1)m^D(n-1, a) \\ & + (n+1)\delta m^D(n+1, a) \\ & - n[x(n) + \delta]m^D(n, a). \end{aligned} \quad (\text{A.58})$$

Define the total measure density of dynamic firms at age a :

$$m^D(a) \equiv \sum_{n=1}^{\infty} m^D(n, a). \quad (\text{A.59})$$

We derive an equation for $\partial_a m^D(a)$ by summing the KF equation (A.57)–(A.58) over each

n . Summing the expansion inflow and outflow terms across all n :

$$\begin{aligned}\text{Expansion inflow: } & \sum_{n=2}^N (n-1)x(n-1)m^D(n-1, a) = \sum_{m=1}^{N-1} mx(m)m^D(m, a) \\ \text{Expansion outflow: } & \sum_{n=1}^N nx(n)m^D(n, a).\end{aligned}$$

The net contribution of expansion is:

$$\sum_{m=1}^{N-1} mx(m)m^D(m, a) - \sum_{n=1}^N nx(n)m^D(n, a) = -Nx(N)m^D(N, a). \quad (\text{A.60})$$

As $N \rightarrow \infty$, the density $m^D(N, a)$ decays to zero (all eigenvalues of the sub-generator have strictly negative real part), so the net expansion contribution vanishes:

$$\lim_{N \rightarrow \infty} Nx(N)m^D(N, a) = 0. \quad (\text{A.61})$$

This telescoping holds for any function $x(n)$, not only for constant x . The reason is purely algebraic: each expansion event moves exactly one unit of mass from state n to state $n+1$, so the in-flow at n from expansion is exactly the out-flow at $n-1$ from expansion. Summing over all n , the interior terms cancel pairwise, leaving only the boundary term at N , which vanishes in the limit. Summing the creative destruction inflow and outflow terms across all n :

$$\begin{aligned}\text{Destruction inflow: } & \sum_{n=1}^N (n+1)\delta m^D(n+1, a) = \delta \sum_{m=2}^{N+1} mm^D(m, a) \\ \text{Destruction outflow: } & \sum_{n=1}^N n\delta m^D(n, a) = \delta \sum_{n=1}^N nm^D(n, a).\end{aligned}$$

The net destruction contribution is:

$$\delta \left[\sum_{m=2}^{N+1} mm^D(m, a) - \sum_{n=1}^N nm^D(n, a) \right] = \delta [(N+1)m^D(N+1, a) - m^D(1, a)].$$

As $N \rightarrow \infty$, the first term vanishes, leaving:

$$\text{Net destruction} = -\delta \cdot m^D(1, a). \quad (\text{A.62})$$

The interpretation is clear: the *only* way a dynamic firm can leave the system is by losing its sole product when at $n = 1$, which occurs at rate $\delta \cdot m^D(1, a)$. All other destruction events merely shift firms from $n + 1$ to n without removing them.

Summing the KF equation (A.57)–(A.58) over n and using (A.61) and (A.62):

$$\partial_a m^D(a) = -\delta \cdot m^D(1, a) \quad \text{where} \quad m^D(0) = e \cdot d. \quad (\text{A.63})$$

This equation holds for *any* expansion rate schedule $x(n)$. It states that the rate at which the surviving dynamic firm density decreases with age equals the exit rate of single-product firms. However, equation (A.63) is *not* a closed equation for $m^D(a)$ alone, because $m^D(1, a)$ depends on the entire product-count distribution. To close it, one must solve the full infinite-dimensional system (A.57)–(A.58).

Stagnant firms enter at flow rate $e(1 - d)$ with a single product and exit at rate δ . Their age measure density is:

$$m^S(a) = e(1 - d)e^{-\delta a} \quad (\text{A.64})$$

with total mass $M^S = \int_0^\infty m^S(a) da = e(1 - d)/\delta$. Since the old-to-young ratio is computed from the ASI (organized sector), we condition on dynamic firms only. Define the dynamic-firm age share:

$$s^D[a_1, a_2] \equiv \frac{\int_{a_1}^{a_2} m^D(a) da}{M^D}. \quad (\text{A.65})$$

The old-to-young ratio targeted in the calibration is $s^D[40, 45]/s^D[1, 5]$. Because M^D cancels:

$$\frac{s^D[40, 45]}{s^D[1, 5]} = \frac{\int_{40}^{45} m^D(a) da}{\int_1^5 m^D(a) da}. \quad (\text{A.66})$$

When the expansion rate $x(n)$ depends on n , the dynamic measure density $m^D(a)$ has no closed-form expression: as noted above, equation (A.63) is not a closed ODE. Computing the ratio (A.66) therefore requires solving the KF system (A.57)–(A.58) numerically. We truncate the product-count state space at $\bar{n}$, discretize the age axis on a non-uniform grid, solve the resulting finite-dimensional linear ODE system forward in age using the Crank–Nicolson scheme, and evaluate the integrals in (A.66) by quadrature.

Innovation DiD (dynamic firms).

The expected number of new products per dynamic firm is:

$$\bar{A}^D \equiv \frac{\sum_{n=1}^{\infty} nx(n)m^D(n)}{M^D} = \sum_{n=1}^{\infty} nx(n)f^D(n). \quad (\text{A.67})$$

Since the innovation regression uses ASI (organized-sector) data, the DiD is conditioned on dynamic firms: $\beta_{\text{DiD}}^{\text{innov}} = \ln(\bar{A}^{D,\text{post}}) - \ln(\bar{A}^{D,\text{pre}})$.

Destruction DiD (dynamic firms).

Products held by dynamic firms are destroyed at rate δ per product. The destruction rate per dynamic firm is $\bar{D}^D \equiv \delta \cdot \mathbb{E}[n \mid \text{dynamic}] = \delta \cdot N^D / M^D$. The DiD is $\beta_{\text{DiD}}^{\text{destr}} = \ln(\bar{D}^{D,\text{post}}) - \ln(\bar{D}^{D,\text{pre}})$.

Dynamic entry DiD.

The flow of dynamic entrants per unit time is $e \cdot d$. This moment does not require sample conditioning since it identifies the entry of innovative (dynamic) firms directly. The entry DiD moment is $\beta_{\text{DiD}}^{\text{entry}} = \ln(e^{\text{post}}d^{\text{post}}) - \ln(e^{\text{pre}}d^{\text{pre}})$.

Exit DiD (dynamic firms).

Among dynamic firms, only those at $n = 1$ exit (by losing their sole product at rate δ). The exit rate among dynamic firms is:

$$\bar{E}^D \equiv \delta \cdot f^D(1) \quad (\text{A.68})$$

where $f^D(1) = m^D(1)/M^D$ is the fraction of dynamic firms with a single product. The exit DiD is a level difference (matching the LPM specification): $\beta_{\text{DiD}}^{\text{exit}} = \bar{E}^{D,\text{post}} - \bar{E}^{D,\text{pre}}$.

MRPLand DiD (dynamic firms).

The marginal revenue product of land for product i held by firm f is:

$$\text{MRPLand}_{ift} \equiv \frac{(1 - \alpha)p_{ift}y_{ift}}{\ell_{ift}} = \frac{(1 - \alpha)Y_t}{\ell_{ift}}.$$

From the cost-minimization first-order condition for land, the price-to-marginal-cost ratio times the effective land cost gives:

$$\text{MRPLand}(n) = \lambda q_t [1 + \tau(n)]^\alpha.$$

Taking logs: $\ln \text{MRPLand}(n) = \ln \lambda + \ln q_t + \alpha \ln[1 + \tau(n)]$. Under the parallel-trends assumption, time fixed effects absorb the common change in $\ln q_t$. Since the regression uses ASI data, the DiD coefficient is the dynamic-firm-weighted average:

$$\beta_{\text{DiD}}^{\text{MRPLand}} = -\alpha \cdot \sum_{n=1}^{\infty} \ln[1 + \tau(n)] f^D(n) = -\alpha \cdot \mathbb{E}_D[\ln(1 + \tau(n))]. \quad (\text{A.69})$$

Loss function and moment weights.

The six parameters $(\lambda, \theta_X, \theta_E, \theta_T, \bar{\tau}, \kappa)$ are estimated by minimizing:

$$\mathcal{L}(\theta) = \sum_{j=1}^{10} w_j \left(\frac{m_j(\theta)}{d_j} - 1 \right)^2 \quad (\text{A.70})$$

where $m_j(\theta)$ is the model-implied moment and d_j is its empirical counterpart. The squared percentage deviation $(m_j/d_j - 1)^2$ provides a scale-free, sign-robust loss that treats all moments in comparable units.

The weight vector w reflects two principles. First, the total weight is split 2/3 to the cross-sectional moments and 1/3 to the DiD moments, proportional to the number of parameters each panel primarily identifies (four structural parameters and two policy parameters, respectively). Second, within each panel, weights reflect the information content of each moment.

Within the cross-sectional panel, the TFP growth rate, average products among dynamic firms, and entry rate each receive weight 1/4, as each primarily identifies a distinct parameter (λ , θ_X , and θ_E). The average products across all firms and the old-to-young ratio each receive weight 1/8, as both primarily identify the type-selection parameter θ_T through their dependence on the stagnant-to-dynamic firm composition.

Within the DiD panel, moments are weighted by $(d_j/\hat{\sigma}_j)^2$, proportional to the squared signal-to-noise ratio of each DiD estimate. Here $\hat{\sigma}_j$ is the clustered standard error from the corresponding regression. Since the loss function uses percentage deviations, $\hat{\sigma}_j/|d_j|$ is the relevant standard error of the percentage miss (by the delta method), and weighting

by its inverse variance $(d_j/\hat{\sigma}_j)^2$ gives more influence to more precisely estimated moments. Table A.11 reports the standard errors and resulting weights.

Moment	$\hat{\sigma}_j$	Within-panel	Overall w_j
<i>Panel A. Cross-sectional (2/3 total weight)</i>			
TFP growth rate	—	1/4	1/6
$\mathbb{E}[n \mid \text{all}]$	—	1/8	1/12
$\mathbb{E}[n \mid \text{dynamic}]$	—	1/4	1/6
Entry rate	—	1/4	1/6
Old-to-young ratio	—	1/8	1/12
<i>Panel B. Difference-in-differences (1/3 total weight)</i>			
Product creation DiD	0.051	0.230	0.077
Product destruction DiD	0.058	0.200	0.067
Entry DiD	0.115	0.154	0.051
Exit DiD	0.004	0.260	0.087
MRPLand DiD	0.048	0.155	0.052

Note: Panel A moments have no reported standard errors; within-panel weights reflect the parameter identification structure. Panel B within-panel weights are proportional to $(d_j/\hat{\sigma}_j)^2$, the squared signal-to-noise ratio, and normalized to sum to one. Overall weights sum to one.

Table A.11: Calibration weights

B Additional figures and tables

B.1 Additional figures

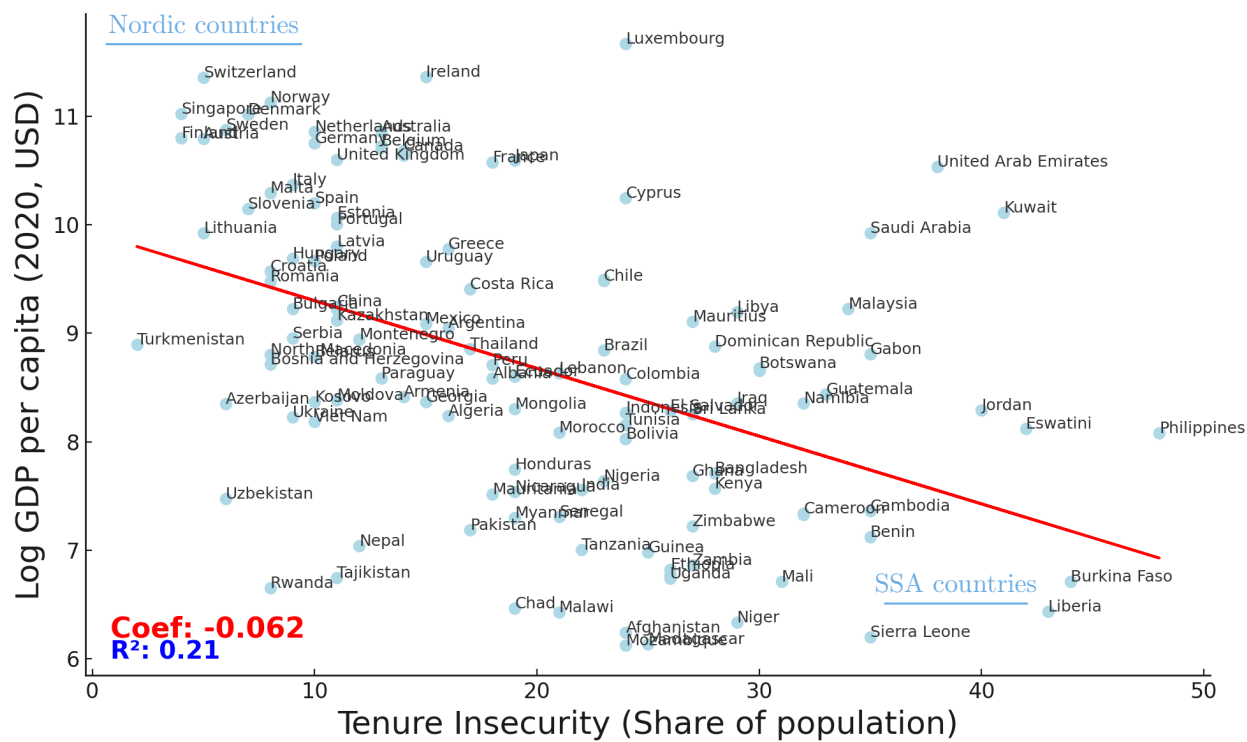


Figure B.7: Cross-country relationship between GDP per capita and tenure insecurity

Note: This figure illustrates the cross-country relationship between log GDP per capita and tenure insecurity, defined as the percentage of the population expecting to lose property rights against their will within 5 years. Tenure insecurity data is sourced from [Prindex \(2020\)](#).

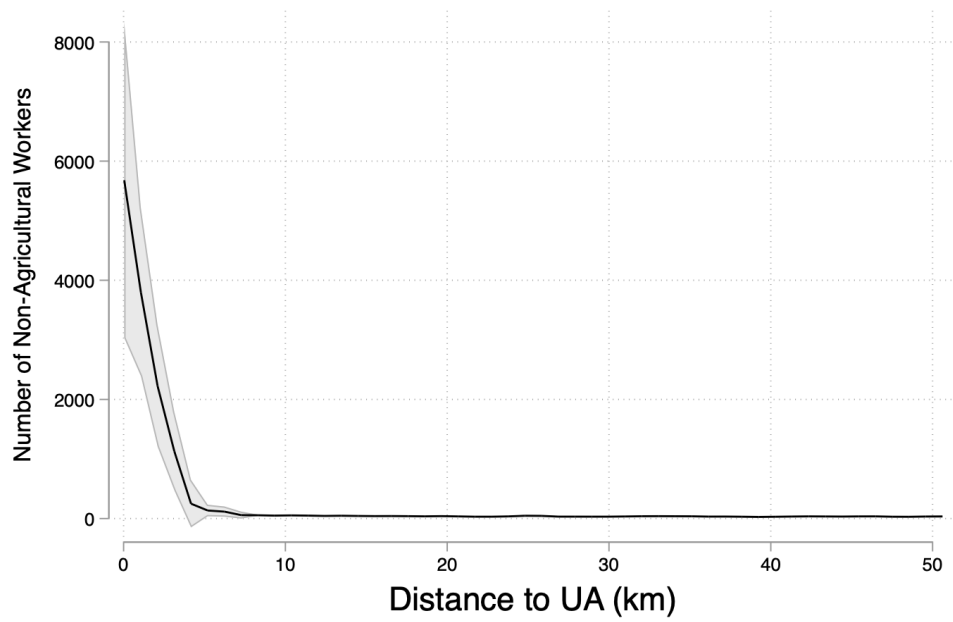
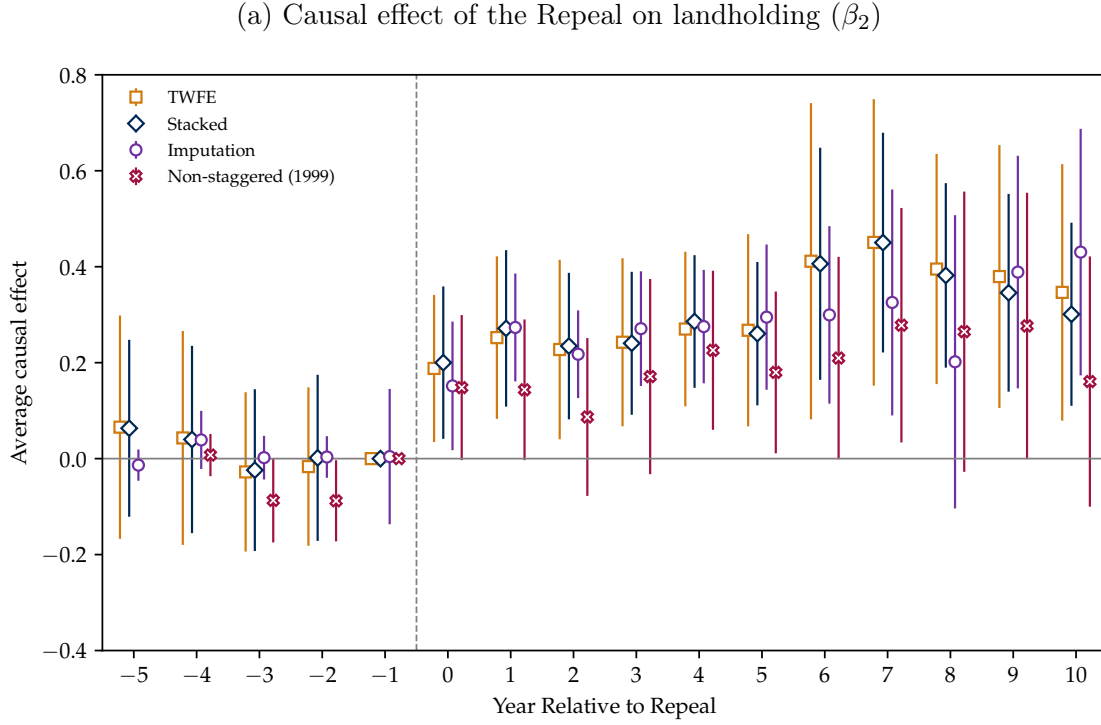
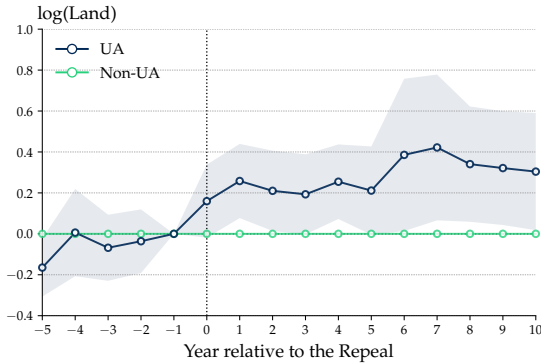


Figure B.8: Non-agricultural employment and distance to UAs

Note: This figure presents the distribution of non-agricultural employment at the village and town level against their distance from the centroids of Urban Agglomerations (UAs), using village-level population census data from 1991. The black line represents a local polynomial fitted line, and the shaded area indicates the 95% confidence interval.



(b) Cengiz et al. (2019)



(c) Borusyak et al. (2023)

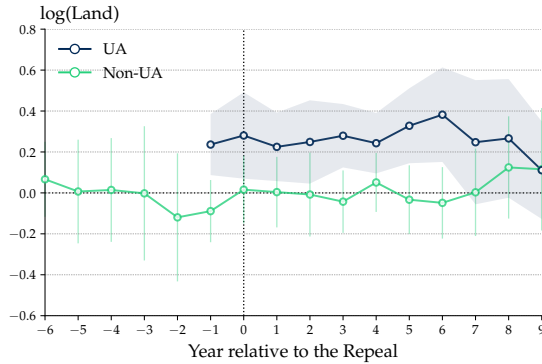


Figure B.9: Event study graphs for landholdings (staggered DiD)

Note: Panel (a) reports event-study estimates of the Repeal’s effect on UA landholdings from equation (1), using five staggered-DiD estimators: two-way fixed effects (orange square), the stacked DiD of Cengiz et al. (2019) (navy diamond), the single-stage imputation estimator of Borusyak et al. (2023) (purple circle), Callaway and Sant’Anna (2020) (green triangle), and a non-staggered single-cohort triple-difference restricted to the 1999 repeal cohort (maroon cross). The two-way fixed-effects, stacked, Callaway and Sant’Anna (2020), and single-cohort estimators identify the UA-versus-non-UA differential β_2 , retaining non-UA plants as an additional comparison group; the imputation estimator instead reports the UA-specific effect (the Repeal’s effect on UA plants), because the Borusyak et al. (2023) routine does not provide a standard error for the UA-versus-non-UA difference. All five series are estimated on the same sample, event window and fixed effects—those of our baseline specification—so they differ only in the estimator. Two qualifications. First, the `csdid` implementation of Callaway and Sant’Anna (2020) cannot absorb high-dimensional categorical controls at this sample size, so that series is estimated without covariates; its agreement with the other differential estimators is therefore reassuring rather than mechanical. Second, all five series are unweighted, whereas the regression tables weight by the ASI sampling multiplier, so the levels here are not directly comparable to Table 1: estimating the pooled specification on this sample gives 0.221 weighted against 0.293 unweighted, the weights shifting the estimate toward smaller plants, whose response is smaller. Neither qualification affects the comparison across estimators, which is what this panel is for. Panels (b) and (c)

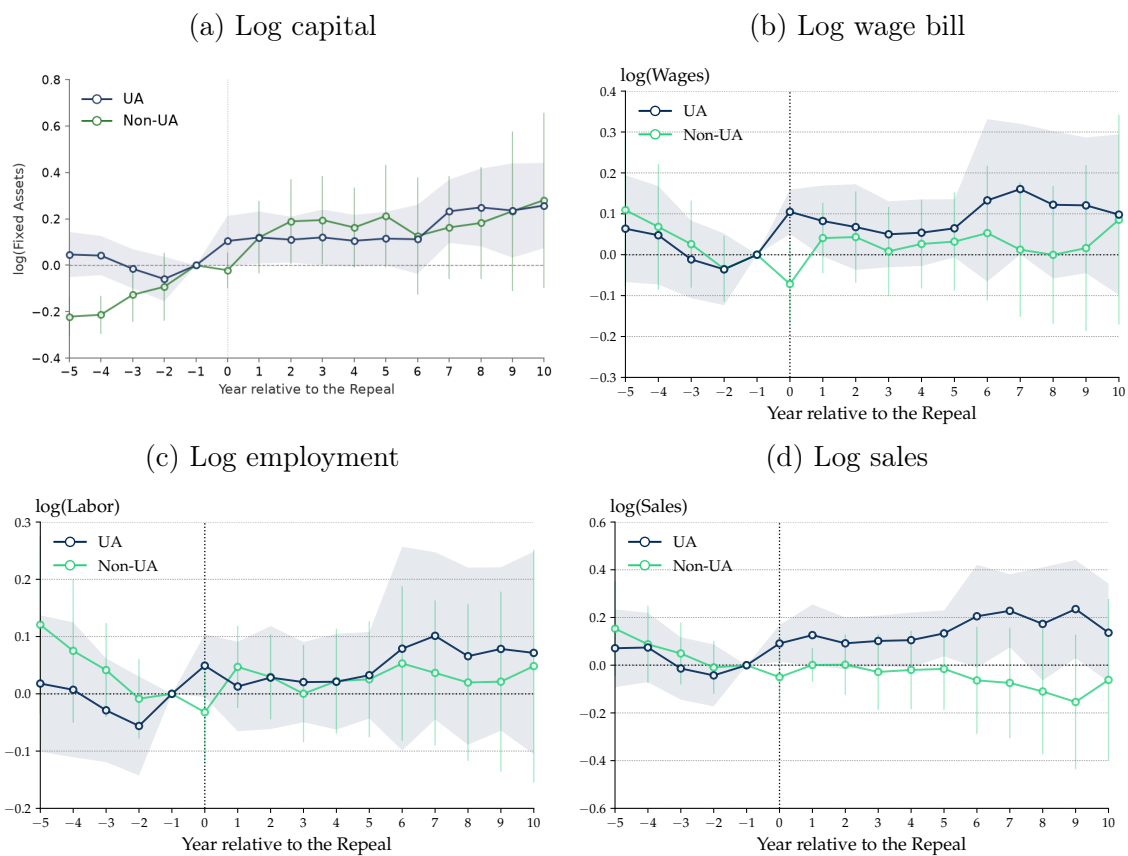


Figure B.10: Event study graphs for capital, wage bill, employment, and sales

Note: All dependent variables are in logarithms. Coefficients are normalized a year before the reform. The confidence interval is at the 95% level.

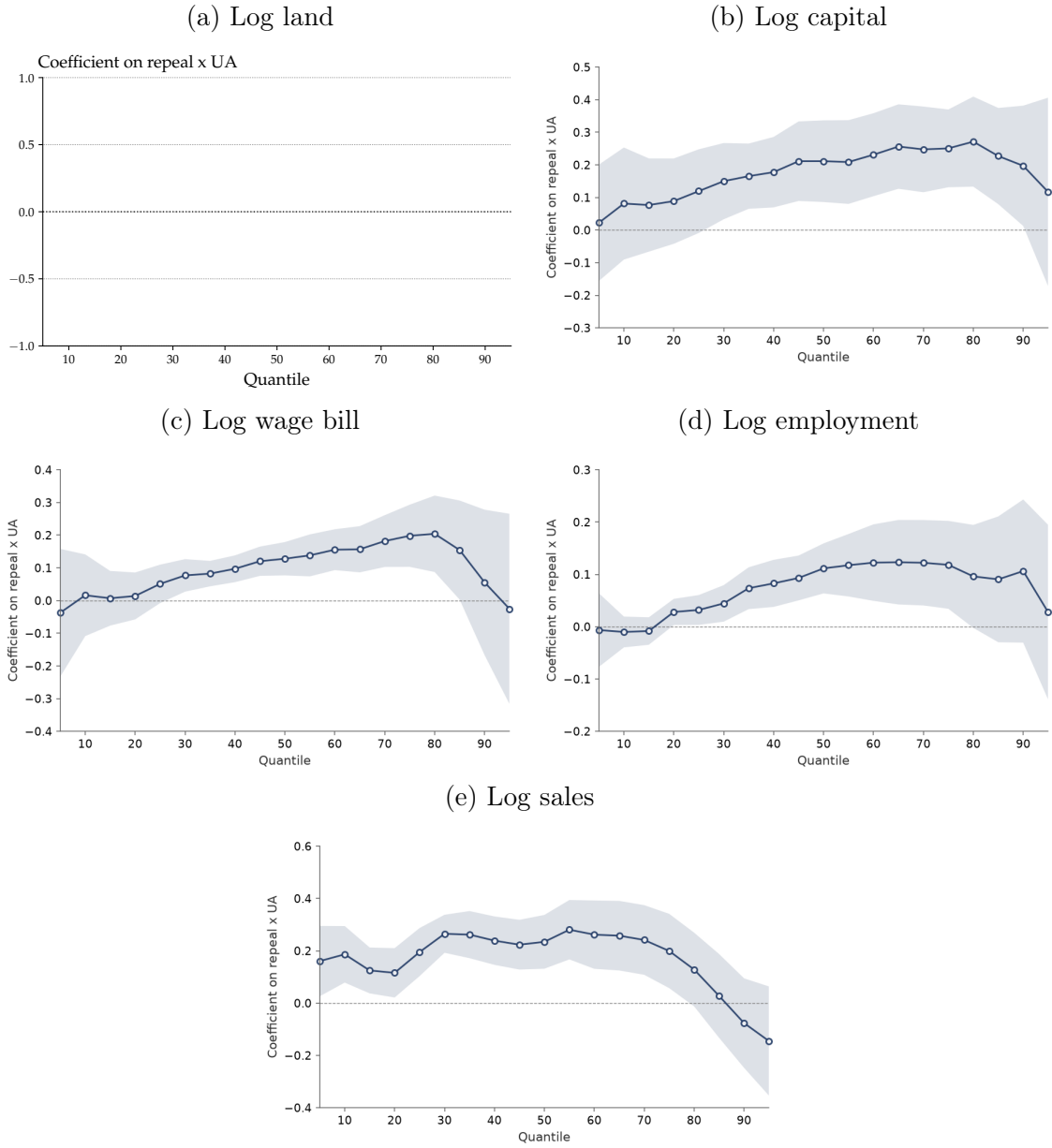


Figure B.11: Quantile treatment effects of the Repeal on firms in UAs

Note: This figure reports the quantile regression estimates for the treatment effect of the Repeal Act on firm-level outcomes, from 5th to 95th quintiles, using the re-centered influence function approach of [Firpo et al. \(2009\)](#). All dependent variables are in logarithms. The confidence interval is at the 95% level, calculated using standard errors clustered at the state level.

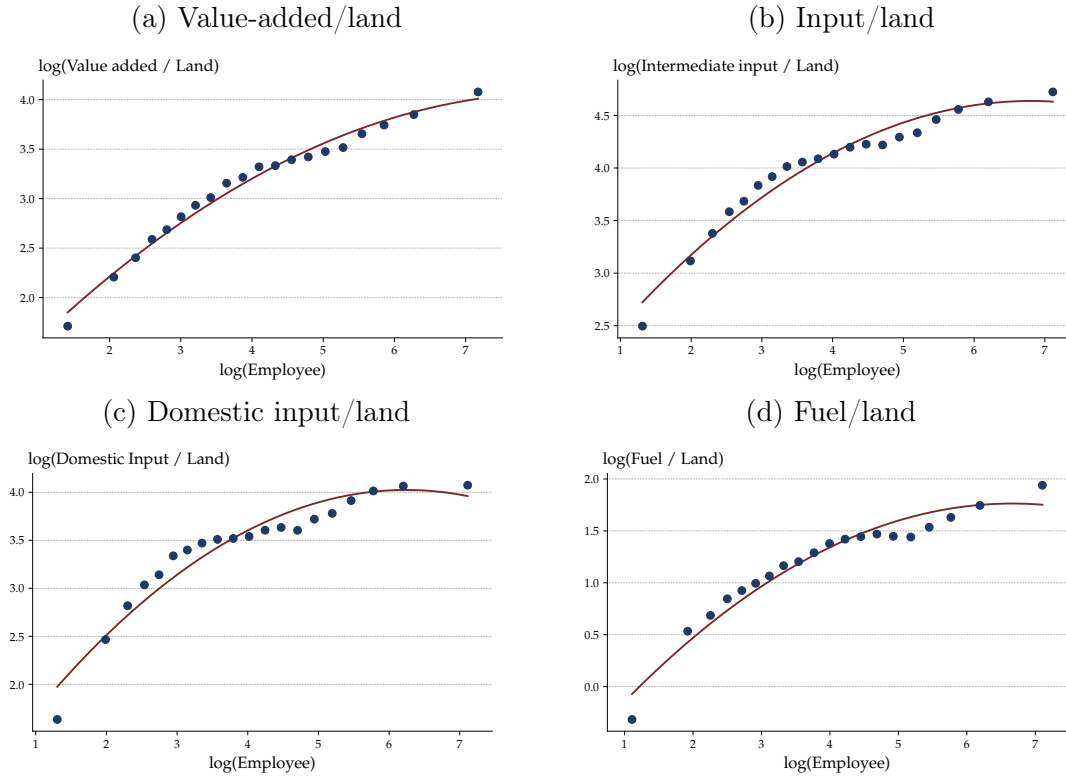


Figure B.12: Average value-added per land and firm size

Note: This figure reports the binned scatter plot and quadratic fitted line of log value-added and inputs per land against log employment: value-added (panel a), all intermediate inputs (panel b), domestic inputs (panel c), and fuels (panel d). All regressions control for 2-digit industry fixed effects and year fixed effects.

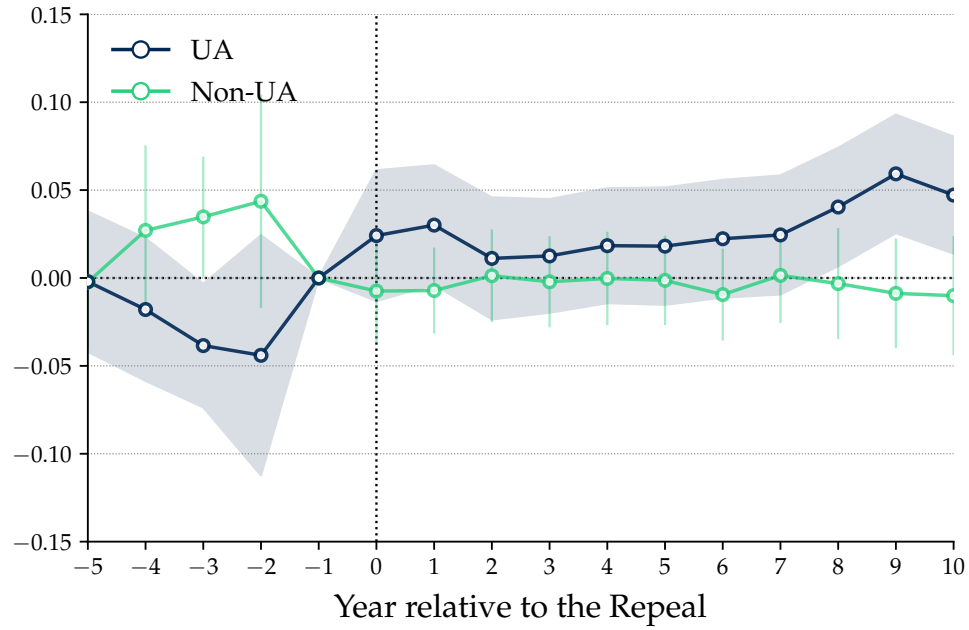


Figure B.13: Event study graphs for TFPQ

Note: The dependent variable (log TFPQ) is measured following the methodology of [De Ridder et al. \(2022\)](#). TFPQ is estimated separately for each 2-digit industry. Coefficients are normalized a year before the reform. The confidence interval is at the 95% level.

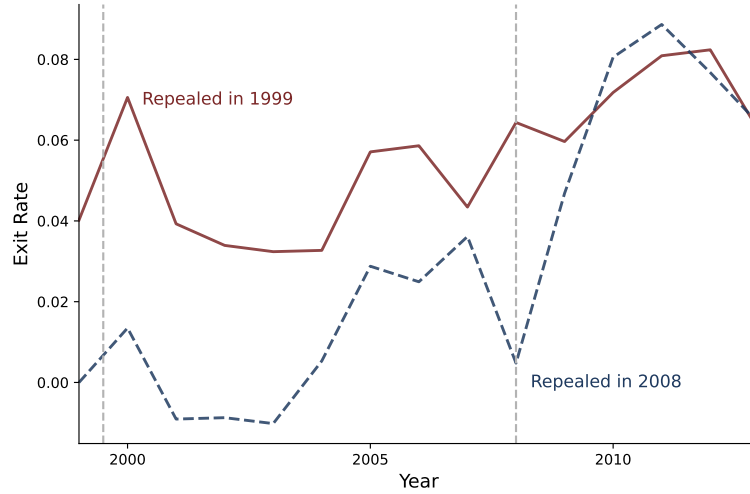


Figure B.14: Exit rates by repeal-timing cohort (descriptive)

Note: Weighted mean plant exit rates in the seventeen states and union territories that repealed the ULCRA in 1999 versus Andhra Pradesh and Maharashtra, the last states to repeal. Both series are shifted by a common constant equal to the late-repealing group's 1999 exit rate, so the vertical axis measures exit rates *relative to the late-repealing group in 1999* rather than exit rates in levels; the early cohort's positive value in 1999 reflects a pre-existing level difference. This exhibit is descriptive and is not the paper's identifying design: it compares cohorts of states rather than UA and non-UA plants within a state, it pools UA and non-UA plants, the later-repealing group contains only two states, and it uses formal (calendar) repeal dates rather than the fiscal-onset coding used in the regressions. Because the ASI status codes that identify exit begin only in 1998 and exit is identically zero in the first panel year, no pre-repeal observation exists for the 1999 cohort, so the figure displays no pre-treatment period and supports no parallel-trends test. Sampling-multiplier weights are used.

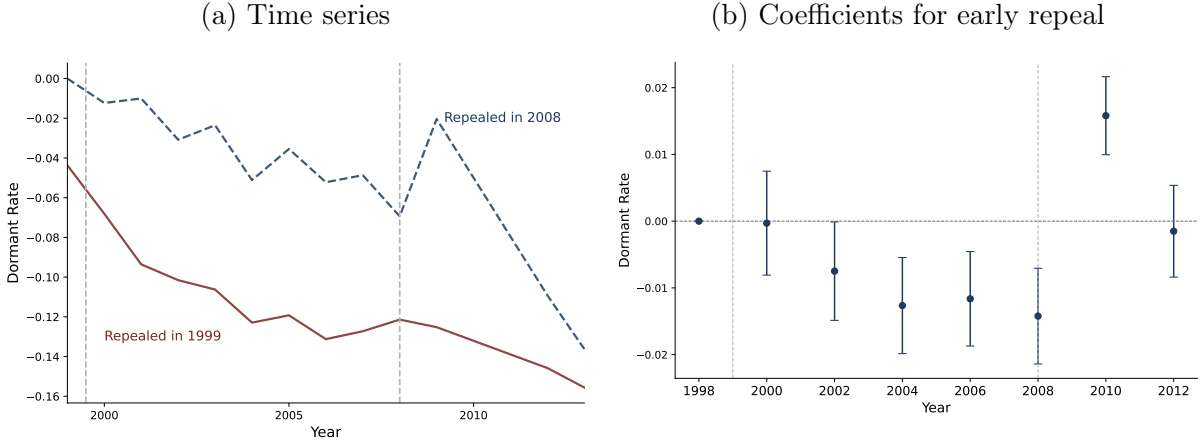


Figure B.15: Dormancy rates in the 1999 repeal cohort versus Andhra Pradesh and Maharashtra (descriptive)

Note: Panel (a) displays the time series of dormancy rates for the seventeen states and union territories that adopted the Repeal in 1999 (navy, solid) versus Andhra Pradesh and Maharashtra, the last states to repeal (green, dashed; labelled “Repealed in 2008” in the legend, their formal repeal dates being 2008 and 2007). Both series are shifted by a common constant equal to the late-repealing group’s 1999 dormancy rate, so the vertical axis measures dormancy *relative to the late-repealing group in 1999* rather than dormancy rates in levels. ‘Dormancy’ is defined by codes 2 and 3 in the variable ‘status of units,’ which include units found during the survey that were present at the location but did not initiate production and/or employ any staff during the reference period. The ASI did not record the dormancy codes in the 2010 and 2011 rounds, so neither series has an observation in those two years; panel (a) simply connects 2009 to 2012 across the gap, and the 2010–11 bin in panel (b) is estimated over rounds in which the indicator is identically zero and should not be read as an effect. Panel (b) plots the regression coefficients of our firm dormancy measure on an early repeal dummy variable, interacted with biennial dummies, controlling for industry-by-year, firm age, and state fixed effects; the pooled 1998–99 bin is the omitted reference. Like Figure B.14, this exhibit is descriptive and is not the paper’s identifying design: it compares cohorts of states rather than UA and non-UA plants within a state, it pools UA and non-UA plants, the later-repealing group contains only two states, and it uses formal (calendar) repeal dates rather than the fiscal-onset coding used in the regressions. Panel (a) uses sampling-multiplier weights; panel (b) is unweighted. The confidence intervals are set at the 95% level.

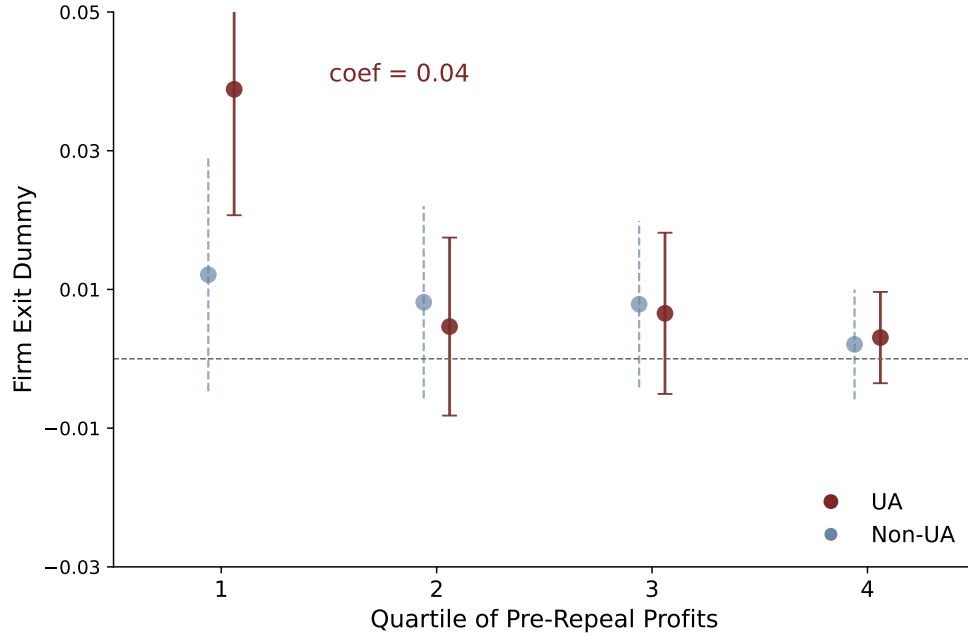


Figure B.16: Treatment effect of the Repeal on plant exit by pre-Repeal profits

Note: The dependent variable is a plant exit dummy, coded as 4 in the ‘status of units’ (as defined in Section 4.3). Quartile 1 represents the least profitable plants prior to the Repeal. Profit is measured as variable profits, calculated as sales revenue minus material and labor costs. Controls include a rural dummy, 2-digit industry fixed effects, firm age fixed effects, and land size quartile-by-year fixed effects.

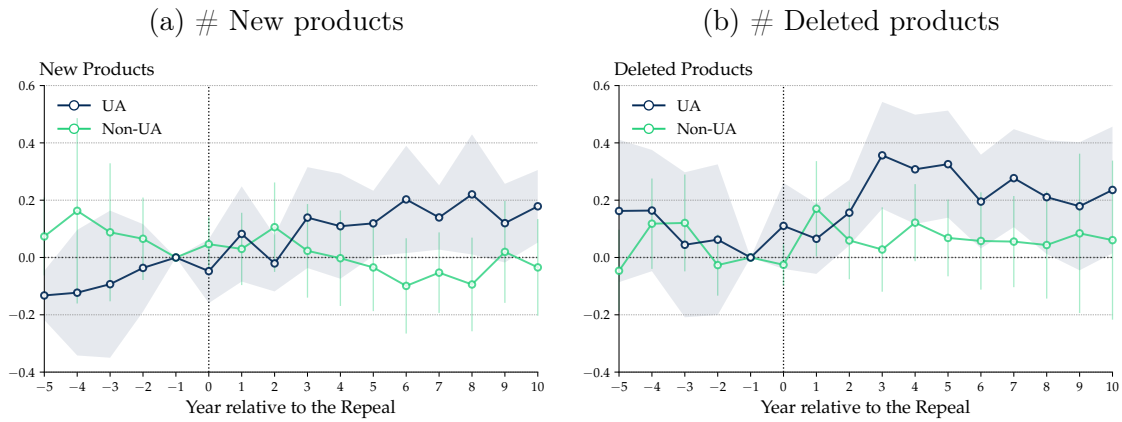


Figure B.17: Event study graphs for creative destruction

Note: The effects on adding/deleting products are estimated using the Poisson Pseudo-Maximum Likelihood regression. Coefficients are normalized a year before the reform. # New products is the number of new 5-digit products added to a firm’s product portfolio in a given year, and # Deleted products is the number of products deleted from the portfolio. The confidence interval is at the 95% level.

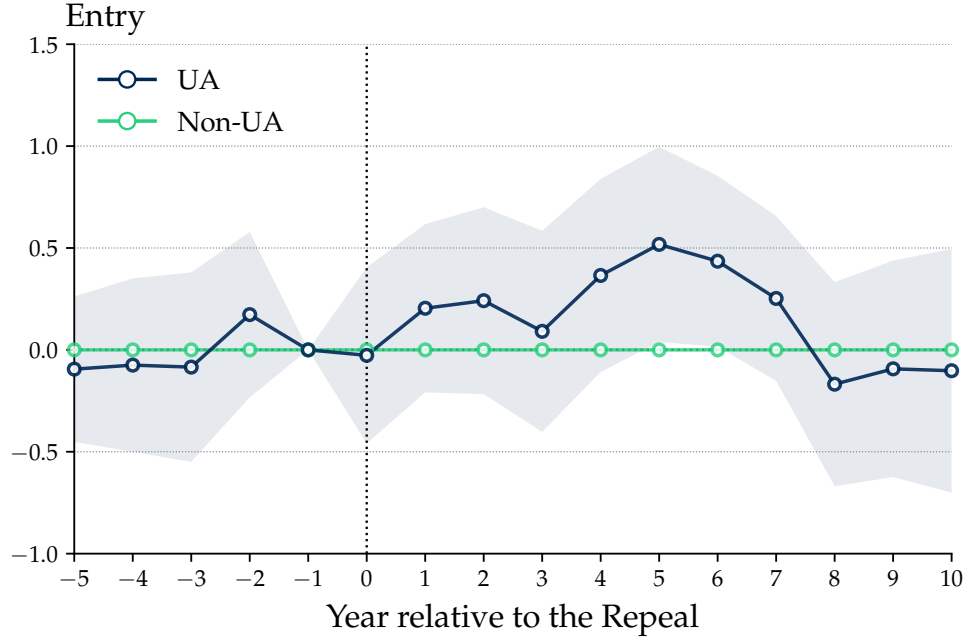


Figure B.18: Event study graphs for the entry of dynamic plants

Note: Dynamic plants are plants that have introduced new products during their life cycle. The data is aggregated to the district-year level using sampling weights. District-years with fewer than 30 plants are excluded. The analysis employs the Poisson Pseudo-Maximum Likelihood method, controlling for district fixed effects, year fixed effects, and district control-by-year fixed effects. These district controls-by-year include quartiles of pre-Repeal metrics: share of plants in urban areas, share of privately owned plants, and number of plants, each interacted with year fixed effects. Confidence intervals are set at the 95% level.

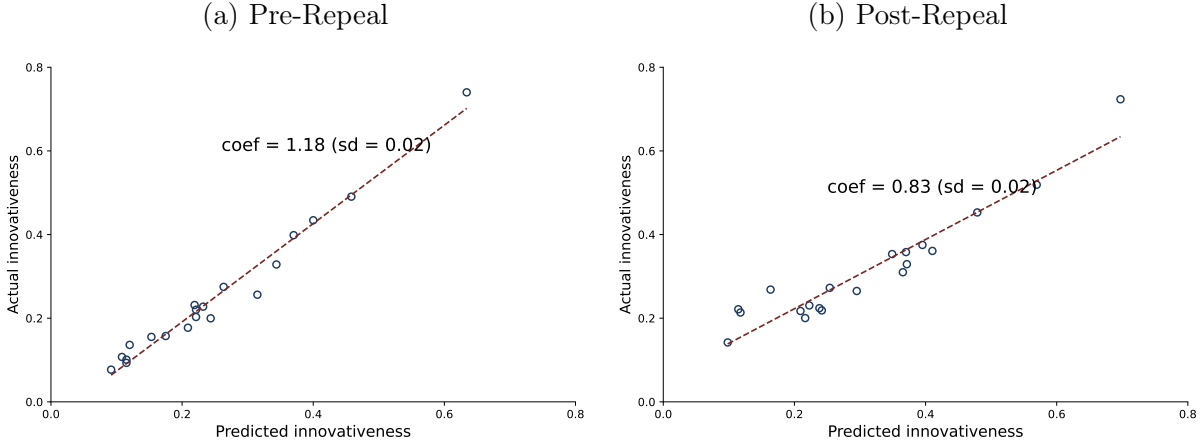
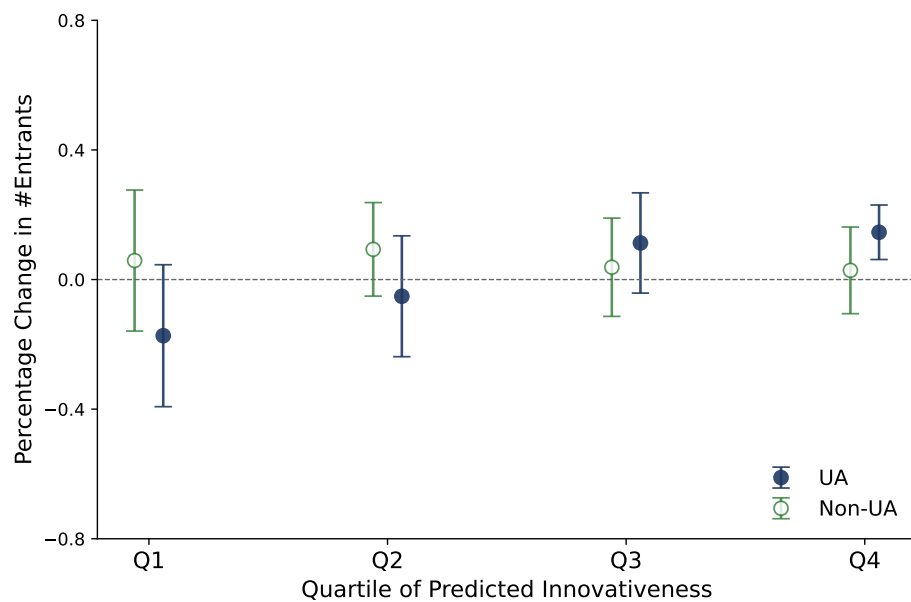


Figure B.19: Relationship between actual and LASSO-predicted dynamism

Note: This figure presents binned scatter plots comparing plants' average number of new products throughout their life cycle with our LASSO-predicted dynamism measure. Panels (a) and (b) show these relationships using pre-Repeal and post-Repeal data, respectively. Since the LASSO model is trained with pre-Repeal data, the left panel provides in-sample predictions, while the right panel offers out-of-sample predictions. Both panels include the correlation coefficients. The LASSO model is trained by regressing the number of new products introduced by plants throughout their life cycle on various plant-level characteristics recorded at entry, using pre-Repeal data. These characteristics encompass a range of dummies for private ownership, urban location, sole proprietorship, census scheme participation, importing foreign intermediates, and using computers, in addition to the total number of products. The model also includes the quadratic and interaction terms of these variables.



Note: This figure illustrates the treatment effects of the Repeal (β_2) on the number of entrants within each quartile of dynamism. Quartile 1 represents the least dynamic plants, while Quartile 4 represents the most dynamic, as determined by the LASSO-predicted dynamism measure.

Figure B.20: Treatment effects of the Repeal on entry across dynamism quartiles

Never pated	partici-	Jammu and Kashmir; Kerala
1999		Gujarat; Haryana; Himachal Pradesh; Karnataka; Manipur; Meghalaya; Punjab; Rajasthan; Tripura; Uttar Pradesh; Uttarakhand; all Union Territories
2000		Chhattisgarh; Madhya Pradesh
2002		Odisha
2003		Assam
2006		Bihar
2007		Maharashtra
2008		Andhra Pradesh; Telangana
2011		Jharkhand
Still in effect		Tamil Nadu (state-specific land ceiling law remains in effect); West Bengal

Note: States are listed by their formal (calendar) repeal year. The regression baseline instead uses fiscal-onset coding (the ASI April–March year containing the repeal) and dates Andhra Pradesh and Maharashtra to the 2006 JNNURM statewide commitment; see Section 2.

Table B.12: Timeline of the Repeal

B.2 Additional tables

Variable	N	Mean	p10	p50	p90
UA share (Land stock (mn Rs.))	344,211	10.3	0.0	0.4	9.4
Fixed Assets (Million Rupee)	537,151	123.9	0.1	2.6	109.6
Wages (mn Rs.)	537,918	14.6	0.2	1.1	26.3
Sales (mn Rs.)	468,778	413.2	1.4	21.0	561.2
Land Adjustment Flows - Addition (Thousand Rupee)	339,093	359.5	0.0	0.0	19.2
$\mathbf{1}\{Add > 0\}$	339,848	11.1	0.0	0.0	100.0
Land Adjustment Flows - Deduction (Thousand Rupee)	338,808	67.2	0.0	0.0	0.0
$\mathbf{1}\{Deduct > 0\}$	339,016	3.0	0.0	0.0	0.0

Note: Reported statistics are weighted plant-level moments from the ASI analysis sample used in the GitHub pipeline.

Table B.13: Summary statistics of the ASI data

	(1)	(2)	(3)	(4)	(5)
	Land	Capital	Labor	Sales	MRPLand
UA	−0.030 (0.034)	−0.269*** (0.030)	−0.131*** (0.016)	−0.141*** (0.026)	0.057 (0.035)
Linear trend	0.055*** (0.008)	0.182*** (0.007)	0.056*** (0.004)	0.108*** (0.006)	0.014* (0.008)
UA × trend	−0.014 (0.012)	−0.010 (0.009)	−0.002 (0.005)	−0.004 (0.008)	−0.001 (0.012)
Observations	92,557	160,436	163,847	138,821	78,769
R^2	0.098	0.169	0.120	0.170	0.097
Industry FE	Yes	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes	Yes

Note: Each column regresses a pre-Repeal outcome on the UA indicator, a linear time trend, and their interaction. Robust standard errors clustered at the state level are in parentheses. Significance levels: * 10%, ** 5%, *** 1%.

Table B.14: Pre-Repeal differences in time trends between UAs and non-UAs

	(1)	(2)
	Total land	Plant count
Repeal	0.079 (0.132)	-0.026 (0.089)
Repeal $\times$ UA	-0.012 (0.096)	-0.101* (0.056)
Observations	6,118	6,503
R^2	0.832	0.911
Year FE	Yes	Yes
District FE	Yes	Yes
Size $\times$ Year FE	Yes	Yes

Note: District-level regressions. Robust standard errors clustered at the state level are in parentheses. Significance levels: * 10%, ** 5%, *** 1%.

Table B.15: Treatment effect of the Repeal on district-level total land and number of plants

	(1)	(2)	(3)
	MRPLand	MRPLand	MRPLand
Repeal	0.017 (0.044)	0.040 (0.040)	
Repeal $\times$ UA	-0.102** (0.048)	-0.137*** (0.045)	-0.185*** (0.040)
Observations	298,863	298,816	298,816
R^2	0.053	0.079	0.084
Year FE	Yes	Subsumed	Subsumed
District FE	Yes	Yes	Yes
Size $\times$ Year FE	Yes	Yes	Yes
3-digit Ind. $\times$ Year FE	No	Yes	Yes
State $\times$ Year FE	No	No	Yes

Note: The dependent variable is the *within-industry* marginal revenue product of land: $\log$ MRPLand (log sales over land) residualized on 2-digit-industry-by-year means, which differences out cross-industry differences in the land elasticity and price levels. Column (1) is the baseline specification; columns (2)–(3) add 3-digit-industry-by-year and state-by-year fixed effects, respectively. ‘Land Size $\times$ Year’ are district average pre-treatment land-size bins interacted with year. The Repeal main effect is absorbed by the state-by-year fixed effects in column (3). Robust standard errors clustered at the state level are in parentheses. Significance levels: * 10%, ** 5%, *** 1%.

Table B.16: Treatment effect of the Repeal on marginal revenue product of land

	(1)	(2)	(3)	(4)
	Land stock	Buy	Sell	Revalue
Repeal $\times$ UA	0.270*** (0.048)	0.244** (0.109)	0.221 (0.158)	-0.397 (0.297)
Observations	338,685	332,819	326,956	250,500
R^2	0.188			
Year FE	Yes	Yes	Yes	Yes
District FE	Yes	Yes	Yes	Yes
Size $\times$ Year FE	Yes	Yes	Yes	Yes
State $\times$ Year FE	Yes	Yes	Yes	Yes

Note: Columns (2)–(4) report PPML estimates for land purchases, sales, and revaluation after adding state $\times$ year fixed effects. Robust standard errors clustered at the state level are in parentheses. Significance levels: * 10%, ** 5%, *** 1%.

Table B.17: Treatment effect of the Repeal on land adjustment: state $\times$ year FE

	Mean MRPLand		MRPLand dispersion		Land dispersion	
	(1)	(2)	(3)	(4)	(5)	(6)
	No FE	+ FE	No FE	+ FE	No FE	+ FE
<i>Panel A. Mean MRPLand</i>						
Lag 1						
Lag 2	0.008	0.003*				
	(0.005)	(0.001)				
Lag 3	−0.006	−0.001				
	(0.006)	(0.005)				
<i>Panel B. MRPLand dispersion</i>						
Lag 1						
Lag 2			−0.005	0.001		
			(0.011)	(0.005)		
Lag 3			0.008	−0.006		
			(0.009)	(0.004)		
<i>Panel C. Land dispersion</i>						
Lag 1						
Lag 2					−0.006	−0.011
					(0.014)	(0.007)
Lag 3					0.013*	0.002
					(0.006)	(0.003)
Observations	2,094	2,094	1,828	1,828	1,929	1,928
R^2	0.001	0.797	0.000	0.815	0.001	0.812
State FE	No	Yes	No	Yes	No	Yes
Year FE	No	Yes	No	Yes	No	Yes
Industry FE	No	Yes	No	Yes	No	Yes

Note: The dependent variable is repeal adoption. Column pairs report lagged MRPLand means, lagged MRPLand dispersion, and lagged land dispersion, with even columns adding state, year, and industry fixed effects. Robust standard errors clustered at the state level are in parentheses. Significance levels: * 10%, ** 5%, *** 1%.

Table B.18: The impact of land misallocation and distribution on repeal timing

	(1)	(2)	(3)	(4)
	Land stock	Buy	Sell	Revalue
Repeal	0.025 (0.048)	0.075 (0.097)	0.129 (0.097)	0.122 (0.259)
Repeal $\times$ UA	0.164*** (0.046)	0.212* (0.112)	0.187 (0.123)	-0.314 (0.344)
Observations	338,681	333,227	324,832	208,705
R^2	0.190			
Year FE	Yes	Yes	Yes	Yes
District FE	Yes	Yes	Yes	Yes
Size $\times$ Year FE	Yes	Yes	Yes	Yes
Ind. $\times$ Year FE	Yes	Yes	Yes	Yes

Note: Columns (2)–(4) report PPML estimates for land purchases, sales, and revaluation after adding industry $\times$ year fixed effects. Robust standard errors clustered at the state level are in parentheses. Significance levels: * 10%, ** 5%, *** 1%.

Table B.19: Treatment effect of the Repeal on land adjustment: sector $\times$ year FE

	OLS		LP		ACF	
	(1)	(2)	(3)	(4)	(5)	(6)
	High	Low	High	Low	High	Low
Repeal	0.010 (0.044)	0.051 (0.056)	0.023 (0.050)	-0.015 (0.048)	-0.032 (0.065)	0.011 (0.048)
Repeal $\times$ UA	0.143** (0.058)	0.097 (0.071)	0.166* (0.088)	0.061 (0.060)	0.100** (0.037)	0.043 (0.041)
Observations	46,621	45,073	43,901	44,576	45,957	47,242
R^2	0.880	0.872	0.869	0.882	0.889	0.890
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Size $\times$ Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Ind. $\times$ Year FE	Yes	Yes	Yes	Yes	Yes	Yes

Note: Alternative TFP estimators for the pre-Repeal split. All specifications include plant fixed effects. Robust standard errors clustered at the state level are in parentheses. Significance levels: * 10%, ** 5%, *** 1%.

Table B.20: Within-plant land growth by high/low pre-Repeal productivity plants: alternative measures of TFP

	(1)	(2)	(3)	(4)
	Capital	Labor	Wages	Sales
Repeal $\times$ UA	0.175*** (0.054)	0.068** (0.028)	0.084*** (0.028)	0.152*** (0.046)
Observations	530,008	533,802	531,119	464,755
R^2	0.257	0.151	0.217	0.246
Year FE	Yes	Yes	Yes	Yes
District FE	Yes	Yes	Yes	Yes
Size $\times$ Year FE	Yes	Yes	Yes	Yes
State $\times$ Year FE	Yes	Yes	Yes	Yes

Note: Columns (1)–(4) are in logarithms. Robust standard errors clustered at the state level are in parentheses. Significance levels: * 10%, ** 5%, *** 1%.

Table B.21: Treatment effect of the Repeal on firm production: state $\times$ year FE

	(1)	(2)	(3)	(4)
	Capital	Labor	Wages	Sales
Repeal	0.091 (0.086)	−0.010 (0.033)	0.009 (0.044)	0.024 (0.049)
Repeal $\times$ UA	0.117** (0.045)	0.033* (0.019)	0.051** (0.019)	0.088*** (0.023)
Observations	530,005	533,799	531,116	464,752
R^2	0.259	0.155	0.220	0.252
Year FE	Yes	Yes	Yes	Yes
District FE	Yes	Yes	Yes	Yes
Size $\times$ Year FE	Yes	Yes	Yes	Yes
Ind. $\times$ Year FE	Yes	Yes	Yes	Yes

Note: Columns (1)–(4) are in logarithms. Robust standard errors clustered at the state level are in parentheses. Significance levels: * 10%, ** 5%, *** 1%.

Table B.22: Treatment effect of the Repeal on firm production: sector $\times$ year FE

	(1)	(2)	(3)
	OLS	LP	ACF
Repeal	0.001 (0.074)	-0.022 (0.080)	-0.002 (0.073)
Repeal $\times$ UA	0.139** (0.066)	0.173*** (0.059)	0.119* (0.066)
Observations	359,848	358,500	359,432
R^2	0.092	0.081	0.080
District FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Size $\times$ Year FE	Yes	Yes	Yes

Note: Columns report alternative TFPQ estimators with district and year fixed effects. Robust standard errors clustered at the state level are in parentheses. Significance levels: * 10%, ** 5%, *** 1%.

Table B.23: Treatment effect of the Repeal on TFPQ: alternative measures

	(1)	(2)	(3)	(4)	(5)
	TFPR OLS	TFPR LP	TFPR ACF	Price	Markup
Repeal	-0.000 (0.010)	-0.014 (0.022)	0.001 (0.012)	-0.001 (0.072)	0.016 (0.021)
Repeal $\times$ UA	0.017** (0.008)	0.047** (0.021)	-0.006 (0.011)	-0.110* (0.062)	-0.016 (0.019)
Observations	430,301	430,300	430,298	389,963	368,953
R^2	0.034	0.156	0.101	0.090	0.111
District FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Size $\times$ Year FE	Yes	Yes	Yes	Yes	Yes

Note: All specifications include district and year fixed effects. Robust standard errors clustered at the state level are in parentheses. Significance levels: * 10%, ** 5%, *** 1%.

Table B.24: Treatment effect of the Repeal on TFPR, firm-level prices, and markup

	(1)	(2)	(3)	(4)	(5)
	Land	Capital	Wages	Labor	Sales
Repeal	0.088 (0.078)	0.009 (0.096)	0.052 (0.059)	0.024 (0.040)	-0.045 (0.088)
Repeal $\times$ UA	0.229** (0.099)	0.205* (0.110)	0.089 (0.065)	0.063 (0.055)	0.129 (0.089)
Observations	335,033	523,973	528,255	527,353	463,775
R^2					
Year FE	Yes	Yes	Yes	Yes	Yes
District FE	Yes	Yes	Yes	Yes	Yes

Note: Did-imputation specifications. The synced local outputs currently contain empty coefficient rows for this table; the layout preserves those blank local results. Robust standard errors clustered at the state level are in parentheses. Significance levels: * 10%, ** 5%, *** 1%.

Table B.25: Accounting for potential bias in staggered difference-in-differences: imputation approach based on [Borusyak et al. \(2023\)](#)

	(1)	(2)	(3)	(4)	(5)
	Land	Capital	Wages	Labor	Sales
Repeal					
Repeal $\times$ UA	0.291*** (0.040)	0.181*** (0.047)	0.091*** (0.029)	0.097*** (0.025)	0.156*** (0.043)
Observations	930,228	1,358,440	1,362,792	1,359,669	1,213,837
R^2	0.241	0.327	0.262	0.217	0.295
Stack $\times$ District FE	Yes	Yes	Yes	Yes	Yes
Stack $\times$ Year FE	Yes	Yes	Yes	Yes	Yes

Note: Stacked DiD specifications. Robust standard errors clustered at the stack $\times$ state-year level are in parentheses. Significance levels: * 10%, ** 5%, *** 1%.

Table B.26: Accounting for potential bias in staggered difference-in-differences: stacked DiD based on [Cengiz et al. \(2019\)](#)

	(1)	(2)	(3)
	Base	+ Size-Year	+ Ind-Year
Repeal	-0.007 (0.008)	-0.006 (0.008)	-0.003 (0.007)
Repeal $\times$ UA	0.011** (0.004)	0.011** (0.004)	0.008** (0.004)
Observations	249,183	249,172	249,166
R^2	0.055	0.059	0.067
Year FE	Yes	Yes	Yes
District FE	Yes	Yes	Yes
Firm controls	Yes	Yes	Yes
Size $\times$ Year FE	No	Yes	Yes
Ind. $\times$ Year FE	No	No	Yes

Note: All specifications include firm controls. Robust standard errors clustered at the state level are in parentheses. Significance levels: * 10%, ** 5%, *** 1%.

Table B.27: Treatment effect of the Repeal on exit among non-dormant plants

	4-digit		3-digit	
	(1)	(2)	(3)	(4)
	New	Deleted	New	Deleted
Repeal	0.004 (0.053)	-0.042 (0.075)	-0.044 (0.072)	-0.071 (0.077)
Repeal $\times$ UA	0.113** (0.057)	0.114* (0.061)	0.148** (0.067)	0.132* (0.073)
Observations	264,424	272,645	264,238	272,253
R^2				
Year FE	Yes	Yes	Yes	Yes
District FE	Yes	Yes	Yes	Yes
Size $\times$ Year FE	Yes	Yes	Yes	Yes

Note: Columns (1)–(2) use 4-digit product definitions; columns (3)–(4) show the corresponding 3-digit robustness checks. Robust standard errors clustered at the state level are in parentheses. Significance levels: * 10%, ** 5%, *** 1%.

Table B.28: Treatment effect of the Repeal on creative destruction: 4-digit product

	(1)	(2)	(3)
	Base	+ State-Year	+ State-Year + Ind-Year
Repeal	0.002 (0.016)		
Repeal $\times$ UA	0.006 (0.011)	0.016 (0.012)	0.008 (0.013)
Observations	422,453	422,453	422,445
R^2	0.265	0.270	0.281
Year FE	Yes	Yes	Yes
District FE	Yes	Yes	Yes
Size $\times$ Year FE	Yes	Yes	Yes
State $\times$ Year FE	No	Yes	Yes
Ind. $\times$ Year FE	No	No	Yes

Note: Log-number-of-products specifications under progressively richer fixed effects. Robust standard errors clustered at the state level are in parentheses. Significance levels: * 10%, ** 5%, *** 1%.

Table B.29: Treatment effect of the Repeal on product scope

	(1)	(2)	(3)
	All	Bin 1	Bin 2
Repeal	0.091 (0.119)	0.119 (0.152)	-0.117 (0.205)
Repeal $\times$ UA	-0.044 (0.151)	-0.177 (0.168)	0.441*** (0.149)
Observations	5,396	5,354	5,082
R^2			
Year FE	Yes	Yes	Yes
District FE	Yes	Yes	Yes
Size $\times$ Year FE	Yes	Yes	Yes

Note: Columns split entry using the paper's LASSO-predicted innovativeness bins. Robust standard errors clustered at the state level are in parentheses. Significance levels: * 10%, ** 5%, *** 1%.

Table B.30: Treatment effect of the Repeal on firm entry: LASSO predicted innovativeness

C Additional empirical results

C.1 Land adjustments

Land market transactions. The main text shows that the Repeal increased land use and land *purchases* without a commensurate increase in land *revaluation*, consistent with lower frictions in transacting land rather than a pure appreciation of existing holdings. Here we decompose the transaction response into extensive (the likelihood of transacting) and intensive (the scale of a transaction) margins, separately for purchases and sales, and find that the Repeal raised activity on both margins for both types of transaction, though three of the four estimates are significant only at the 10 percent level.

Table C.31 reports this decomposition. The extensive-margin columns (1) and (3) employ a Probit model; the intensive-margin columns (2) and (4) set all zero values to the smallest positive land adjustment before applying a logarithmic transformation, following [Chen and Roth \(2023\)](#), which isolates intensive-margin variation by shutting down the extensive movement between 0 and y_{min} . Treated UA plants raised the probability of *buying* land by 0.9 percentage points (average marginal effect, relative to a sample mean of 10.9%), and the intensive-margin estimate implies an 8.1% increase in purchase volumes. The Repeal also increased land *sales* on both margins: the likelihood of selling rose by 0.5 percentage points (relative to a 3.0% mean), with a 3.1% intensive-margin increase in volumes.⁴² That the reform raised turnover on *both* sides of the market—acquisitions and disposals alike—is the signature of lower transaction frictions rather than a one-sided asset revaluation.

⁴²Columns (1) and (3) of Table C.31 report Probit index coefficients; the percentage-point effects quoted here are the corresponding average marginal effects. The intensive-margin regressions are run on the full sample with zeros recoded to the smallest positive adjustment, so they isolate variation above y_{min} rather than conditioning on the subsample of plants that transact; they are not conditional-on-transacting estimates.

	Add		Sell	
	(1)	(2)	(3)	(4)
	Extensive	Intensive	Extensive	Intensive
Repeal	0.017 (0.045)	0.008 (0.042)	−0.035 (0.035)	−0.016 (0.016)
Repeal $\times$ UA	0.052* (0.029)	0.078* (0.039)	0.070** (0.035)	0.031* (0.016)
Observations	334,588	334,583	334,307	334,302
R^2	0.039	0.030	0.033	0.010
Year FE	Yes	Yes	Yes	Yes
District FE	Yes	Yes	Yes	Yes
Size $\times$ Year FE	Yes	Yes	Yes	Yes

Note: Columns (1)–(2) report land purchases (“Add”); columns (3)–(4) report land sales. Within each pair, the first column is the extensive margin (a Probit for the likelihood of transacting) and the second is the intensive margin (log adjustment with zeros set to the smallest positive value). Robust standard errors clustered at the state level are in parentheses. Significance levels: * 10%, ** 5%, *** 1%.

Table C.31: Treatment effect of the Repeal on land adjustment: extensive vs. intensive

Table C.32 corroborates this pattern with PPML estimates on the full sample of plant–years, the vast majority of which record no transaction. Land *purchases* increase significantly (Column 2), while land *revaluations* show no detectable change (Column 4). Land *sales* carry a positive point estimate (Column 3) but are imprecisely estimated in levels.⁴³ The extensive/intensive decomposition above, which pools information across the many zero and small transactions, recovers the sell-side response that the level specification is underpowered to detect. On balance, the reform is associated with greater activity in land markets on both sides—most sharply in acquisitions—rather than a pure appreciation of existing holdings, reinforcing the interpretation that the Repeal eased the *frictions* of transacting land rather than merely raising its price.

⁴³Land transactions are sparse—about 3% of plant–years record a positive land sale and fewer than 0.3% a revaluation—so the level-PPML sales and revaluation columns are estimated with limited power.

	(1)	(2)	(3)	(4)
	Land	Buy	Sell	Rev.
Repeal	0.005 (0.057)	0.092 (0.089)	0.148 (0.104)	−0.176 (0.279)
Repeal × UA	0.202*** (0.056)	0.224** (0.104)	0.183 (0.133)	−0.034 (0.317)
Observations	338,686	333,805	330,681	297,070
R^2	0.183			
Year FE	Yes	Yes	Yes	Yes
District FE	Yes	Yes	Yes	Yes
Size × Year FE	Yes	Yes	Yes	Yes

Note: Column (1) reports log-landholdings. Columns (2)–(4) report PPML estimates for land purchases, sales, and revaluation. Robust standard errors clustered at the state level are in parentheses. Significance levels: * 10%, ** 5%, *** 1%.

Table C.32: Treatment effect of the Repeal on landholdings and adjustments

Flexible land inputs. We further demonstrate that the Repeal reduced land adjustment costs, making it easier for firms to divest land in response to profitability shocks. Following Guiso et al. (2005), Fagereng et al. (2018), and Chatterjee et al. (2023), we measure plant-level profitability shocks by unanticipated changes in gross value added that cannot be explained by plant-specific characteristics and aggregate fluctuations. Specifically, we regress the logarithm of value-added (VA_{it}) on plant (δ_i), industry-year (δ_{jt}), and state-year (δ_{st}) fixed effects, as shown below:

$$\ln(VA_{it}) = \delta_i + \delta_{jt} + \delta_{st} + \epsilon_{ijst}.$$

The regression residuals, ϵ_{ijst} , thus capture shocks to a plant’s “profitability”, including variation in both technical efficiency and product appeal.⁴⁴ We define a dummy variable $Shock_{i,t-1} = 1$ if the residual is negative in the previous year.⁴⁵ We then interact this measure of negative shocks with our baseline DiD regressors, while controlling for plant, industry-year, and state-year fixed effects, with standard errors clustered at the state level. Our identification leverages within-plant variation in responsiveness to negative shocks be-

⁴⁴An alternative method for measuring productivity shocks involves estimating the residual of the revenue function, or Total Factor Productivity (TFP), by deducting the cost of inputs from sales revenue. However, as noted by Decker et al. (2020), this TFP measurement is susceptible to errors due to omitted variables. In contrast, the plant fixed effects in our approach effectively absorb any plant-specific factors that do not change over time.

⁴⁵Note that all the estimates would be symmetric if we defined a positive shock as having positive residuals in the previous year.

fore and after the Repeal, compared to control firms in the same industry-year and state-year cells.

Table C.33 reports the regressions for landholdings (Column 1), number of employees (Column 2), and fixed assets (excluding land) in Column 3. Two patterns are precisely estimated. First, land adjustments were considerably less flexible than adjustments in labor and other fixed assets such as machines and plants: facing negative shocks, plants divest their landholdings by 9.1%, compared with a 14.6% and 16.4% decline in labor and capital, respectively, as indicated by the first row. Second, plants in UAs were, if anything, even less responsive to negative shocks before the Repeal (second row of Column 1, +3.7%), though this differential is not statistically significant.

The triple interaction in the last row of Column (1) is the coefficient of interest: it measures whether treated UA plants became more willing to shed land in response to a negative shock after the Repeal. Its sign is consistent with that interpretation, but the estimate is imprecise (-0.056 , standard error 0.040) and not significant at conventional levels, so we read it as suggestive rather than conclusive. Taken at face value, the implied post-Repeal response of treated plants—the sum of the four shock-interacted terms, -0.102 —would narrow the gap between land and the other inputs without closing it, leaving land still less flexible than labor (-0.158) or capital (-0.179). We therefore treat the flexible-land-inputs evidence as corroborative of, rather than independent proof of, the transaction-cost interpretation developed above.

The corresponding triple interactions for labor and capital (last row of Columns 2 and 3) are close to zero and likewise insignificant, so the reform shows no sign of having changed the flexibility of those inputs. The contrast is therefore one of sign and magnitude rather than of statistical significance: only land carries a point estimate pointing toward greater post-Repeal flexibility, which is consistent with the reform having relaxed land-market constraints specifically rather than frictions in labor and capital markets. More firmly, the $\text{Repeal} \times \text{UA}$ coefficient in Column (1) implies a statistically significant 9.2% within-plant increase in landholdings for plants that did not experience a negative shock, suggesting that the Repeal may have reallocated land toward more profitable plants.

Land misallocation. An alternative approach to test the impacts on land misallocation is to examine the heterogeneous treatment effects on firms with high versus low marginal revenue product of land (MRPLand) prior to the Repeal, as in [Bau and Matray \(2023\)](#). Post-Repeal, we expect plants with higher pre-Repeal MRPLand to experience faster land growth and a subsequent decline in MRPLand. The Repeal reduced land misallocation if it

	(1)	(2)	(3)
	Land	Labor	Capital
Negative shock _{t-1}	-0.095*** (0.028)	-0.158*** (0.024)	-0.179*** (0.016)
Negative shock _{t-1} × UA	0.036 (0.030)	-0.039 (0.025)	0.005 (0.025)
Negative shock _{t-1} × Repeal	0.013 (0.027)	0.009 (0.024)	0.005 (0.016)
Repeal × UA	0.088** (0.034)	-0.038 (0.026)	0.043 (0.046)
Negative shock _{t-1} × Repeal × UA	-0.056 (0.040)	0.030 (0.026)	0.006 (0.026)
Observations	79,470	106,935	107,046
R ²	0.924	0.949	0.971
Firm FE	Yes	Yes	Yes
State × Year FE	Yes	Yes	Yes
Ind. × Year FE	Yes	Yes	Yes

Note: All regressions include firm, state × year, and industry × year fixed effects. Robust standard errors clustered at the state level are in parentheses. Significance levels: * 10%, ** 5%, *** 1%.

Table C.33: Input adjustments following negative shocks

compressed the MRPLand dispersion.

However, a concern with this approach is the slow adjustment of land, e.g., firms may need years to build up their land stock. Consequently, firms with higher MRPLand (i.e., those more constrained by land according to this approach) might be the firms that are actively increasing their land stock and would thus experience accelerated land growth even without the Repeal. Moreover, previously high or low MRPLand firms may experience mean reversion, leading to a smaller MRPLand dispersion even without the Repeal.

Table C.34, Columns 1-2, present the Repeal's effects on within-plant land growth for previously high/low MRPLand plants, where the split is by whether a plant's pre-Repeal average MRPLand lies above or below the district median. Consistent with our concern, high MRPLand firms saw a 13.1% land increase even in non-UAs, likely reflecting slow-moving land adjustments or strong mean reversion forces. However, high MRPLand firms in UAs experienced an additional 16.1% land increase relative to high MRPLand plants in non-UAs. Together with the null effects on low MRPLand firms, these results suggest that the Repeal

	Land growth		MRPLand	
	(1)	(2)	(3)	(4)
	High	Low	High	Low
Repeal	0.123*	−0.045	−0.096	0.161***
	(0.070)	(0.046)	(0.066)	(0.044)
Repeal × UA	0.149**	−0.007	−0.182**	−0.022
	(0.061)	(0.040)	(0.071)	(0.043)
Observations	65,975	58,551	63,242	53,517
R^2	0.860	0.927	0.676	0.753
District FE	Yes	Yes	Yes	Yes
Size × Year FE	Yes	Yes	Yes	Yes
Ind. × Year FE	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes

Note: Columns (1)–(2) report land-growth specifications. Columns (3)–(4) report log-MRPLand specifications for the same pre-Repeal MRPLand split. Robust standard errors clustered at the state level are in parentheses. Significance levels: * 10%, ** 5%, *** 1%.

Table C.34: Treatment effects of the Repeal on land growth and MRPLand by high/low pre-Repeal MRPLand plants

improved land allocative efficiency by allowing previously land-constrained (high MRPLand) firms to expand. Figure C.21 presents the event study graphs for the different impacts on UA plants. The pre-Repeal coefficients are small and show no systematic trend, though one lead in each panel is individually distinguishable from zero, so we read the pre-period as broadly flat rather than as a sharp placebo test.

Columns 3-4 examine the impacts on MRPLand for previously high/low MRPLand plants. Two key patterns emerge. First, previously high (low) MRPLand plants experienced a decrease (increase) in MRPLand post-Repeal in non-UAs, indicating a mean reversion effect. Second, high MRPLand plants in UAs saw a 16.6% larger decline in MRPLand compared to their non-UA counterparts. Low MRPLand plants in UAs experienced a marginally smaller (2.2%) increase in MRPLand than those in non-UAs, though this differential is not statistically significant. Before the Repeal, high MRPLand plants had far higher MRPLand than low MRPLand plants—a pre-Repeal gap of roughly an order of magnitude (about a 16-fold difference in levels, or $\Delta \log \approx 2.8$)—suggesting that the liberalization reduced land misallocation by compressing the MRPLand dispersion.

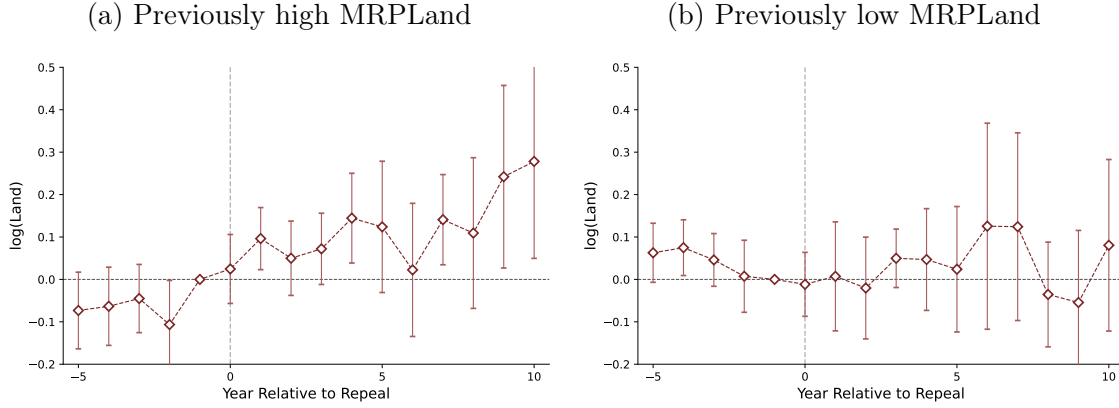


Figure C.21: Event study graphs for land use among high/low MRPLand plants in UAs

Note: The dependent variable is log land used in production. MRPLand is calculated as the ratio of sales revenue to land use. All regressions include plant fixed effects, year fixed effects, and controls incorporate both district pre-Repeal land endowment size bins and firm-specific land size bins, interacted with year fixed effects. Standard errors are clustered at the state level. The confidence interval is at the 95% level.

C.2 Heterogeneity across the plant size distribution

We next examine the distributional consequences of the Repeal using the (re-centered) influence function approach proposed by [Firpo et al. \(2009\)](#). The quantile regression estimates in Figure B.11 indicate that, for land, capital, employment, and the wage bill, the Repeal had small and insignificant, albeit positive, effects on plants at the lower end of the size distribution. These effects gradually increase, peaking between the 60th and 80th percentiles, and then fade at the very top of the distribution; sales are the exception, with effects that are already positive and significant at the bottom. This pattern is consistent with plants in the upper-middle of the size distribution benefiting most from the easing of land-market frictions, since they had greater need to buy, sell, and adjust landholdings.

To further probe this hypothesis, we conduct an analysis in the spirit of [Hsieh and Olken \(2014\)](#), examining the relationship between firm size and the average revenue product of land. Panel (a) of Figure B.12 shows that, within each 2-digit industry and year, larger plants—as measured by employment—exhibit a systematically higher average revenue product of land. Under a Cobb–Douglas production technology, a higher *average* revenue product directly implies a higher *marginal* revenue product of land. This pattern is therefore consistent with larger plants facing higher effective land costs, reflected in their elevated returns to land.⁴⁶

⁴⁶A potential alternative explanation is that larger plants charge higher markups rather than face higher land costs. Panel (b) of Figure B.12 addresses this concern by showing that larger plants also have higher input expenditures per unit of land. Another concern is that larger plants may rely more heavily on higher-quality or imported inputs, as emphasized by [Kugler and Verhoogen \(2012\)](#). To mitigate this possibility,

panel (c) shows that the positive relationship between input value per unit of land and plant size persists when restricting attention to domestic inputs. Finally, panel (d) documents a similar pattern for fuel inputs, which are plausibly homogeneous across plants.

D Socioeconomic impacts of the Repeal

In the main text, we demonstrated that the Repeal of the Urban Land Ceiling and Regulation Act (ULCRA) unleashed manufacturing growth and improved the allocative efficiency of land. In this section, drawing on village-level data from three rounds of the Population Census and the 2012 Socio-Economic and Caste Census, we systematically investigate the socioeconomic impacts of the Repeal. Using detailed employment information from the Population Census, we explore how the Repeal Act facilitated the reallocation of agricultural laborers to non-agricultural sectors. The rich data on income, wealth, and housing in the Socio-Economic and Caste Census enables us to determine whether the industrial development facilitated by the Repeal came at the expense of the poor. Utilizing household expenditure data from the National Sample Survey, we also demonstrate that the Repeal had a negligible impact on residential rents in Urban Agglomerations. We conclude this section by noting the development effects of the Repeal on poverty reduction.

D.1 Data

Our labor market data are sourced from the Socioeconomic High-resolution Rural-Urban Geographic Dataset (SHRUG),⁴⁷ which integrates data from multiple rounds of the population and economic census. Specifically, we utilize the 1991, 2001, and 2011 village-level Population Census data from the Primary Census Abstract and Village Directory tables. This dataset offers comprehensive information on village infrastructure, demographics, employment, occupation, and population, which we use to construct variables related to labor supply in the agricultural, manufacturing, and service sectors, as well as self-employment. Additionally, we gather village-level income, wealth, and housing data from the Socioeconomic and Caste Census (SECC), a universal enumeration of household assets conducted in 2011. The SECC details household assets that are readily assessable, including roof and wall materials, number of rooms, and assets like agricultural equipment, vehicles, and mobile phones. By mapping data from the population census and the SECC to time-invariant municipalities—including both villages ($n=590,648$) and towns ($n=7,528$)—the SHRUG enables us to trace the evolution of these municipalities both before and after the Repeal. We discuss variable construction along with the results.

We supplement this municipality-level data with a manually collected ULCRA dataset,

⁴⁷Asher and Novosad (2020) provides details on the data construction, accessible at <https://www.devdatalab.org/shrug>.

which includes the repeal time for each state and the geographic coverage of the 1976 Act. The Act specifies numerous Urban Agglomerations (UAs) along with their associated towns and villages. We systematically gathered coordinates for over six hundred locales listed in these delineations through extensive online research. To ensure the accuracy of our data, we cross-referenced each locale with the Indian Village Directory to verify their existence and correct spellings, particularly for those not readily found online.

We overlay the map of UA locales with the map of municipalities to delineate groups of municipalities that were more or less exposed to the Repeal, determined by their proximity to UAs. Since the exact geographic boundaries of UAs are not known, we assume they are defined by the encompassing set of UA locales. We calculate the distance between each municipality and these UA locales. Figure D.22 illustrates this mapping for the Hyderabad UA. Red dots represent the UA locales under Hyderabad, while the colored polygons depict municipalities with available Population Census data. Darker colors indicate greater distances from the municipalities to the these locales—up to 50 km. Larger polygons at the center of the circle identify towns around the Hyderabad metro—areas likely within the boundaries of the Hyderabad UA.

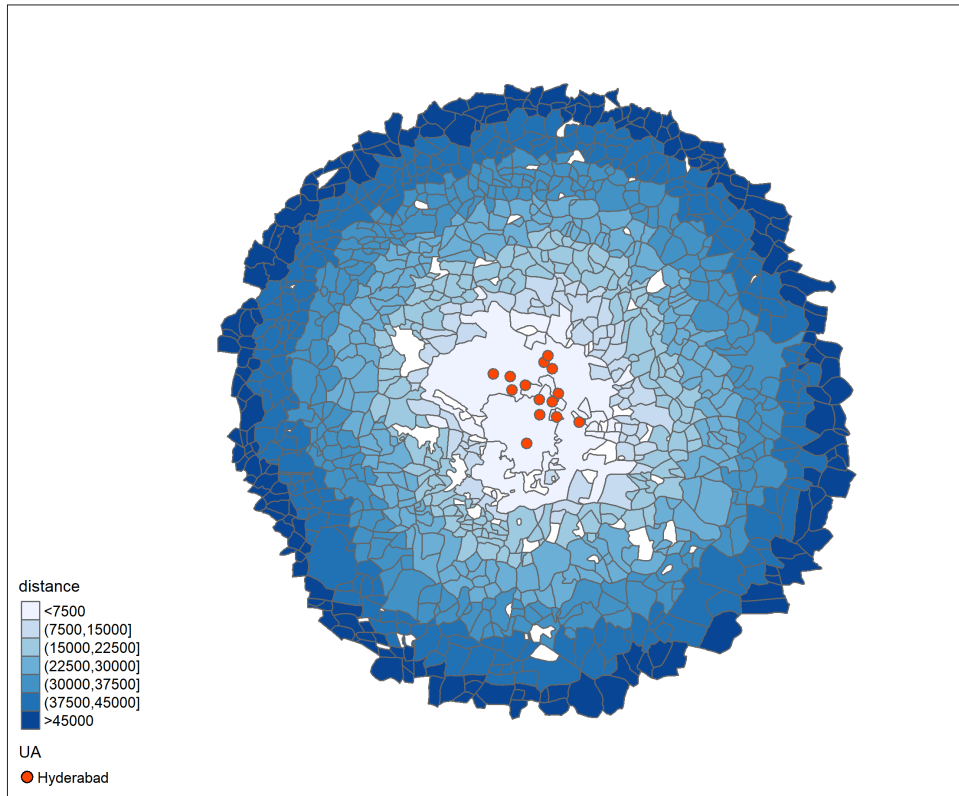


Figure D.22: UA locales under Hyderabad UA and nearby municipalities

D.2 Identification strategy

To estimate the causal effects of the Repeal and to capture potential spatial spillover to nearby municipalities, we implement a Triple-Difference design following [Gechter and Tsivanidis \(2023\)](#). This design compares changes in outcome variables before and after the Repeal, between municipalities that are closer to UAs and those that are further away. Specifically, we group municipalities into 5km-distance rings around their closest UA, up to a distance of 50km. This non-parametric approach allows us to study the spatial effects of the policy. Municipalities outside the 50km radius around a UA are excluded from the analysis.

The main analysis relies on a spatial Triple-Difference model as described below:

$$y_{ijt} = \sum_{d=1}^{d=9} \beta_d (D[d_i = d] \times \text{POST}_{jt}) + X_i \times \text{Year}_t + \sigma_t + \sigma_i + \sigma_{jt} + \epsilon_{ijt} \quad (\text{D.71})$$

where y_{ijt} represents outcomes such as agricultural employment or the share of homeless people in municipality i , around UA j in year t . $D[d_i = d]$ indicates whether a municipality i is in distance bin d to the nearest UA, with $d = 1$ representing municipalities within a 5km distance to the UA—likely municipalities within the confines of the UA—and extending up to 50km. Distance bin $d = 10$ (45-50km) is omitted and serves as the reference category. Our final sample has approximately 120,000 municipalities in a given year. We interact the distance dummies with a post-Repeal dummy, POST_{jt} , which is set to 1 if UA_j is located in a state that had repealed the ULCRA by year t . The model includes municipality fixed-effects σ_i and year fixed effects σ_t . We also include UA-by-year fixed effects σ_{jt} to ensure that we are comparing relative growth among municipalities—those closer and further away—from the same Urban Agglomeration over time. Additional controls $X_i \times \text{Year}_t$ include various municipality-level baseline covariates interacted with year fixed effects. ϵ_{ijt} is the error term clustered at the UA level to account for spatial correlation. The β_d are the parameters of interest, capturing the triple-difference estimates. They capture the differential changes in outcome variables across municipalities in distance bin d that experienced the Repeal, relative to those in the reference category, and contrast these changes with municipalities in states that have not yet implemented the Repeal.

Our identification assumption is that the relative growth rates of municipalities near UAs compared to those farther away do not differ significantly between states that adopted the Repeal early and those that adopted it later. This assumption could be violated if, for example, municipalities closer to urban centers in early-adopter states grow differentially faster than those in late-adopter states, even in the absence of the Repeal. While this

assumption is untestable, we address potential concerns in two ways. First, we control for various baseline covariates—such as population size, distance to the nearest town, and proximity to roads—interacted with year fixed effects. These controls are intended to mitigate any pre-existing differential growth trends between closer and farther municipalities across early and late adopter states. Secondly, we perform a placebo test by substituting the distance-to-UA dummies with distance-to-town dummies. If the differential growth patterns were primarily driven by proximity to urban centers rather than the Repeal, we would expect to see similar patterns in the placebo test.

D.3 Results from the Population Census

Effects on structural transformation. We begin our analysis by examining the effects of the Repeal on structural transformation, which could be facilitated through both labor supply and demand channels. On one hand, the expansion of manufacturing firms post-Repeal is likely to create more job opportunities in the industrial sector. On the other hand, by removing arbitrary building and construction regulations, the Repeal could decrease urban rent, thereby increasing the labor supply in non-agricultural sectors.

Using data from three rounds of the Population Census, we construct municipality-level shares of cultivators, agricultural hired laborers, other workers (include both manufacturing and service workers), and workers in household enterprises, among main workers. Main workers are those who work for more than six months in their jobs. Figure D.23 plots the coefficients β_d of our empirical equation (D.71) with each dot indicating the coefficient for a specific distance bin. Panels (b) and (c) show that municipalities closer to UAs experience an increase in the share of other workers and a decrease in the share of agricultural hired laborers. For more distant municipalities ($> 25\text{km}$), the estimates for β_d turn out to be small and statistically insignificant from the reference group (45-50km), suggesting that employment changes between municipalities in further distance do not differ systematically.

The spatial spillovers on structural change are substantial in magnitude. For example, municipalities that are 10-15 km away from UAs witness an increase in the share of other workers by about 10 percentage points, which corresponds to approximately 30% of the sample mean (31.5%). Similarly, these municipalities observe a decrease in agricultural hired laborers by 4 percentage points, amounting to about 17% of the sample mean (25%). By contrast, Panels (a) and (d) suggest that changes in the shares of cultivators and workers in household enterprises are smaller in magnitude and often insignificant. Moreover, Panels (e) and (f) demonstrate that the Repeal not only changes the employment shares but also accel-

erates the growth of the total population and the number of main workers in municipalities close to UAs.

Effects on amenities. One might wonder how amenities—such as roads, schools, and electricity—respond endogenously to structural changes. These responses are particularly interesting as they could serve as potential mechanisms for the industrial growth results reported in our main text. For example, if the Repeal removed land constraints, it could allow developers in UAs to construct more roads, which in turn could boost manufacturing growth. To investigate this, we utilize municipality-level infrastructure data to construct various indicators of amenities, including the number of primary and high schools, the presence of paved and unpaved roads, and the availability of electricity for agricultural and general purposes. We then apply our spatial difference-in-differences (DiD) specification to assess the impact of the Repeal on these amenities. Note that, unlike our baseline specification, which controls for baseline amenities-by-year fixed effects, this analysis includes only baseline quartiles of population and distance to towns, interacted with year fixed effects. Interestingly, in Figure D.24 we do not observe accelerated growth of amenities in municipalities closer to UAs. In fact, municipalities near UAs experience somewhat slower growth in paved roads, although this effect is relatively modest in magnitude.

D.4 Results from the 2011 Socio-Economic and Caste Census

We now turn to the effects of the Repeal on household income, housing conditions, wealth, and poverty, using data from the 2011 Socio-Economic and Caste Census (SECC). Note that this is the only round of the SECC that contains household income and wealth information. Consequently, we are unable to estimate our baseline spatial DiD specification, as all UAs were treated by 2011. As an alternative, we examine the relative differences in the *levels* of outcomes between places closer to UAs and those further away. Our estimates should be interpreted with caution in light of this potential limitation.

Effects on housing conditions. We begin by examining the impacts on housing conditions. As noted by [Gechter and Tsivanidis \(2023\)](#), urban redevelopment could have adverse effects on the local population, particularly if local dwellers are forcibly relocated. In our context, if residential land near UAs is converted to industrial land following the Repeal, this might lead to a deterioration of housing conditions in nearby municipalities. To assess this, we consider several indicators of housing conditions, including the share of households

that own or rent a house, the share of homeless individuals, the average number of rooms per house, and the proportion of houses with solid walls and roofs.

Figure D.25 demonstrates that municipalities closer to UAs tend to have a smaller share of households that own a house, but a higher share renting. The impacts on the share of homeless individuals and the average number of rooms are negligible and statistically insignificant. Additionally, municipalities nearer to UAs are more likely to have houses with solid walls and roofs. Taken together, these findings suggest that there is no evidence of a deterioration in housing conditions attributable to the Repeal.

Income and poverty. We then examine the effects on municipality-level income, poverty rate, and estimated consumption. Although consumption was not directly surveyed by the SECC, the SHRUG provides imputed consumption information, for each municipality, based on another household survey (IHDS-II, 2011-12), which includes similar asset, income, and land data as the SECC. SHRUG reports mean predicted consumption and the share of the population below the poverty line (27 Rupees/day) for each municipality. Figure D.26 shows that municipalities closer to UAs tend to have higher income, more likely to have salaried wage, consume more, and exhibit a lower poverty rate. We observe significant spatial spillovers in the effect on poverty rate—municipalities 25 km away from UAs show a reduction in poverty rate by approximately 2 percentage points compared to the reference group, which constitutes about 6.5% of the sample mean (31%). By contrast, the effects on income and salaried wage from the government tend to be more localized and diminish more rapidly with distance. In fact, the effects on earning more than 10,000 Rupees per month and the effects on having a government salaried wage are only significant within the 0-5 km distance bin, suggesting that these impacts are primarily driven by municipalities *inside* UAs. By contrast, the effects on private sector salaried wages exhibit stronger spatial spillover and are substantial in magnitude. For instance, municipalities that are 20-25 km away tend to have 1 percentage point more households earning a salaried wage from the private sector, which constitutes approximately 25% of the sample mean (3.9%).

We then examine the impacts of proximity to UAs on the main sources of household income. Consistent with our findings on structural transformation, Figure D.27 demonstrates that municipalities closer to UAs are less likely to report cultivation as the main source of income, and more likely to derive income from manual labor. Additionally, we do not observe any effects on the share of households who report begging or foraging as the main source of income, suggesting that the Repeal may not have displaced people.

D.5 Placebo test

As previously noted, our identification assumption is that the relative growth rates of municipalities near Urban Agglomerations (UAs) compared to those farther away do not differ significantly between states that adopted the Repeal early and those that adopted it later. This assumption could be violated if, for instance, municipalities closer to urban centers in early-adopter states exhibit differentially faster growth than those in late-adopter states, even in the absence of the Repeal. In our baseline regression, we always control for various baseline covariates—such as population size, distance to the nearest town, and proximity to roads—interacted with year fixed effects, to mitigate any pre-existing differential growth trends.

As an additional strategy, we perform a placebo test by substituting the distance-to-UA dummies with distance-to-town dummies. Should our previous results be driven primarily by proximity to urban centers rather than the Repeal itself, we would expect to observe similar patterns in the placebo test. Figures D.28, D.29, D.30, D.31, and D.32 present the results of this placebo test, replicating our earlier analyses using the distance to the nearest town. Collectively, these results are significantly noisier, exhibit no systematic patterns, and tend to be considerably smaller in magnitude. This placebo test lends further support to our main findings, bolstering the causal interpretation of them.

D.6 Rent and housing prices

Another question left unaddressed in the main text concerns whether the increase in the value of firms' landholdings is driven by the increase in the amount of land or in land prices. Unfortunately, there is no systematic data on property prices across districts in India. To address this, we construct several measures of residential rent expenditure using household-level consumption data. The rationale behind this approach is that if urban land prices increased dramatically following the Repeal, we would expect a corresponding rise in rent expenditures. This examination is important in its own right. Suppose the Repeal facilitated the conversion of residential land to industrial land, thereby pushing up rents; this could potentially have negative effects on urban renters.

The consumption data are sourced from the Indian National Sample Survey (NSS), which provides pooled cross-sectional household consumer expenditure data. The full-scale quinquennial ('thick') rounds survey approximately 100,000 households every five years.⁴⁸ For our

⁴⁸In addition to the quinquennial rounds, there are also annual NSS rounds that survey fewer households,

analysis, we utilize the data from the 50th (1993-94), 55th (1999-2000), and 61st (2004-05) rounds, corresponding to the fifth, sixth, and seventh quinquennial rounds.

The NSS dataset contains information on household consumption expenditure and quantity consumed across approximately 300 product categories. Rents are reported in four categories:

- House and garage actual rent, which includes rent for residential buildings and garages for private vehicles.
- Hotel lodging charges, paid by any household member for staying in a hotel.
- Other consumer rent, which includes charges for consumer goods (like furniture) and decorations on ceremonial occasions.
- Imputed house and garage rent for urban households living in owned or otherwise occupied houses without paying rent, based on the prevailing rate of rent for similar accommodations in surrounding areas.

To capture the overall rent expenditure, we use the sum of the first three categories for the actual rent paid by the household over the last 30 days. However, for those urban households who already owned a house, they may not incur actual rents. Therefore, we supplement our rent expenditure measure with the imputed rent to construct a measure of combined rent whenever the actual expenditure information is missing.

Both rent expenditure and imputed rent are prorated to reflect monthly rent amounts. One limitation of our analysis is the lack of data on floor area, which is available only from the 62nd round of the NSS onward. Consequently, we are unable to calculate rent per square meter as in [Harari \(2020\)](#). This limitation may restrict our ability to standardize rent expenditures across household sizes and wealth. Therefore, we derive two outcome variables for each measure: the share of rent over the household’s total monthly expenditure over the last 30 days, and the rent expenditure per household member.

We construct a crosswalk to map districts in each NSS round to the time-invariant 1991 population census district boundaries, along with our policy and Repeal data. Subsequently, we aggregate household-level rent information to the district level by calculating the average rent expenditure using the sampling weights. Using the district-round level panel, we examine the treatment effects of the Repeal on our rent measures using the DiD design. This involves comparing rents in districts that had already adopted the Repeal to those that had not, and

ranging from 20,000 to 50,000 households.

	(1)	(2)	(3)	(4)
	Rent share	Log rent p.c.	Combined share	Combined log p.c.
Repeal	−0.007 (0.008)	−0.032 (0.113)	−0.011 (0.010)	−0.004 (0.106)
Repeal × UA	−0.007 (0.008)	−0.041 (0.132)	0.009 (0.010)	−0.016 (0.104)
Observations	2,034	2,034	2,065	2,065
R^2	0.460	0.587	0.580	0.659
Year FE	Yes	Yes	Yes	Yes
District FE	Yes	Yes	Yes	Yes
Pop. bin × Year FE	Yes	Yes	Yes	Yes

Note: District-level rent regressions. Robust standard errors clustered at the state-year level are in parentheses. Significance levels: * 10%, ** 5%, *** 1%.

Table D.35: Treatment effect of the Repeal on rents

considering districts that contain any Urban Agglomerations versus those that do not. Our regressions include district fixed effects, year fixed effects, and baseline district population quartile bin-by-year fixed effects.

Table D.35 reports the estimates of the treatment effects on four measures of rents. Columns (1) and (3) present the share of actual and combined rents over the household's total monthly expenditure over the last 30 days, respectively. Columns (2) and (4) present the actual and combined rents per household member, respectively. Overall, our results indicate a very small and insignificant effect on rent across all four measures. Additionally, the interaction term $Repeal_{st} * I_d^U$ is not significant in any specification, further confirming that the Repeal did not have a differential impact on households in districts that contain Urban Agglomerations. Taken together, we conclude that the Repeal had minimal impacts on residential rents in UAs. This provides suggestive evidence that our findings regarding firms' landholdings are unlikely to be driven solely by changes in land prices.

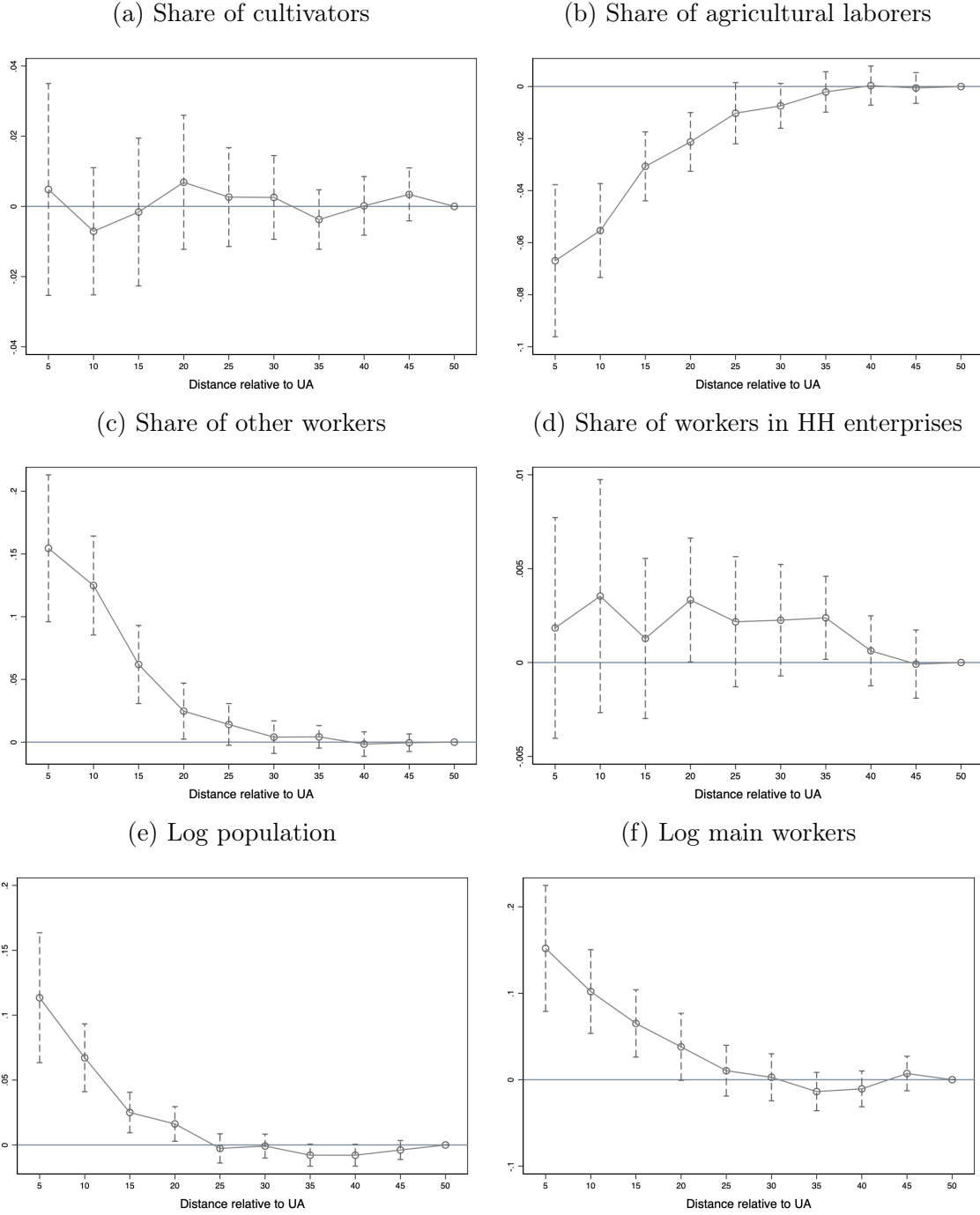


Figure D.23: Spatial DiD effects on employment shares

Note: This figure displays the spatial difference-in-differences (DiD) effects of the Repeal on the shares of cultivators, agricultural hired laborers, other workers, and workers in household enterprises among main workers (panels a-d), and logged population and main workers (panels e-f). The dots represent the estimated parameters β_d from our empirical equation (D.71), where each subscript d corresponds to a distance on the horizontal axis; for example, the coefficient at $d = 1$ indicates the municipality is within 5 km of a UA. The model accounts for municipality fixed effects, year fixed effects, and UA-by-year fixed effects. Controls include quartiles of baseline population, road presence, electrification, and distance to towns, each interacted with year fixed effects. Standard errors are two-way clustered at the UA-year level. Dashed lines represent 95% confidence intervals.

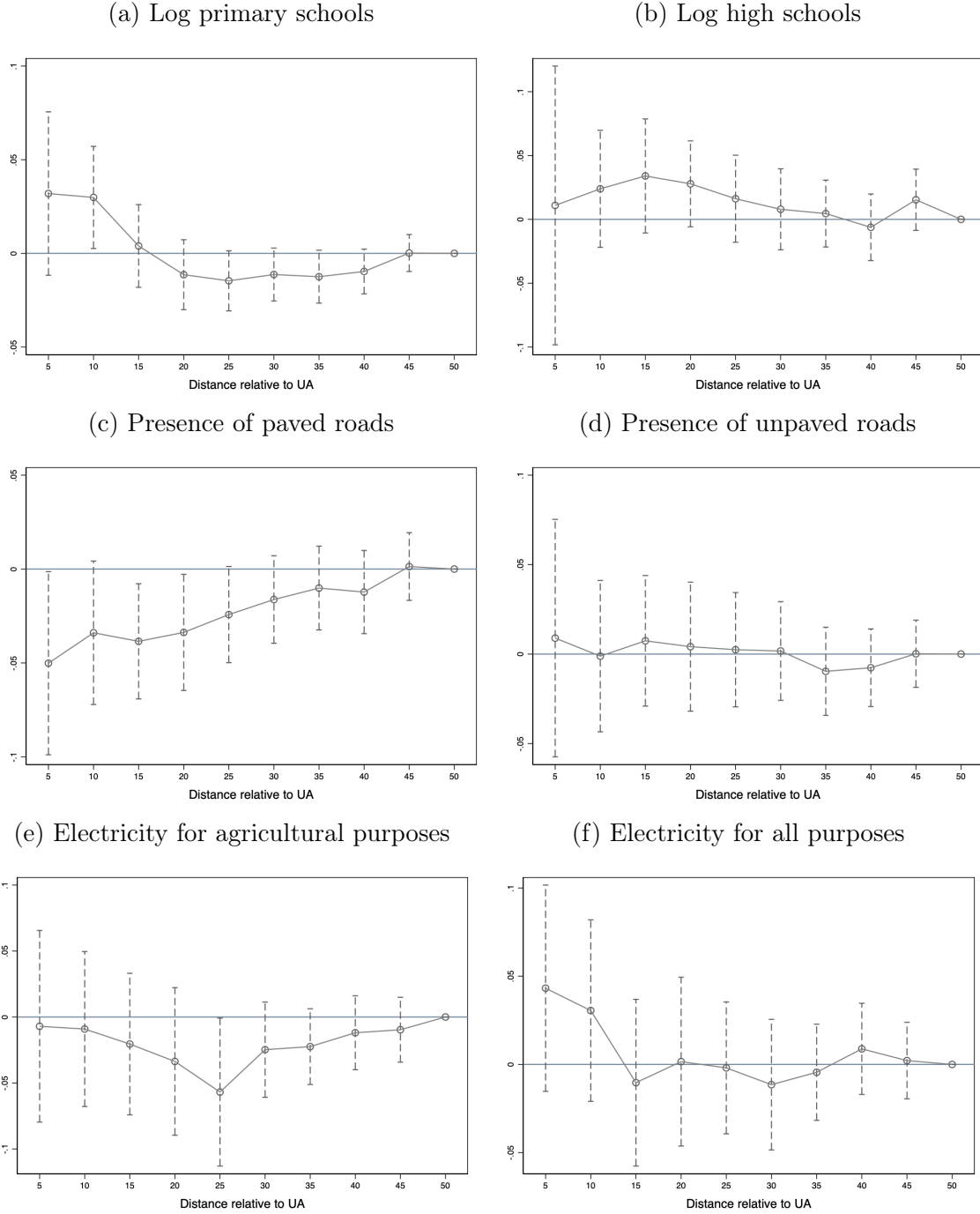


Figure D.24: Spatial DiD effects on amenities

Note: This figure displays the spatial difference-in-differences (DiD) effects of the Repeal on logged number of primary and high schools, the presence of paved and unpaved roads, and the availability of electricity for agricultural and general purposes. The dots represent the estimated parameters β_d from our empirical equation (D.71), where each subscript d corresponds to a distance on the horizontal axis; for example, the coefficient at $d = 1$ indicates the municipality is within 5 km of a UA. The model accounts for municipality fixed effects, year fixed effects, and UA-by-year fixed effects. Controls include quartiles of baseline population and distance to towns, each interacted with year fixed effects. Standard errors are two-way clustered at the UA-year level. Dashed lines represent 95% confidence intervals.

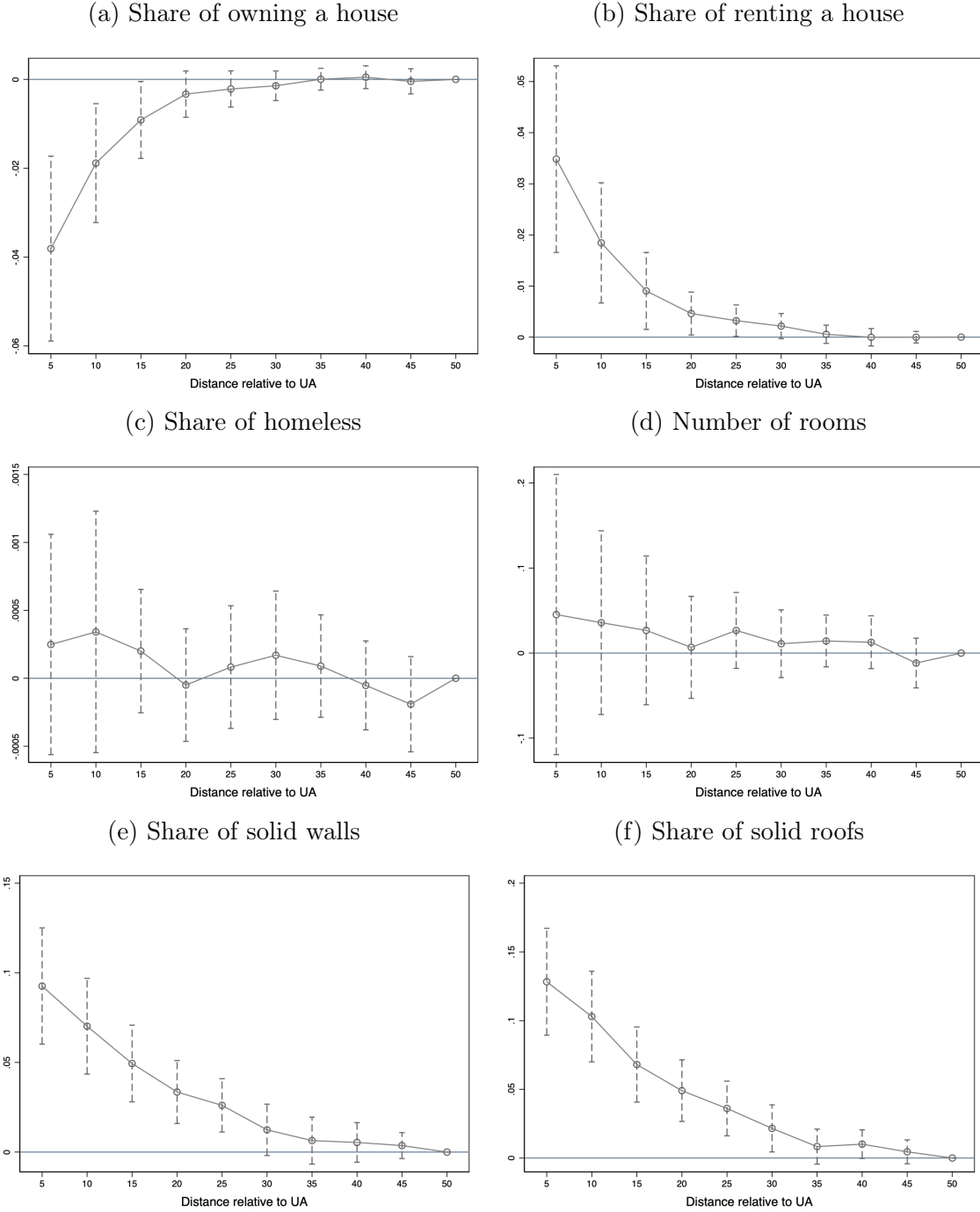


Figure D.25: Proximity to UAs and housing conditions

Note: This figure displays the effects of proximity to UAs on several housing indicators: the share of households that own or rent a house, the share of homeless individuals, the average number of rooms per household, and the proportion of houses with solid walls and roofs. Each dot represents the estimated parameter β_d , where each subscript d corresponds to a specific distance along the horizontal axis. For instance, a coefficient at $d = 1$ suggests that the municipality is within 5 km of a UA. The model incorporates UA fixed effects, and controls include quartiles of baseline population, road presence, electrification, and distance to towns. Standard errors are clustered at the UA level. Dashed lines indicate 95% confidence intervals.

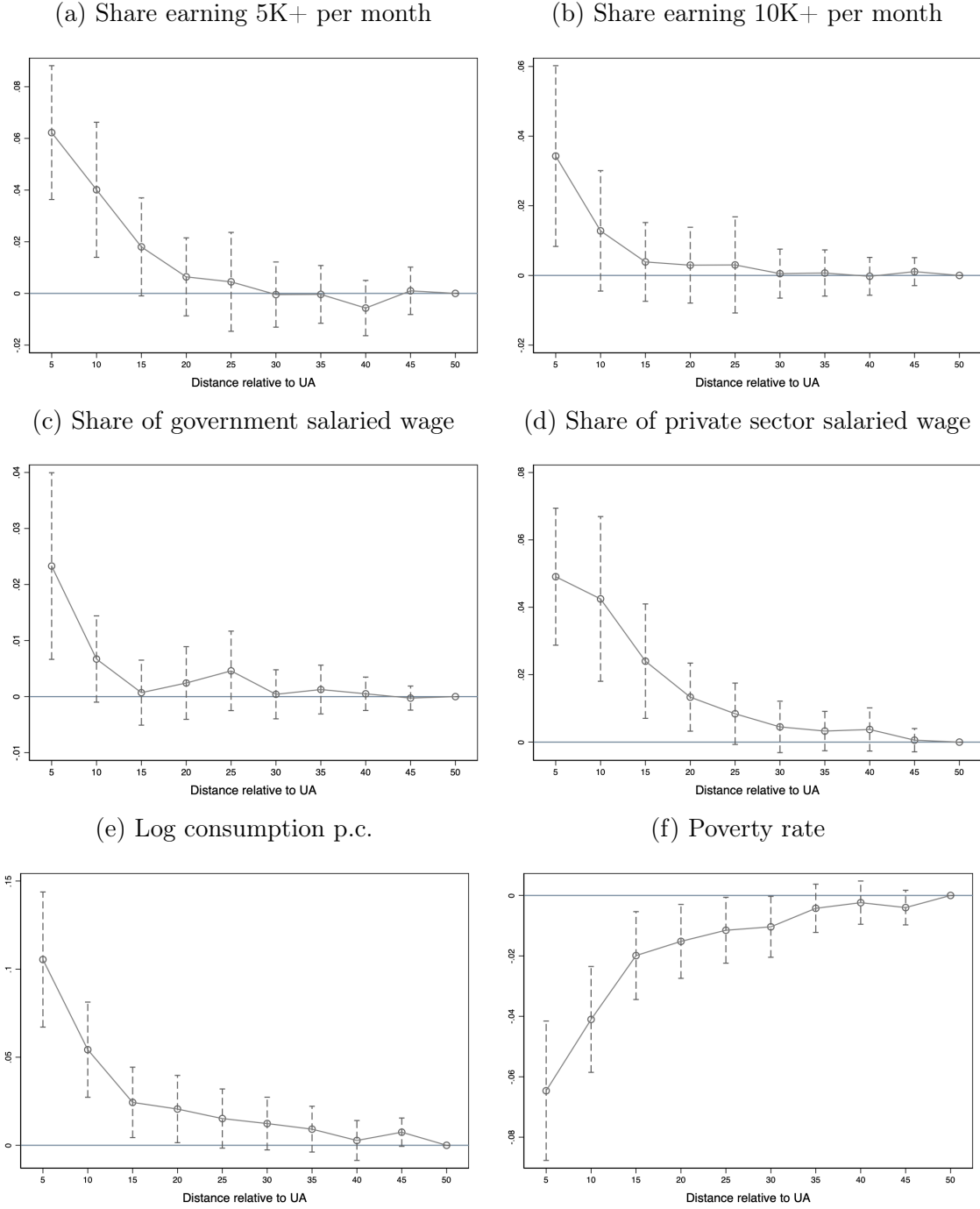


Figure D.26: Proximity to UAs, income, and poverty

Note: This figure displays the effects of proximity to UAs on municipality-level income, poverty rate, and estimated consumption. Consumption information is imputed based on another household survey (IHDS-II, 2011-12). SHRUG reports mean predicted consumption and the share of the population below the poverty line (27 Rupees/day) for each municipality. Each dot represents the estimated parameter β_d , where each subscript d corresponds to a specific distance along the horizontal axis. For instance, a coefficient at $d = 1$ suggests that the municipality is within 5 km of a UA. The model incorporates UA fixed effects, and controls include quartiles of baseline population, road presence, electrification, and distance to towns. Standard errors are clustered at the UA level. Dashed lines indicate 95% confidence intervals.

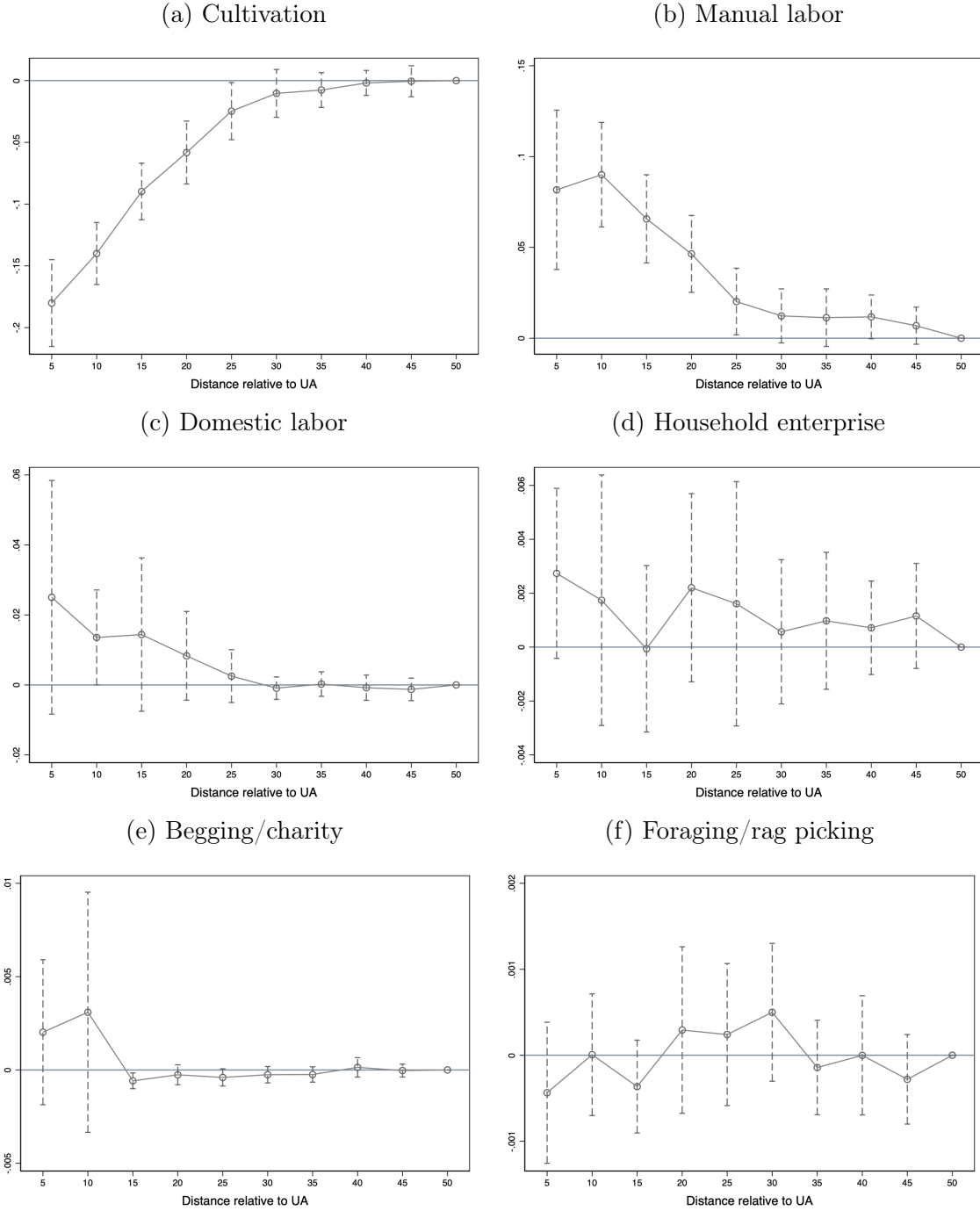


Figure D.27: Proximity to UAs and income sources

Note: This figure displays the effects of proximity to UAs on the main source of household income. Each dot represents the estimated parameter β_d , where each subscript d corresponds to a specific distance along the horizontal axis. For instance, a coefficient at $d = 1$ suggests that the municipality is within 5 km of a UA. The model incorporates UA fixed effects, and controls include quartiles of baseline population, road presence, electrification, and distance to towns. Standard errors are clustered at the UA level. Dashed lines indicate 95% confidence intervals.

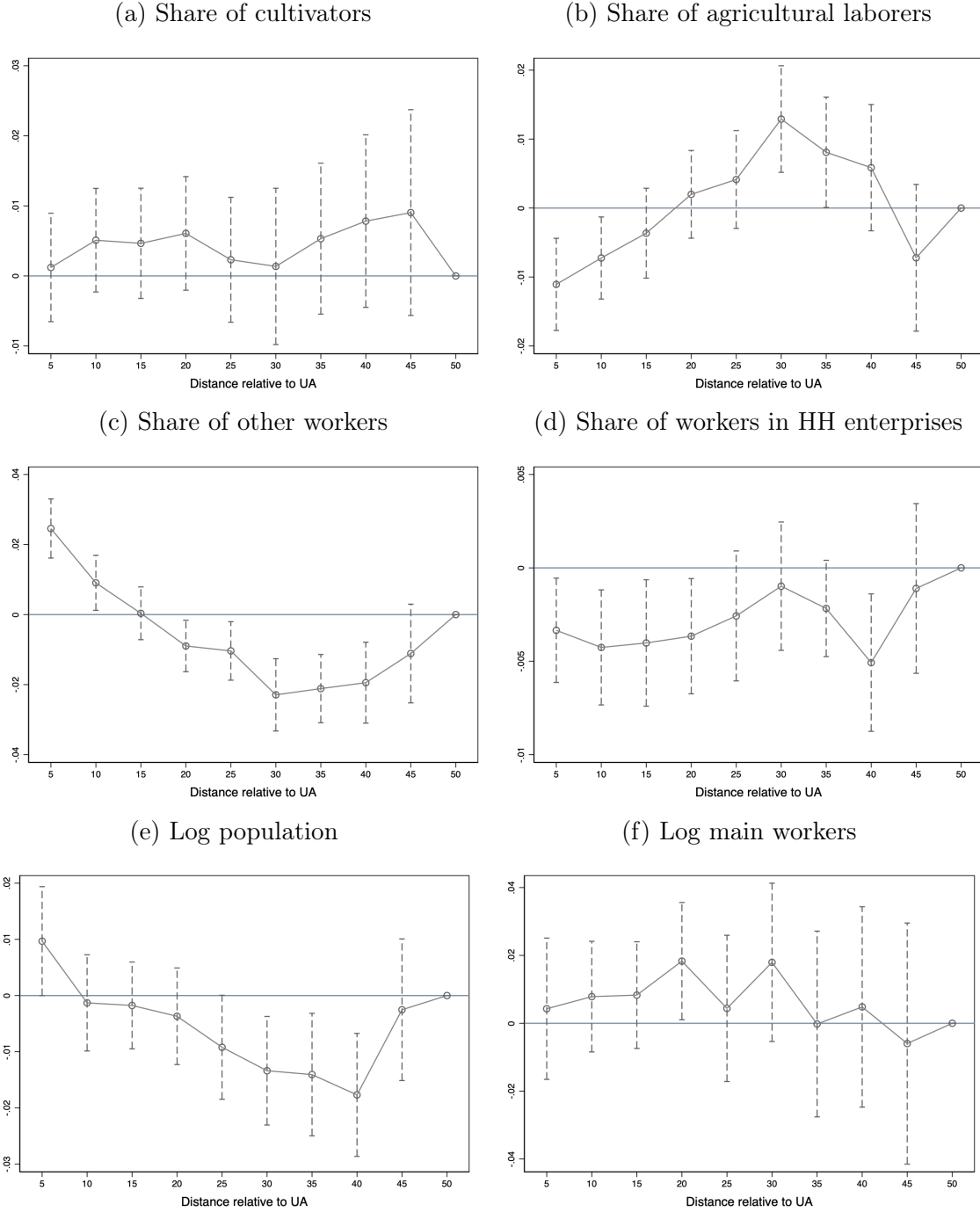


Figure D.28: Placebo: spatial DiD effects on employment shares

Note: This figure displays the spatial difference-in-differences (DiD) effects of the Repeal on the shares of cultivators, agricultural hired laborers, other workers, and workers in household enterprises among main workers (panels a-d), and logged population and main workers (panels e-f). The dots represent the estimated parameters β_d from our empirical equation (D.71), where each subscript d corresponds to a distance on the horizontal axis; for example, the coefficient at $d = 1$ indicates the municipality is within 5 km of a town. The model accounts for municipality fixed effects, year fixed effects, and UA-by-year fixed effects. Controls include quartiles of baseline population, road presence, electrification, and distance to towns, each interacted with year fixed effects. Standard errors are two-way clustered at the UA-year level. Dashed lines represent 95% confidence intervals.

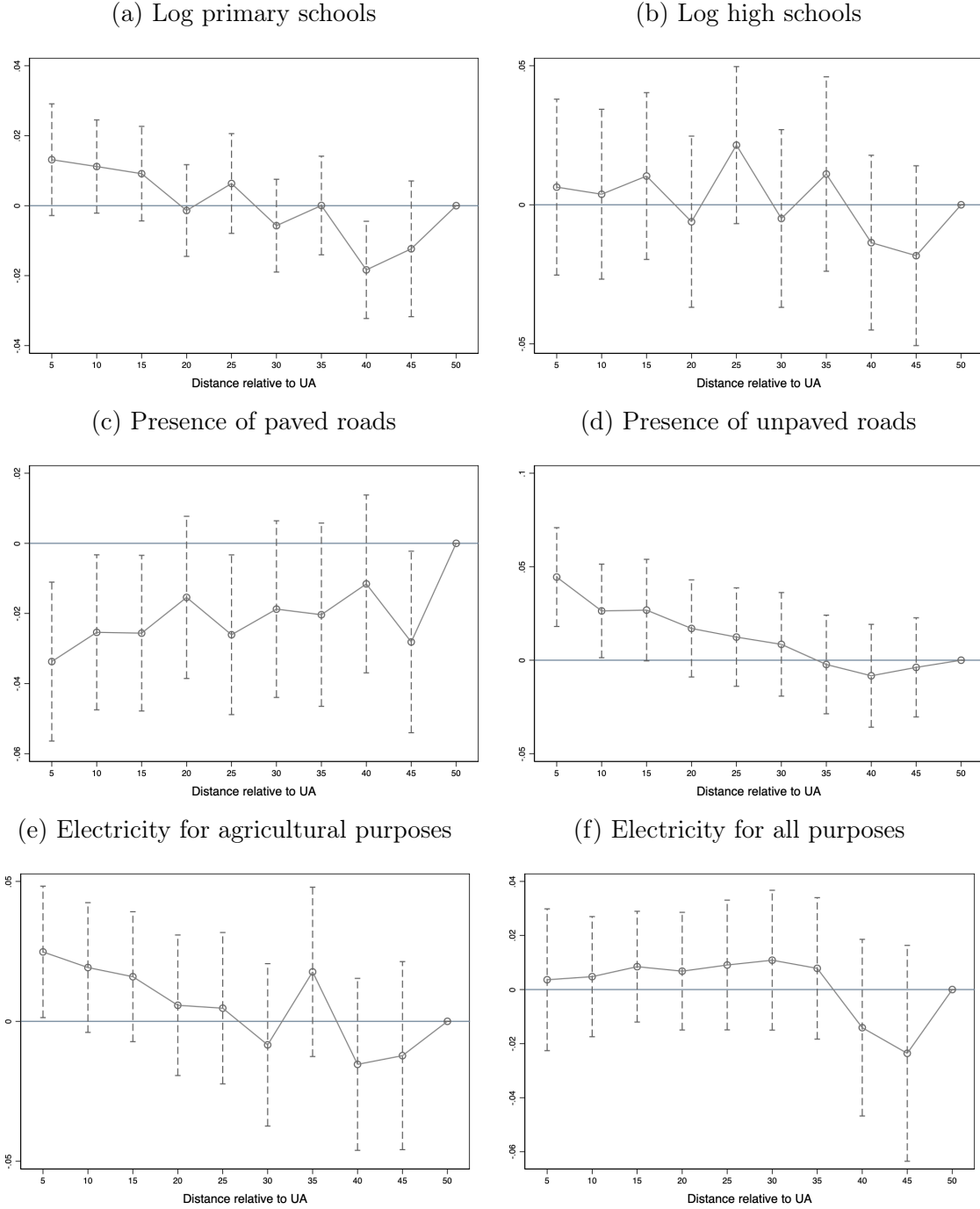


Figure D.29: Placebo: spatial DiD effects on amenities

Note: This figure displays the spatial difference-in-differences (DiD) effects of the Repeal on logged number of primary and high schools, the presence of paved and unpaved roads, and the availability of electricity for agricultural and general purposes. The dots represent the estimated parameters β_d from our empirical equation (D.71), where each subscript d corresponds to a distance on the horizontal axis; for example, the coefficient at $d = 1$ indicates the municipality is within 5 km of a town. The model accounts for municipality fixed effects, year fixed effects, and UA-by-year fixed effects. Controls include quartiles of baseline population and distance to towns, each interacted with year fixed effects. Standard errors are two-way clustered at the UA-year level. Dashed lines represent 95% confidence intervals.

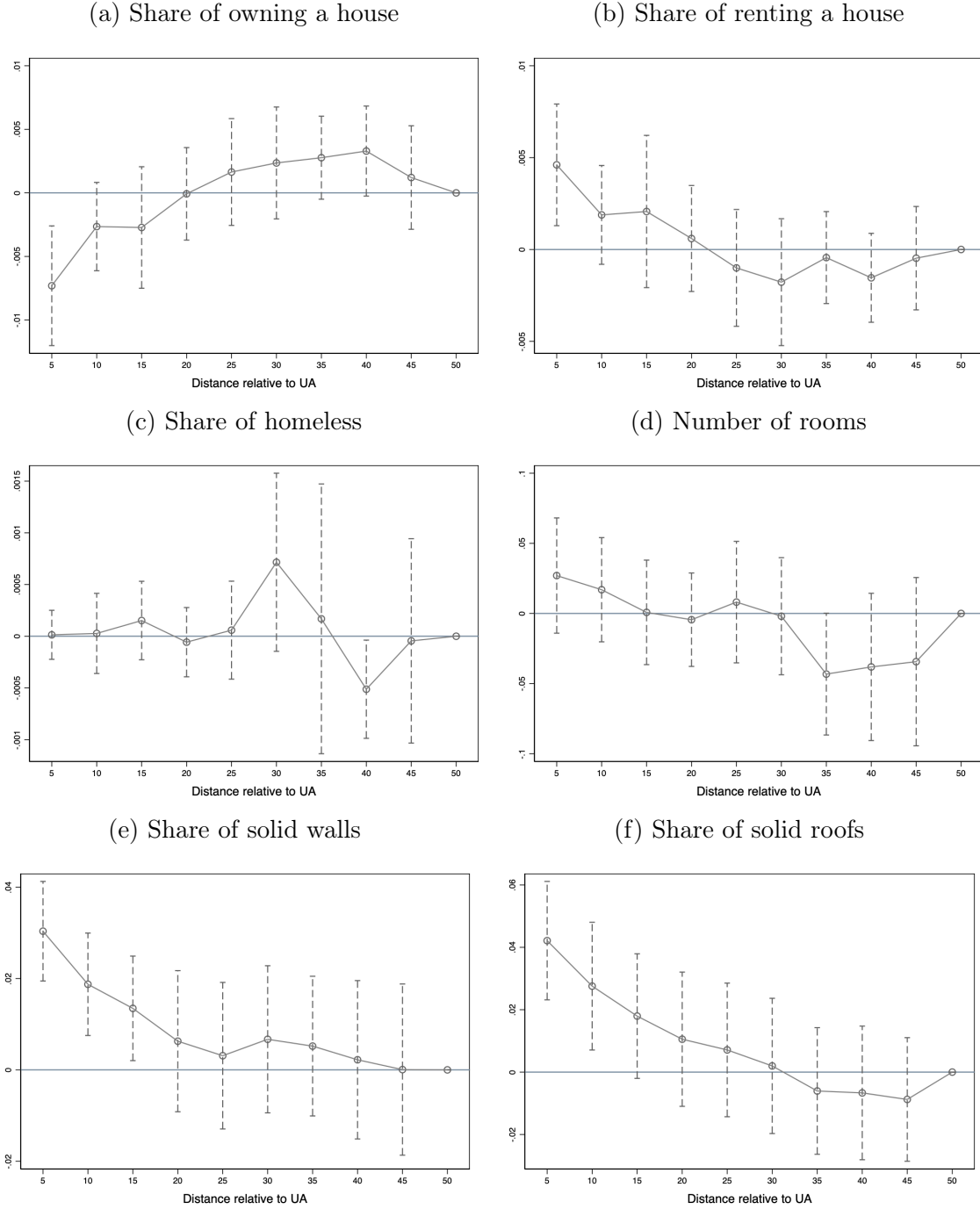


Figure D.30: Placebo: proximity to towns and housing conditions

Note: This figure displays the effects of proximity to towns on several housing indicators: the share of households that own or rent a house, the share of homeless individuals, the average number of rooms per household, and the proportion of houses with solid walls and roofs. Each dot represents the estimated parameter β_d , where each subscript d corresponds to a specific distance along the horizontal axis. For instance, a coefficient at $d = 1$ suggests that the municipality is within 5 km of a town. The model incorporates UA fixed effects, and controls include quartiles of baseline population, road presence, electrification, and distance to towns. Standard errors are clustered at the UA level. Dashed lines indicate 95% confidence intervals.

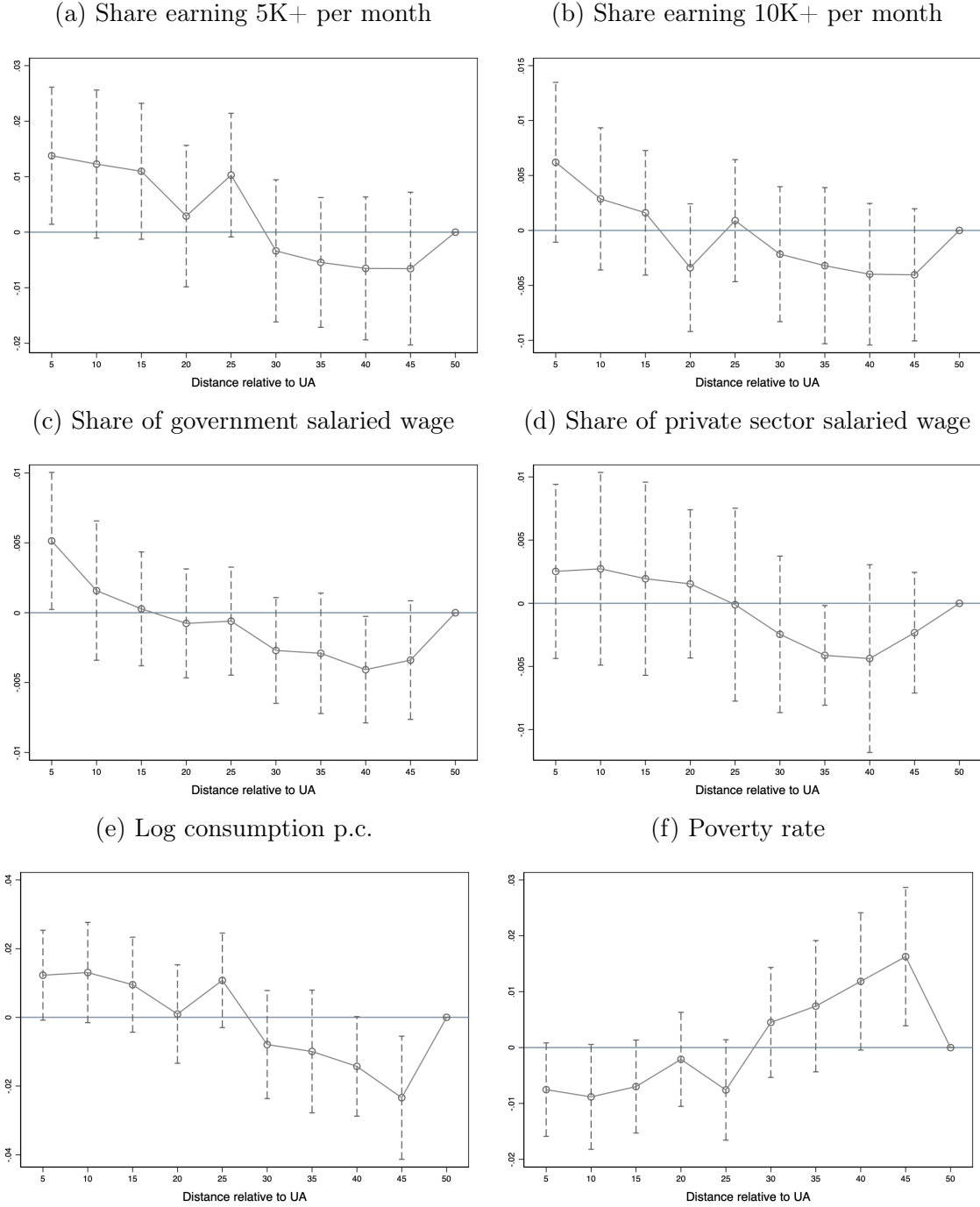


Figure D.31: Placebo: proximity to towns, income, and poverty

Note: This figure displays the effects of proximity to towns on municipality-level income, poverty rate, and estimated consumption. Consumption information is imputed based on another household survey (IHDS-II, 2011-12). SHRUG reports mean predicted consumption and the share of the population below the poverty line (27 Rupees/day) for each municipality. Each dot represents the estimated parameter β_d , where each subscript d corresponds to a specific distance along the horizontal axis. For instance, a coefficient at $d = 1$ suggests that the municipality is within 5 km of a town. The model incorporates UA fixed effects, and controls include quartiles of baseline population, road presence, electrification, and distance to towns. Standard errors are clustered at the UA level. Dashed lines indicate 95% confidence intervals.

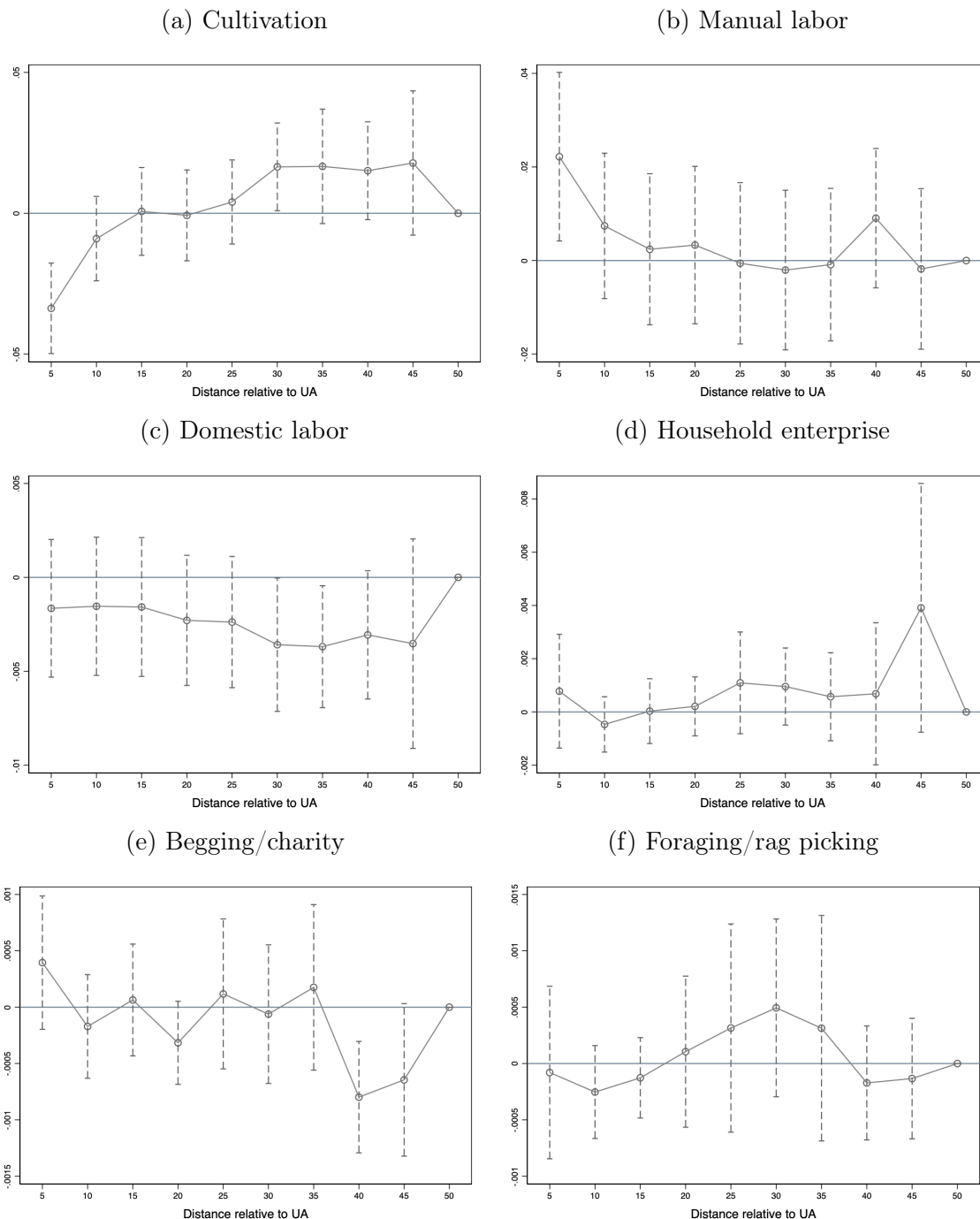


Figure D.32: Placebo: proximity to towns and income sources

Note: This figure displays the effects of proximity to towns on the main source of household income. Each dot represents the estimated parameter β_d , where each subscript d corresponds to a specific distance along the horizontal axis. For instance, a coefficient at $d = 1$ suggests that the municipality is within 5 km of a town. The model incorporates UA fixed effects, and controls include quartiles of baseline population, road presence, electrification, and distance to towns. Standard errors are clustered at the UA level. Dashed lines indicate 95% confidence intervals.

E JNNURM

In this section, we assess the impacts of the Jawaharlal Nehru National Urban Renewal Mission (JNNURM) on manufacturing firms, utilizing detailed data on the program’s implementation. Our primary data source is the audit report by [Grant Thornton \(2011\)](#), which served as the appraisal agency for the JNNURM. The report is based on information collected from a pre-approved sample of 41 mission cities, representing two-thirds of all JNNURM cities. The data includes city-level information on the number of projects approved and completed, as well as both approved and actual expenditures for the Urban Infrastructure and Governance (UIG) and Basic Services to the Urban Poor (BSUP) schemes, as of 2011.

To measure the implementation of the JNNURM, we construct four city-level metrics: logged per capita expenditure and the number of sanctioned projects in each scheme. It is important to note that using the number of sanctioned projects likely overestimates the true impacts of the JNNURM, as only 17.8% of UIG projects and 1.5% of BSUP projects were completed by 2011 ([Comptroller and Auditor General of India, 2012](#)). To estimate the causal effects of the JNNURM, we interact these four measures of JNNURM implementation with a post-2006 dummy variable, employing a Difference-in-Difference framework. We assess whether our estimated effects of the Repeal, as reported in the main text, remain robust when controlling for JNNURM implementation. To the best of our knowledge, our study is the first to examine the short-run causal impacts of the JNNURM on manufacturing firms.

Tables [E.36](#), [E.37](#), [E.38](#), [E.39](#), and [E.40](#) report our results on firms’ landholdings, fixed assets, number of employees, sales revenue, and TFPQ. Column (1) reports the baseline estimates as a benchmark. Column (2) report the estimates of the Repeal’s effects by dropping firms in UAs that were covered by the JNNURM. In Columns (3)-(4), we use the full sample of data and directly control for the implementation of the JNNURM. Our estimates in Column (2) become smaller by roughly 15% on average (except for TFPQ) when dropping UAs covered by the JNNURM, but generally remain statistically significant—with the exception of employment, whose estimate becomes imprecise. Importantly, the estimated Repeal’s effects in Columns (3)-(4) are very close to the baseline estimates reported in Column (1). Moreover, the effects of JNNURM implementation on almost all firm-level outcomes are small and insignificant. Collectively, the evidence indicates that the JNNURM is unlikely to bias our results.

	(1)	(2)	(3)	(4)
	Baseline	Drop JNNURM UAs	Projects	Expenditure
Repeal	0.005 (0.057)	0.026 (0.057)	0.016 (0.055)	0.012 (0.056)
Repeal $\times$ UA	0.202*** (0.056)	0.185*** (0.051)	0.185*** (0.052)	0.194*** (0.050)
BSUP projects			0.018 (0.021)	
UIG projects			-0.002 (0.011)	
BSUP expenditure p.c.				-0.046 (0.053)
UIG expenditure p.c.				0.043 (0.043)
Observations	338,686	235,346	338,686	338,686
R^2	0.183	0.212	0.183	0.183
District FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
Size $\times$ Year FE	Yes	Yes	Yes	Yes

Note: Column (1) is the baseline specification. Column (2) drops JNNURM-covered UAs. Columns (3)–(4) add JNNURM project and expenditure controls, respectively. Robust standard errors clustered at the state level are in parentheses. Significance levels: * 10%, ** 5%, *** 1%.

Table E.36: Effects of the Repeal on landholdings: account for JNNURM

	(1)	(2)	(3)	(4)
	Baseline	Drop JNNURM UAs	Projects	Expenditure
Repeal	0.081 (0.104)	0.155 (0.104)	0.090 (0.104)	0.093 (0.104)
Repeal $\times$ UA	0.145** (0.058)	0.124* (0.064)	0.135** (0.059)	0.128** (0.057)
BSUP projects			−0.001 (0.009)	
UIG projects			0.003 (0.003)	
BSUP expenditure p.c.				−0.005 (0.025)
UIG expenditure p.c.				0.015 (0.021)
Observations	530,008	340,990	530,008	530,008
R^2	0.254	0.310	0.254	0.254
District FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
Size $\times$ Year FE	Yes	Yes	Yes	Yes

Note: Column (1) is the baseline specification. Column (2) drops JNNURM-covered UAs. Columns (3)–(4) add JNNURM project and expenditure controls, respectively. Robust standard errors clustered at the state level are in parentheses. Significance levels: * 10%, ** 5%, *** 1%.

Table E.37: Effects of the Repeal on fixed assets: account for JNNURM

	(1)	(2)	(3)	(4)
	Baseline	Drop JNNURM UAs	Projects	Expenditure
Repeal	−0.029 (0.038)	0.000 (0.038)	−0.030 (0.040)	−0.033 (0.039)
Repeal × UA	0.067** (0.026)	0.055 (0.042)	0.069** (0.027)	0.073** (0.027)
BSUP projects			−0.003 (0.006)	
UIG projects			0.001 (0.003)	
BSUP expenditure p.c.				0.005 (0.025)
UIG expenditure p.c.				−0.007 (0.019)
Observations	533,802	343,519	533,802	533,802
R^2	0.148	0.172	0.148	0.148
District FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
Size × Year FE	Yes	Yes	Yes	Yes

Note: Column (1) is the baseline specification. Column (2) drops JNNURM-covered UAs. Columns (3)–(4) add JNNURM project and expenditure controls, respectively. Robust standard errors clustered at the state level are in parentheses. Significance levels: * 10%, ** 5%, *** 1%.

Table E.38: Effects of the Repeal on employees: account for JNNURM

	(1)	(2)	(3)	(4)
	Baseline	Drop JNNURM UAs	Projects	Expenditure
Repeal	-0.021 (0.049)	0.022 (0.054)	-0.014 (0.050)	-0.018 (0.051)
Repeal $\times$ UA	0.160*** (0.046)	0.119* (0.058)	0.152*** (0.046)	0.156*** (0.045)
BSUP projects			-0.003 (0.006)	
UIG projects			0.004 (0.003)	
BSUP expenditure p.c.				-0.015 (0.027)
UIG expenditure p.c.				0.015 (0.020)
Observations	464,755	301,786	464,755	464,755
R^2	0.244	0.287	0.244	0.244
District FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
Size $\times$ Year FE	Yes	Yes	Yes	Yes

Note: Column (1) is the baseline specification. Column (2) drops JNNURM-covered UAs. Columns (3)–(4) add JNNURM project and expenditure controls, respectively. Robust standard errors clustered at the state level are in parentheses. Significance levels: * 10%, ** 5%, *** 1%.

Table E.39: Effects of the Repeal on sales: account for JNNURM

	(1)	(2)	(3)	(4)
	Baseline	Drop JNNURM UAs	Projects	Expenditure
Repeal	0.004 (0.073)	0.002 (0.092)	0.003 (0.073)	0.000 (0.073)
Repeal $\times$ UA	0.120** (0.056)	0.182** (0.065)	0.122** (0.056)	0.124** (0.052)
BSUP projects			−0.002 (0.009)	
UIG projects			0.000 (0.003)	
BSUP expenditure p.c.				0.017 (0.024)
UIG expenditure p.c.				−0.017 (0.017)
Observations	365,512	241,142	365,512	365,512
R^2	0.489	0.514	0.489	0.489
District FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
Size $\times$ Year FE	Yes	Yes	Yes	Yes

Note: Column (1) is the baseline specification. Column (2) drops JNNURM-covered UAs. Columns (3)–(4) add JNNURM project and expenditure controls, respectively. Robust standard errors clustered at the state level are in parentheses. Significance levels: * 10%, ** 5%, *** 1%.

Table E.40: Effects of the Repeal on TFPQ: account for JNNURM